AI VIET NAM
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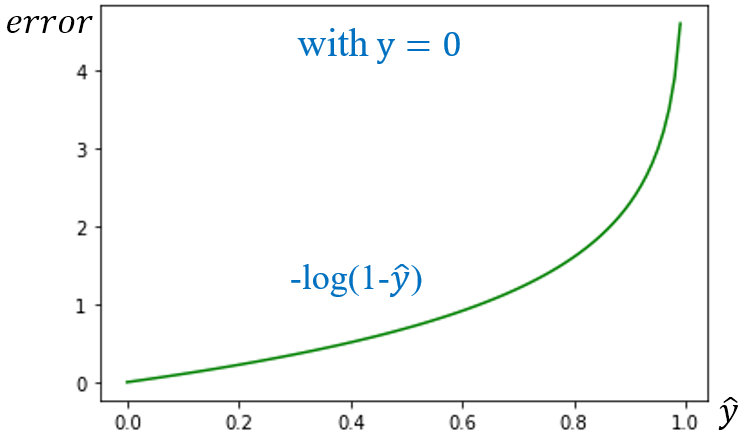
Logistic Regression

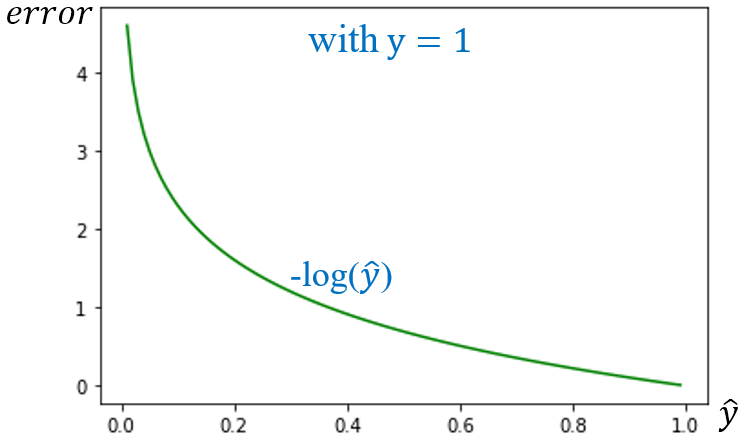
Quang-Vinh Dinh
PhD in Computer Science

Proofreaders: Cao Bá Hoàng - QA/QC: Nguyễn Thông

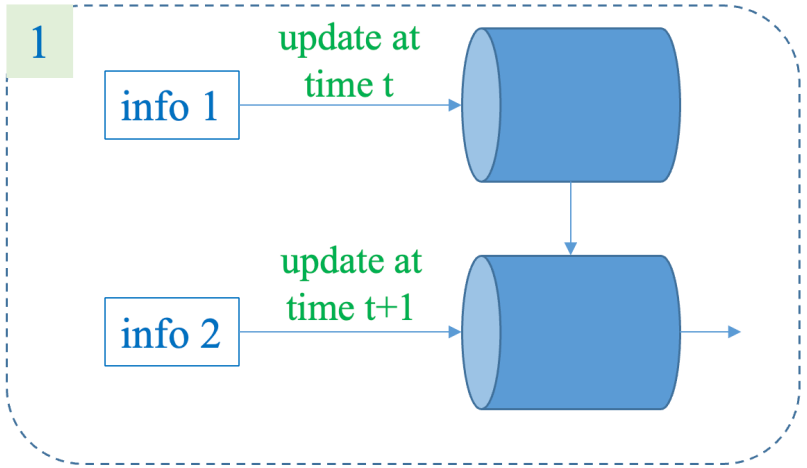
Objectives

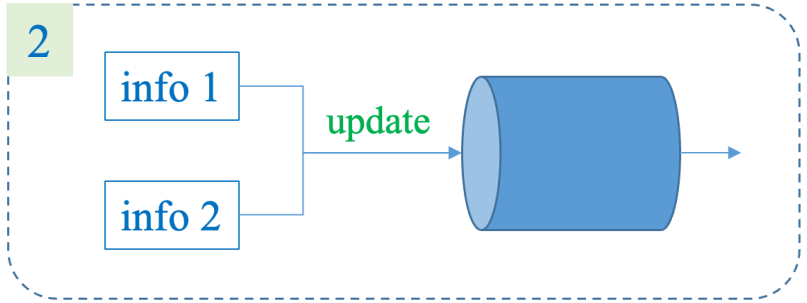
Logistic Regression



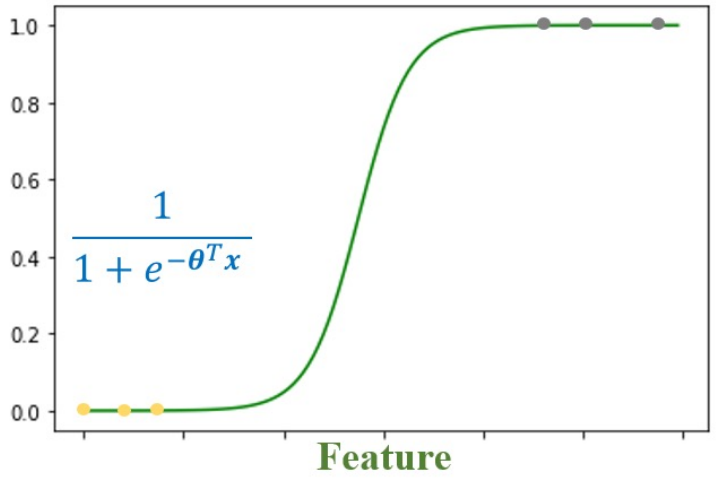


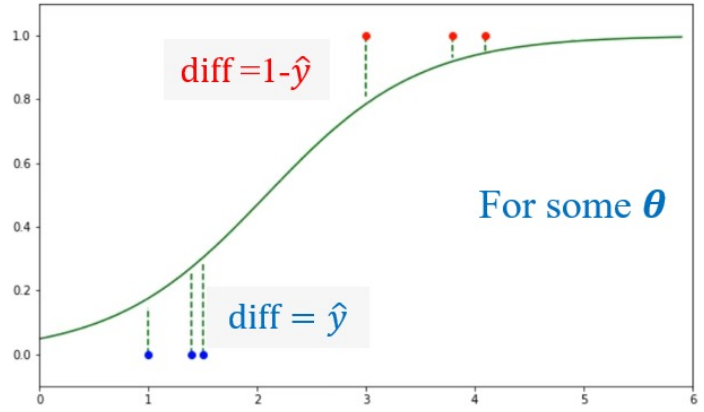
Using Multi-samples



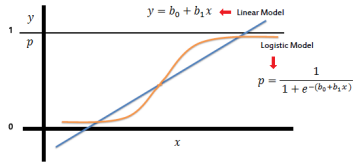


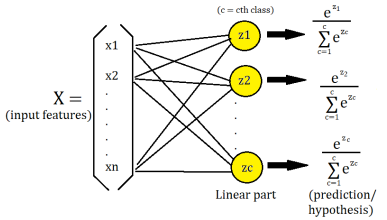
Discussion

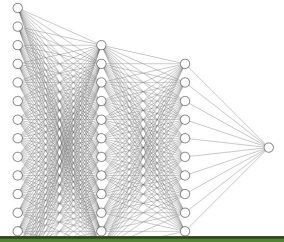


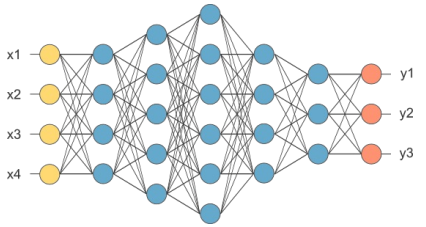


Motivation

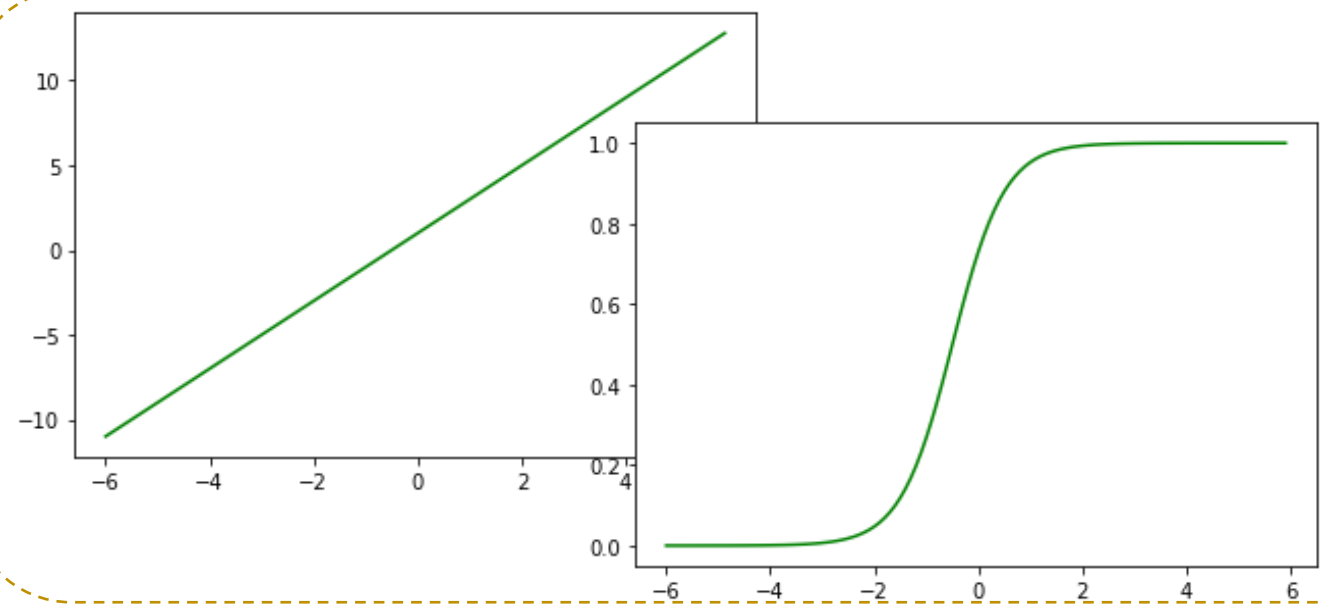


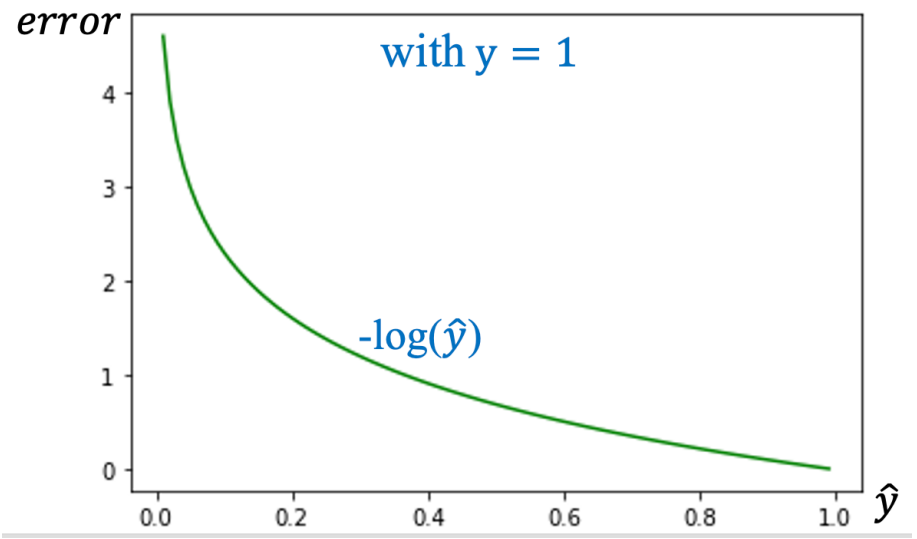






Linear Regression → Logistic Regression → Softmax Regression → Multilayer Perceptron → Deep Learning





Outline

SECTION 1

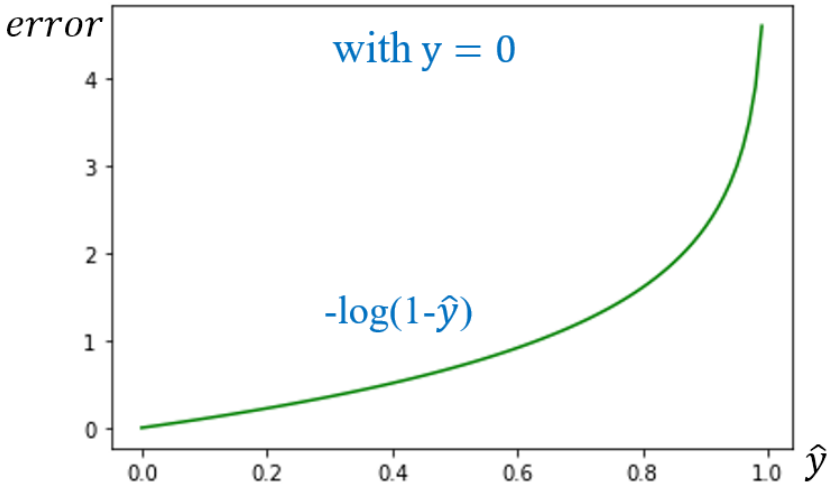
Logistic Regression

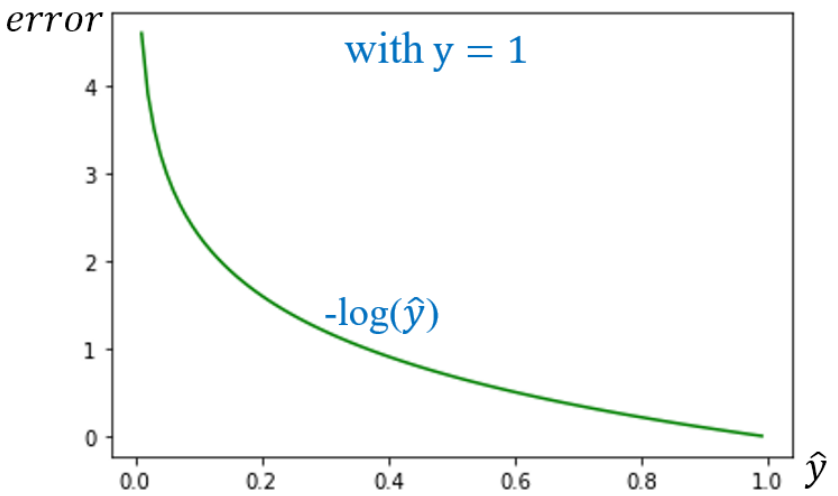
SECTION 2

Using Multi-samples

SECTION 3

Discussion





Linear Regression

❖ Area-based house price prediction

$$\text{predicted_price} = w * \text{area} + b$$

$$\text{error} = (\text{predicted_price} - \text{real_price})^2$$

$$\hat{y} = wx + b$$

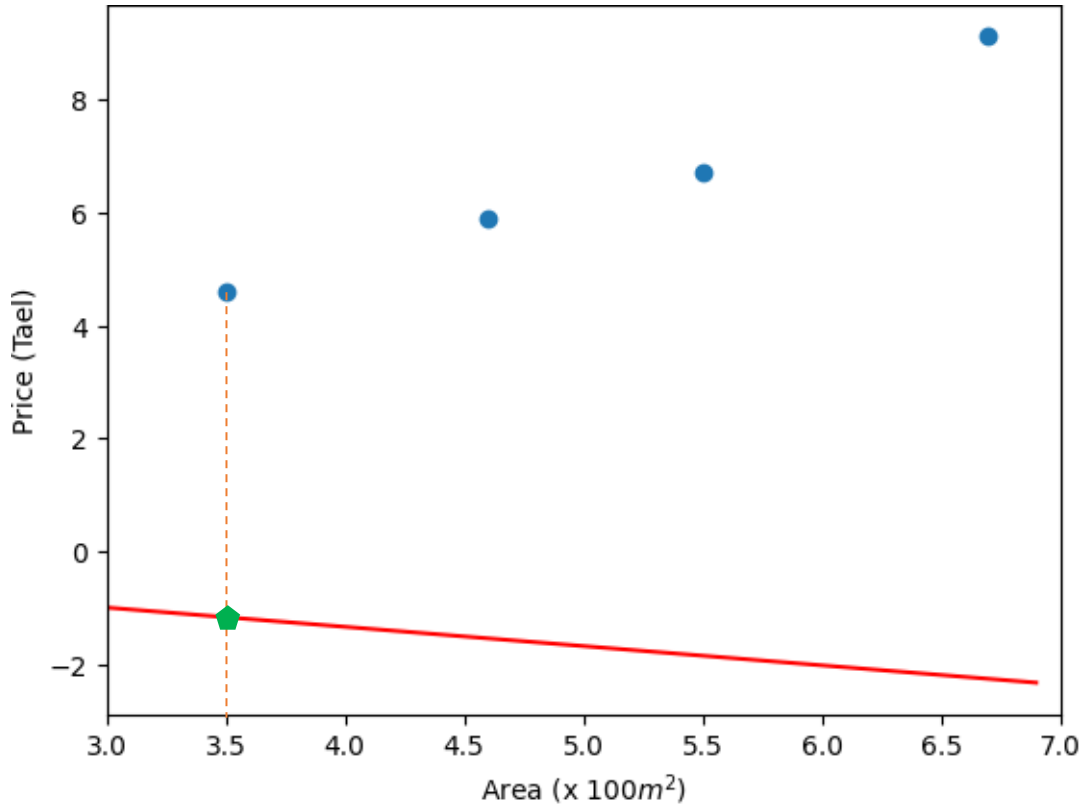
$$L(\hat{y}, y) = (\hat{y} - y)^2$$

area	price	predicted	error
6.7	9.1	-2.238	128.55
4.6	5.9	-1.524	55.11
3.5	4.6	-1.15	33.06
5.5	6.7	-1.83	72.76

$w = -0.34$

$b = 0.04$

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7



Linear Regression

❖ Area-based house price prediction

$$\text{predicted_price} = w * \text{area} + b$$

$$\text{error} = (\text{predicted_price} - \text{real_price})^2$$

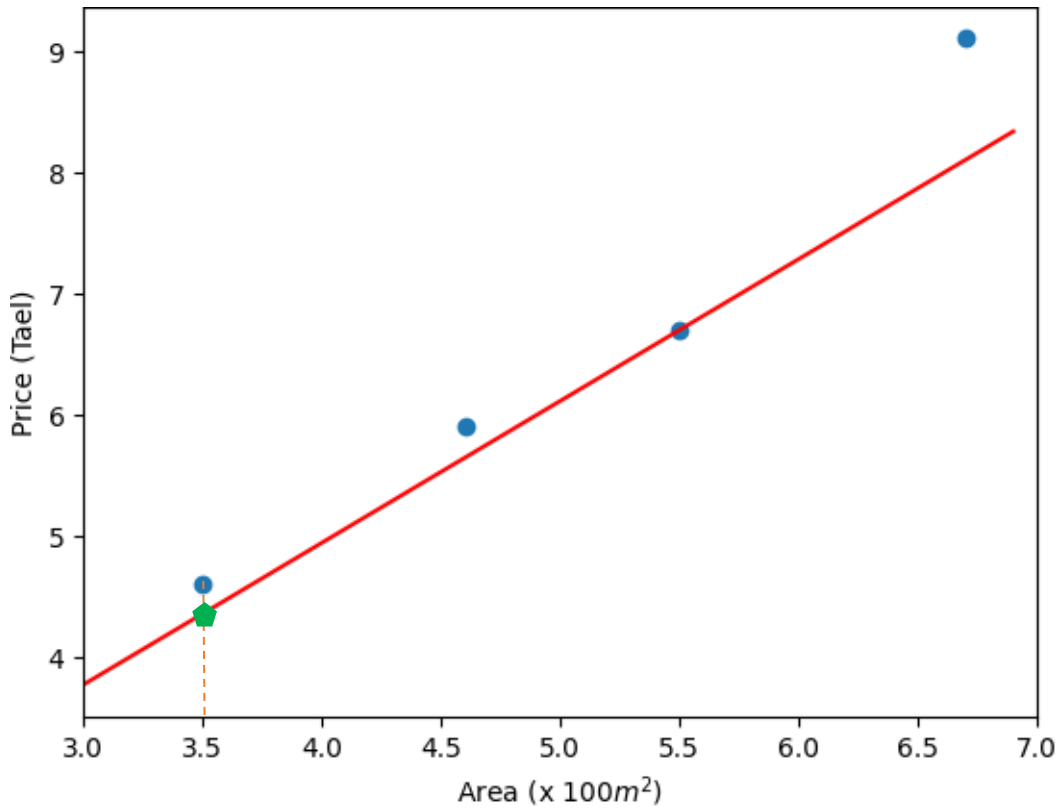
$$\hat{y} = wx + b$$

$$L(\hat{y}, y) = (\hat{y} - y)^2$$

area	price	predicted	error
6.7	9.1	8.099	1.002
4.6	5.9	5.642	0.066
3.5	4.6	4.355	0.06
5.5	6.7	6.695	0.00002

$w = 1.17$

$b = 0.26$



Linear Regression

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7

Linear equation

$$\hat{y} = wx + b$$

where $\hat{y}$ is a predicted value,

w and b are parameters

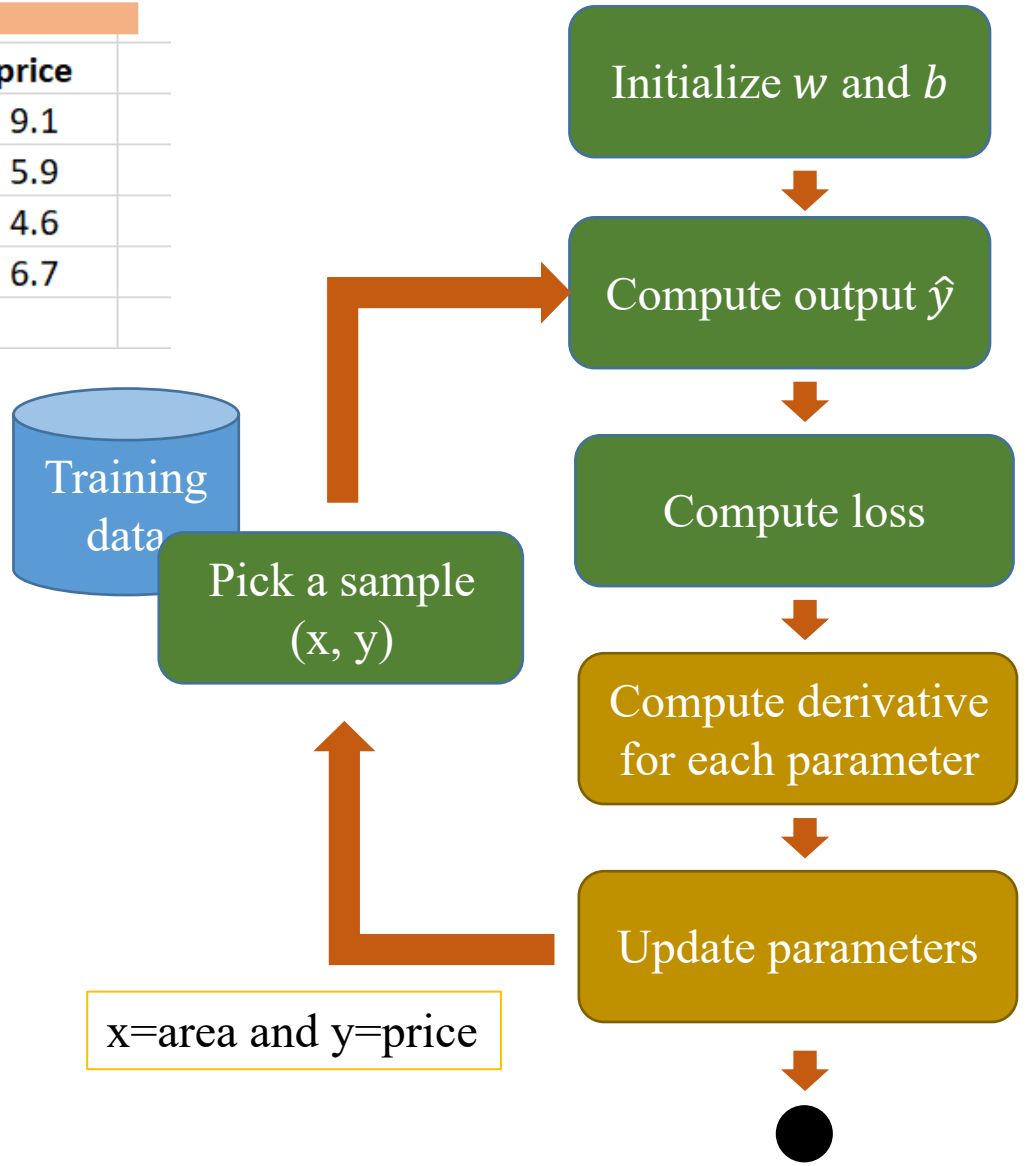
and x is input feature

Error (loss) computation

Idea: compare predicted values $\hat{y}$ and label values y

Squared loss

$$L(\hat{y}, y) = (\hat{y} - y)^2$$



Linear Regression

Linear equation

$$\hat{y} = wx + b$$

where $\hat{y}$ is a predicted value,

w and b are parameters

and x is input feature

Error (loss) computation

Idea: compare predicted values $\hat{y}$ and label values y

Squared loss

$$\mathrm{L}(\hat{y}, y) = (\hat{y} - y)^2$$

Find better w and b

Use gradient descent to minimize the loss function

Compute derivate for each parameter

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial w} = 2x(\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial b} = 2(\hat{y} - y)$$

Update parameters

$$w = w - \eta \frac{\partial L}{\partial w} \qquad b = b - \eta \frac{\partial L}{\partial b}$$

η is learning rate

Idea of Logistic Regression

❖ Using the approach of Linear regression

Area-based House Price Data

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7

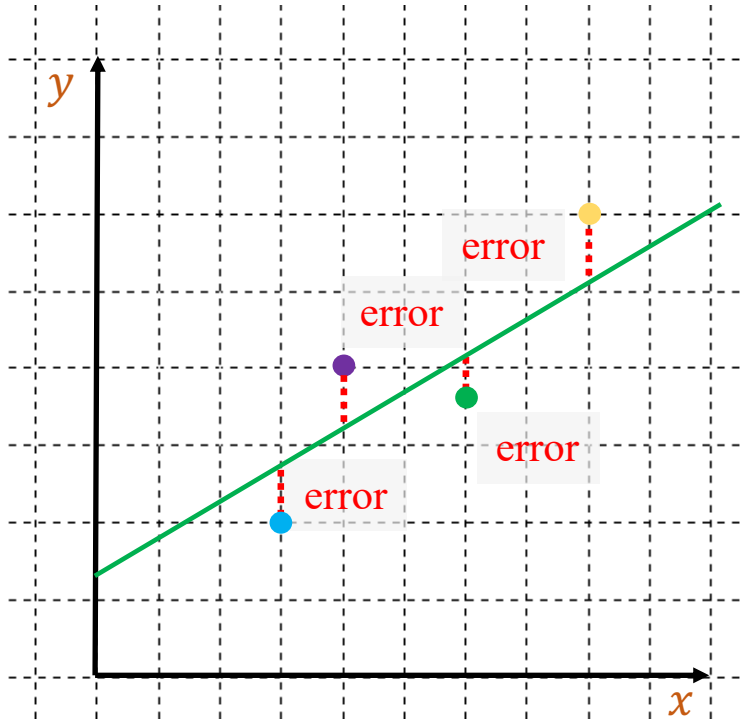
Training data

construct →

$$\hat{y} = \boldsymbol{\theta}^T \boldsymbol{x} = wx + b$$

$$\hat{y} \in (-\infty \quad +\infty)$$

Model



Find the line $\hat{y} = \boldsymbol{\theta}^T \boldsymbol{x}$ that is best fitting to given data, then use $\hat{y}$ to predict for new data

Idea of Logistic Regression

❖ Given a new kind of data

Feature	Label
Petal_Length	Category
1.4	Flower A
1	Flower A
1.5	Flower A
3	Flower B
3.8	Flower B
4.1	Flower B

Category 0 (Flower A rows)

Category 1 (Flower B rows)

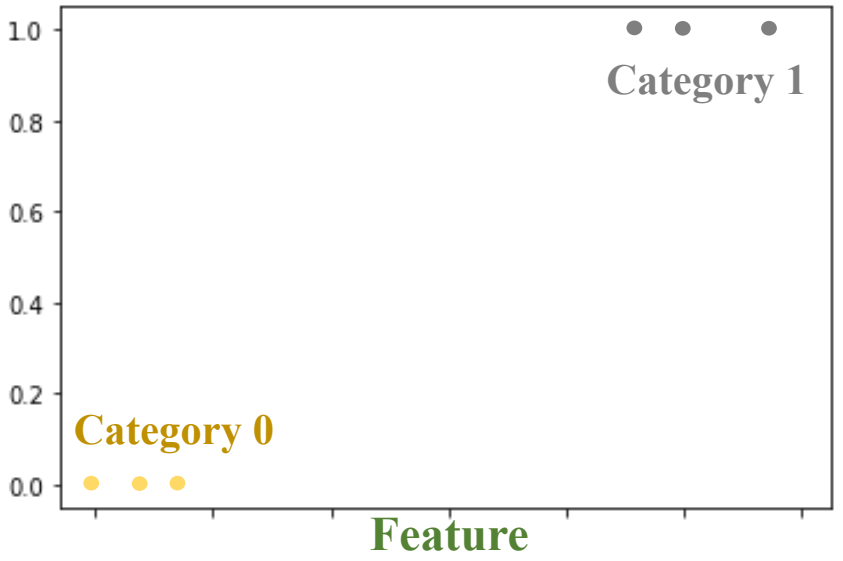
Assign numbers to categories

Feature	Label
Petal_Length	Category
1.4	0
1	0
1.5	0
3	1
3.8	1
4.1	1

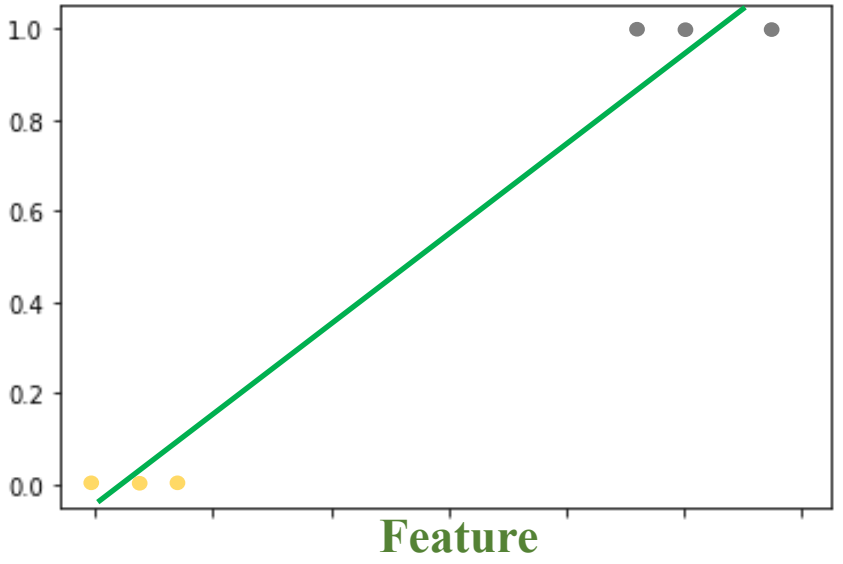
Category 0 (0 rows)

Category 1 (1 rows)

Plot data



A line is not suitable for this data

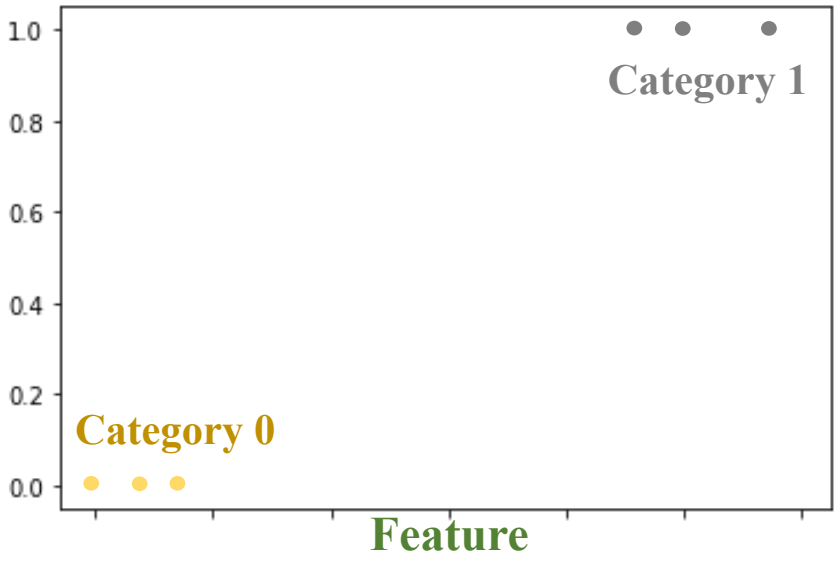


What function?

Discussion

1) If cannot fit, how to make a dicision?

2) Is it ok to work with infinity?



$$\hat{y} = \boldsymbol{\theta}^T \boldsymbol{x} = wx + b$$

$$\hat{y} \in (-\infty \quad +\infty)$$

Mô hình Logistic Growth

❖ Look up what science has done!

Phương trình vi phân của mô hình logistic growth có dạng:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = ry\left(1 - \frac{y}{K}\right)$$

- ✓ $y^{(t)}$: Kích thước quần thể tại thời gian t
- ✓ r: Tốc độ tăng trưởng nội tại
- ✓ K: Sức chứa tối đa

Giải phương trình vi phân ta được:

$$y(t) = \frac{K}{1 + \mathrm{e}^{-r(\mathrm{t}-y_0)}}$$

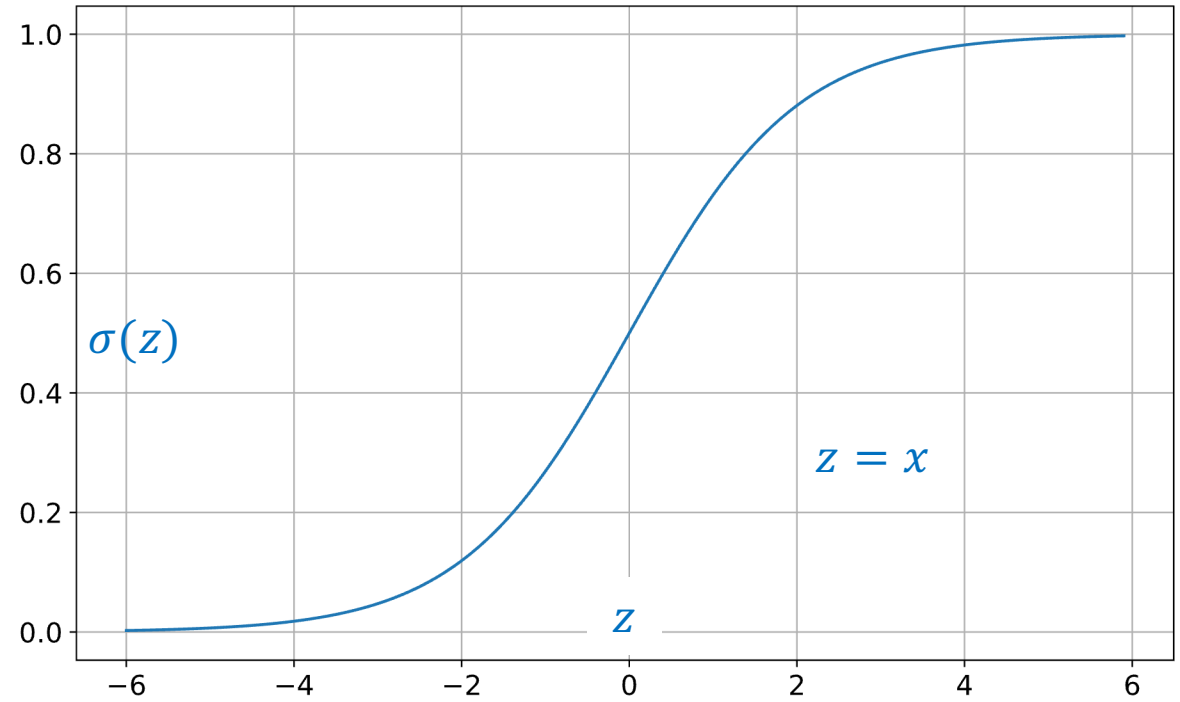


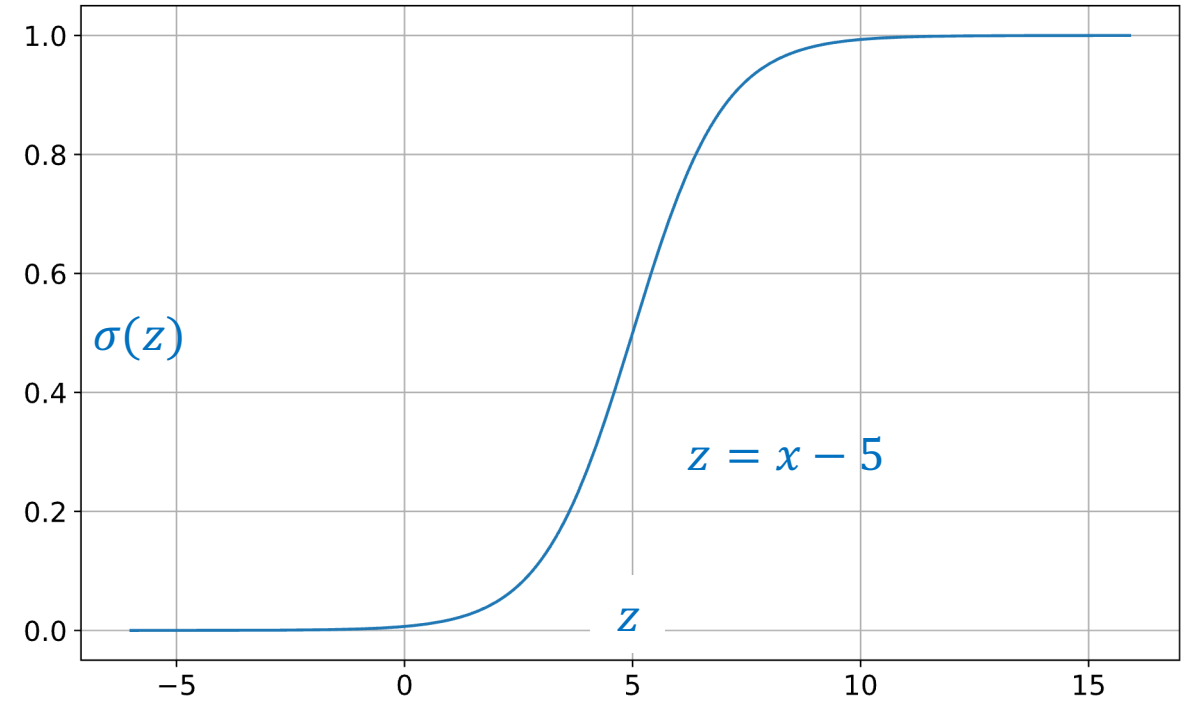
Pierre-François Verhulst (1804–1849)

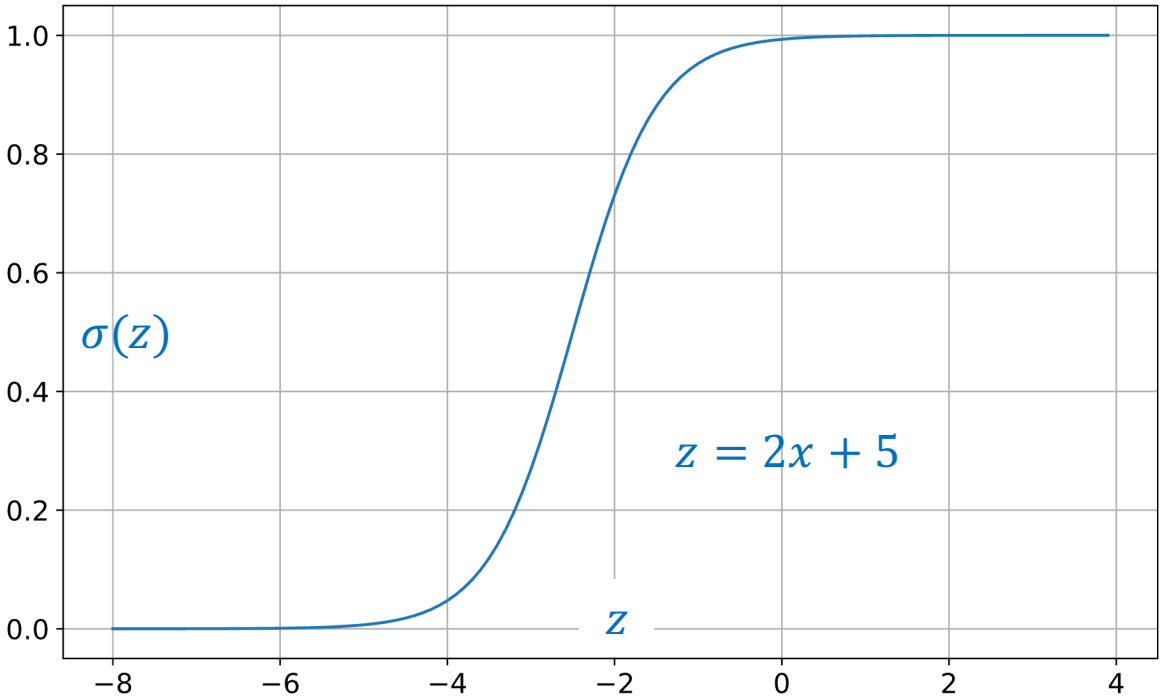
https://en.wikipedia.org/wiki/Logistic_function

Sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



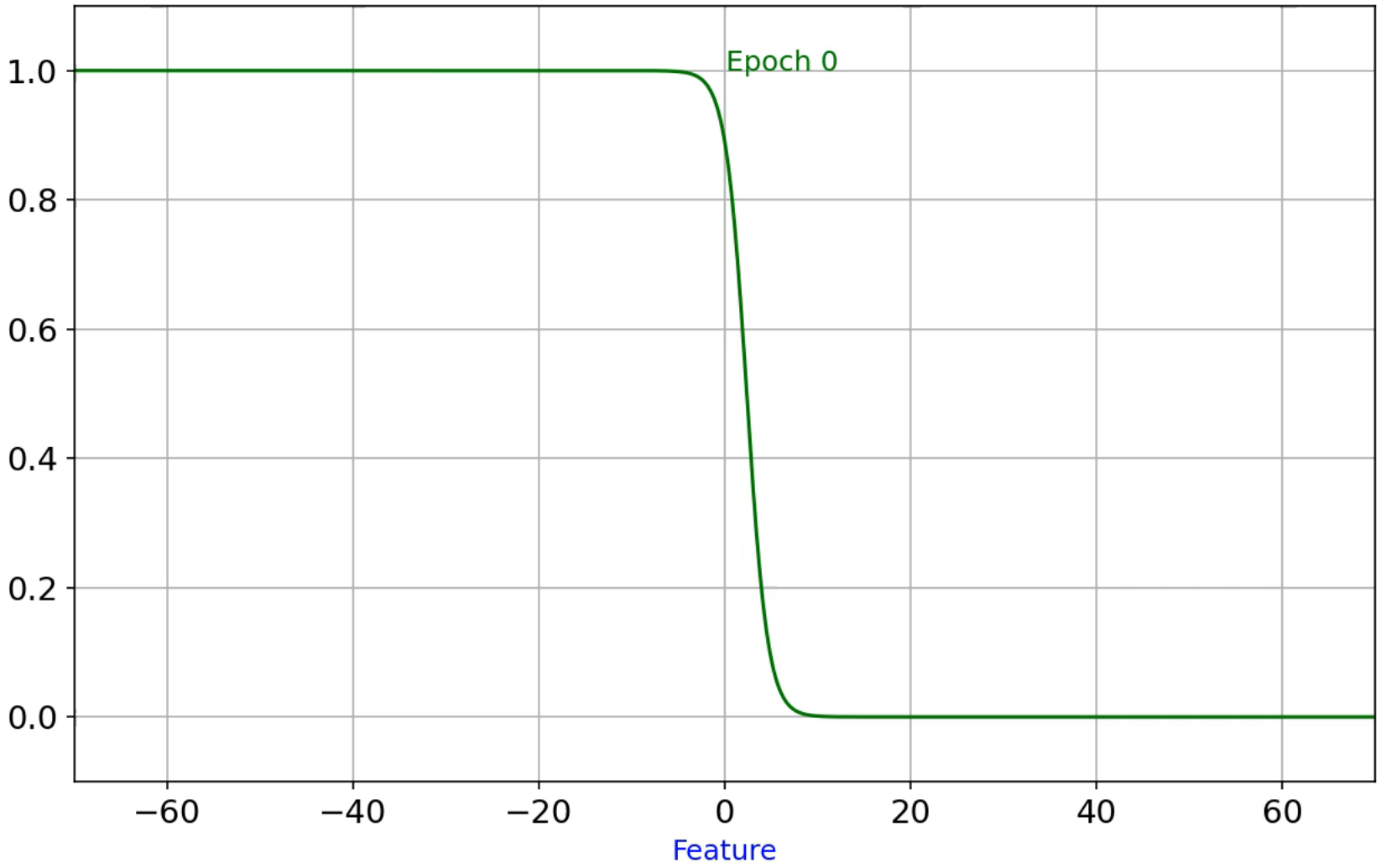




Sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

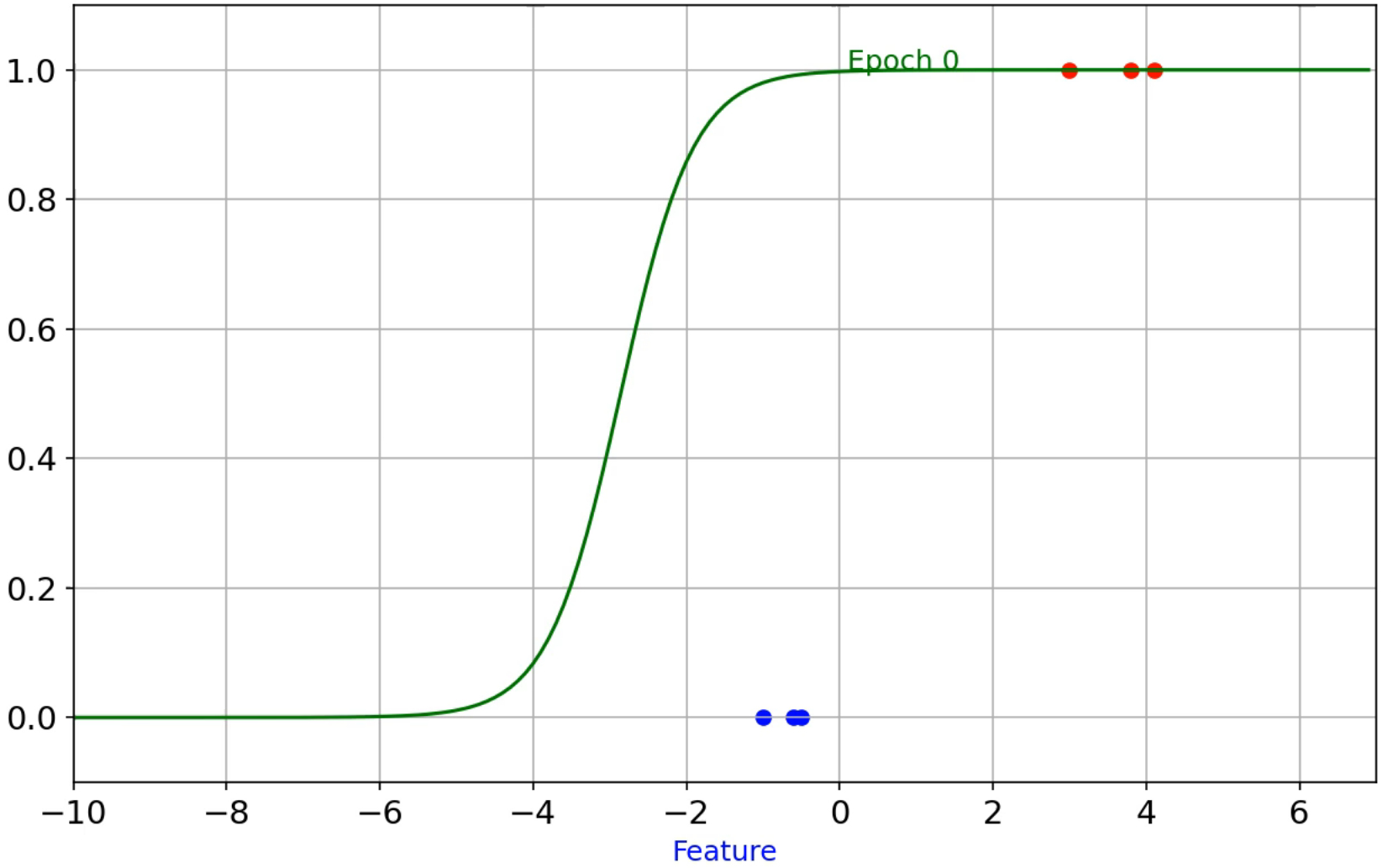
$$z = wx + b$$



Sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$z = wx + b$$



Logistic Regression

❖ Using the Sigmoid function

Sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

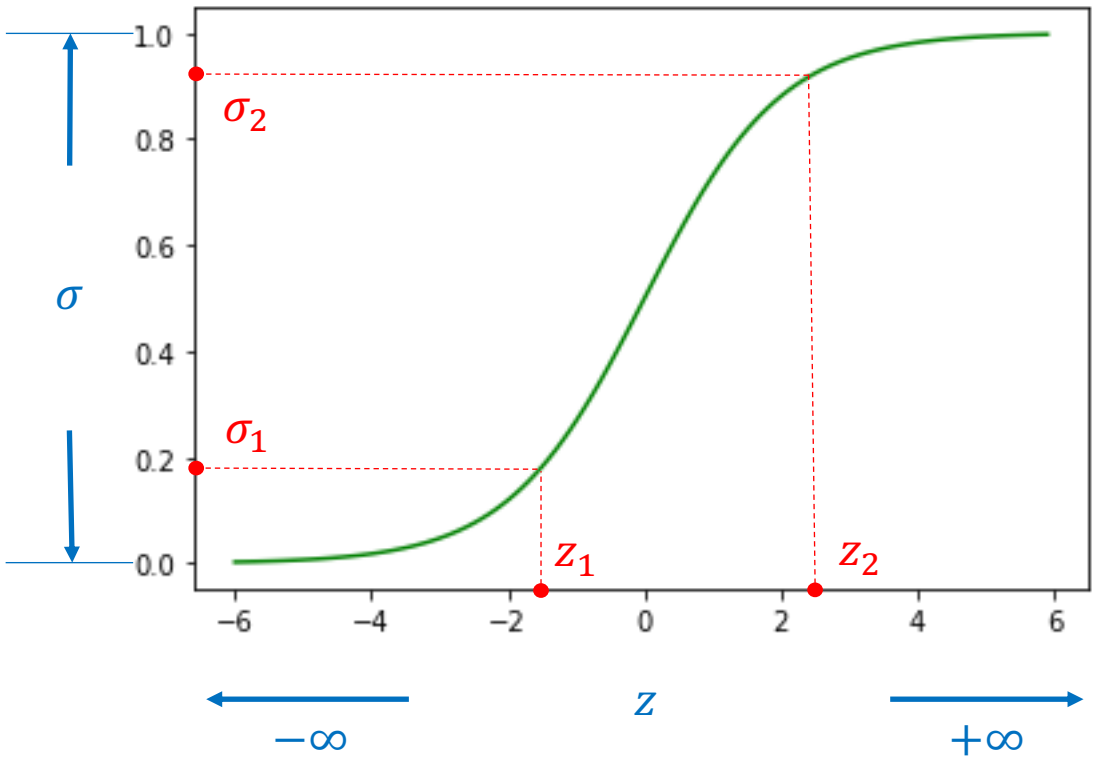
$$z \in (-\infty \quad +\infty)$$

$$\sigma(z) \in (0 \quad 1)$$

Property

$$\forall z_1 z_2 \in [a \quad b] \text{ and } z_1 \leq z_2$$

$$\rightarrow \sigma(z_1) \leq \sigma(z_2)$$

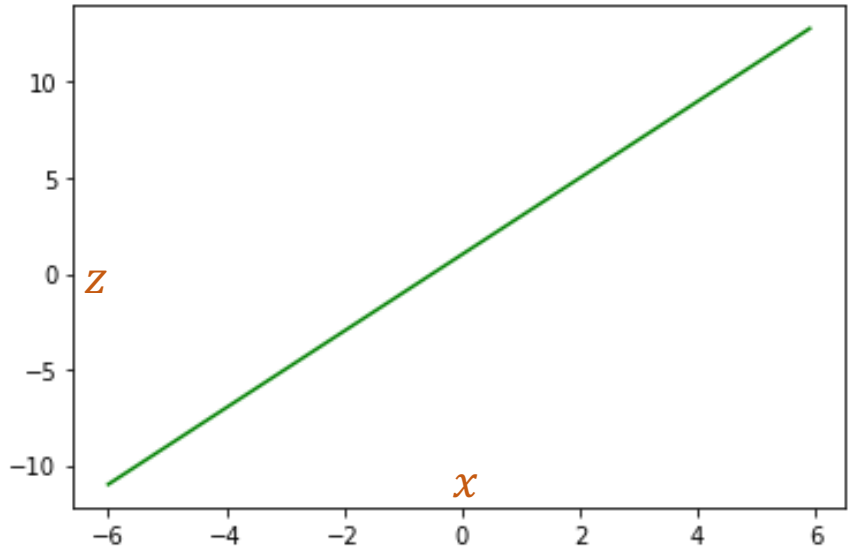


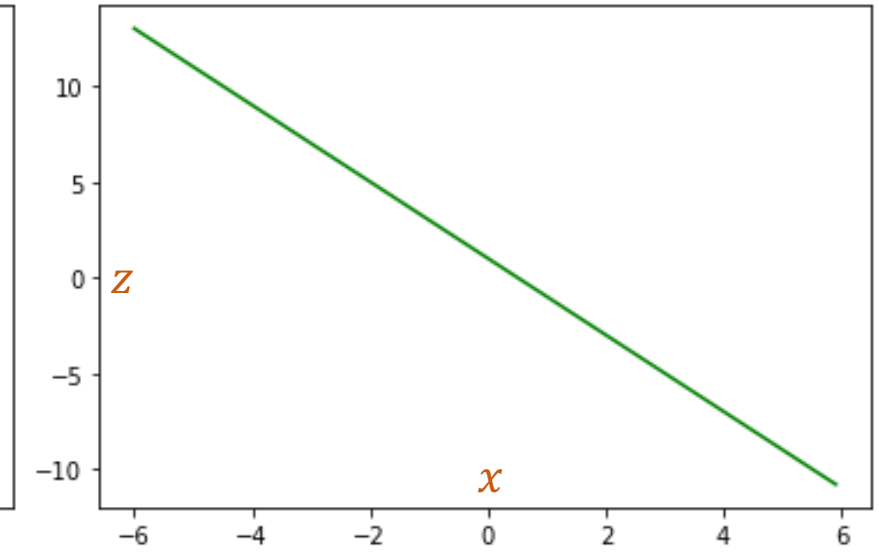
Logistic Regression

❖ Using the Sigmoid function

$$z = wx + b$$

$$z \in (-\infty \quad +\infty)$$

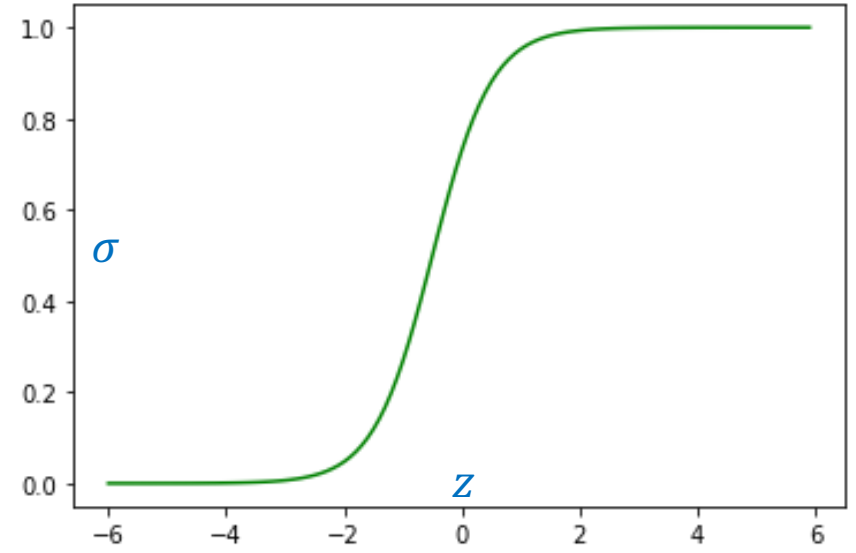


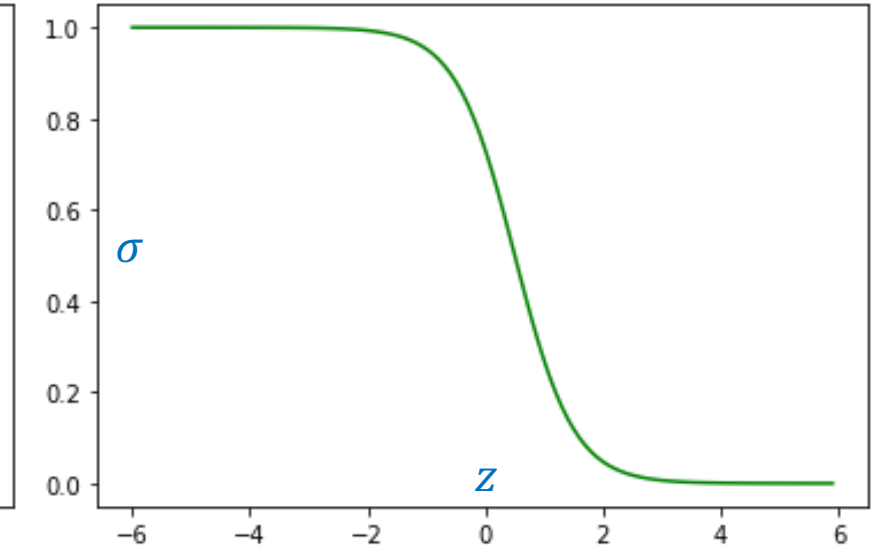


$$z = wx + b$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma(z) \in (0 \quad 1)$$



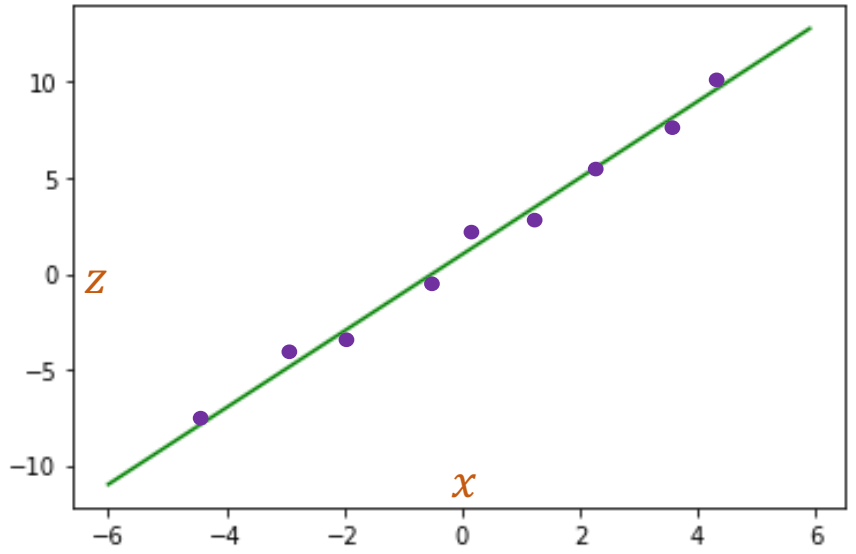


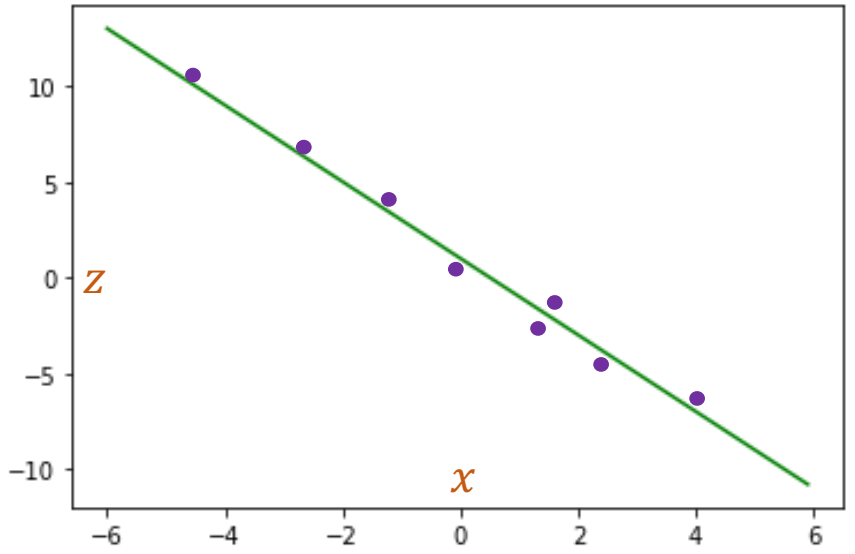
Logistic Regression

❖ Using the Sigmoid function

$$z = wx + b$$

$$z \in (-\infty \quad +\infty)$$

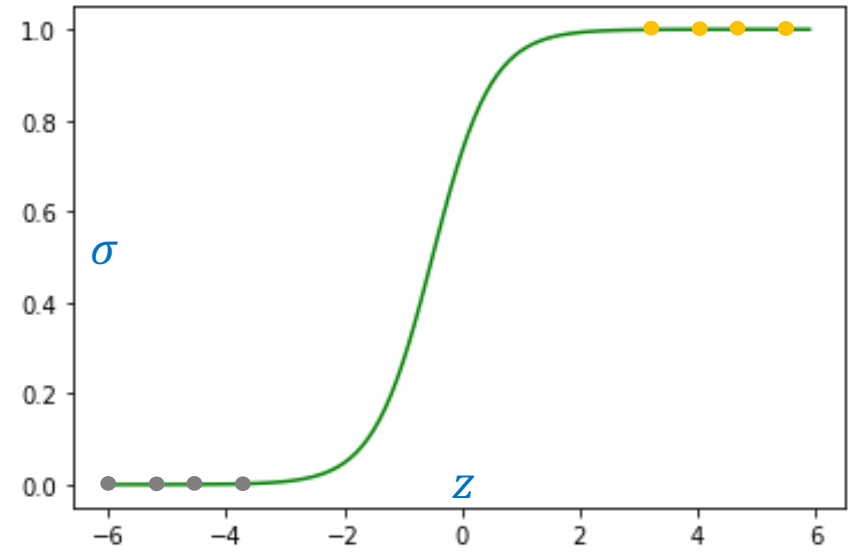


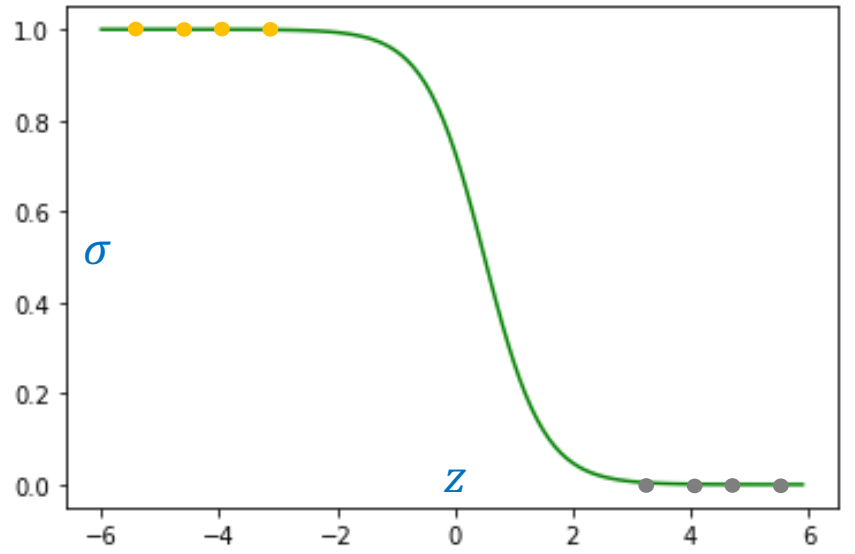


$$z = wx + b$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma(z) \in (0 \quad 1)$$





Logistic Regression

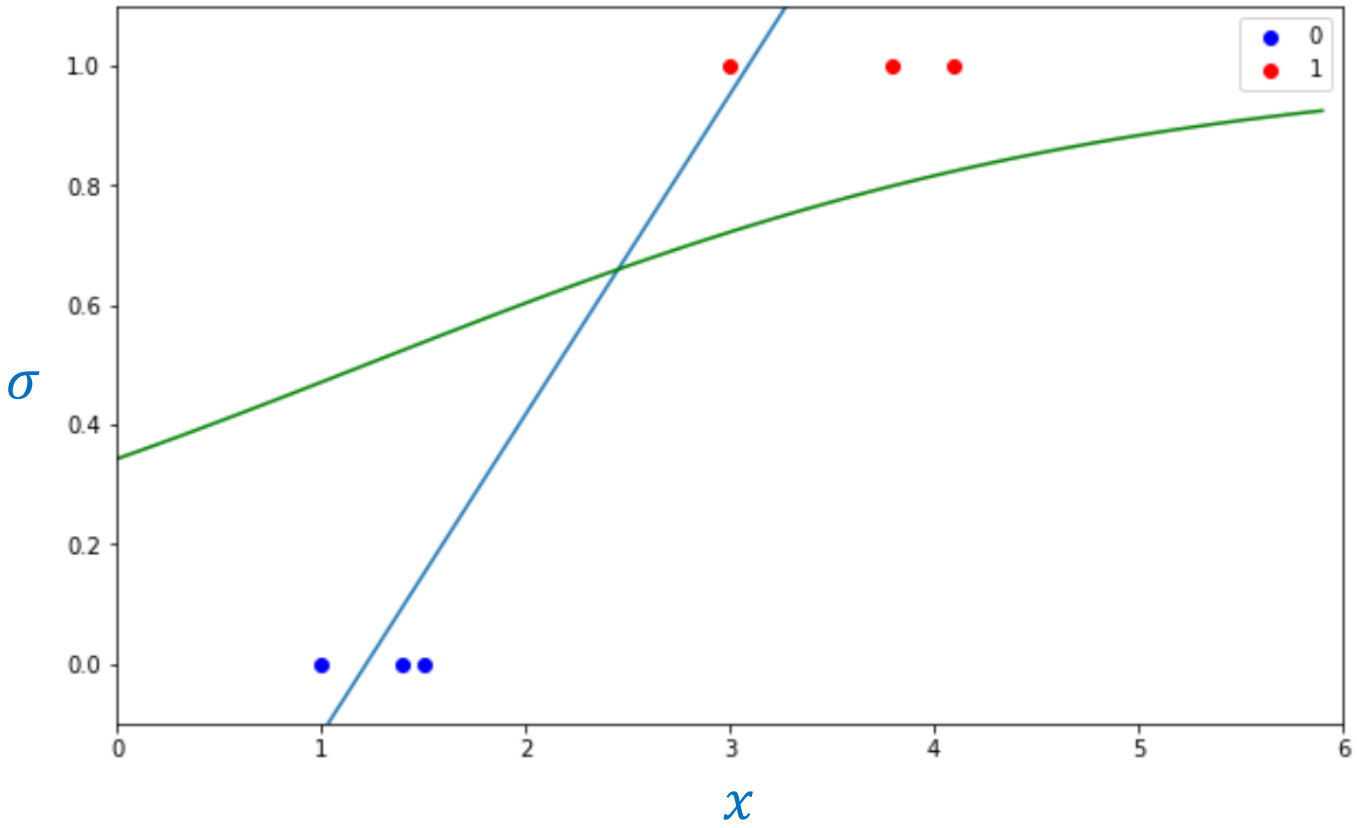
❖ Using the Sigmoid function

Feature	Label
Petal_Length	Category
1.4	0
1	0
1.5	0
3	1
3.8	1
4.1	1

Category 0

Category 1

z	$\sigma(z)$
0.095	0.52
-0.119	0.47
0.1485	0.53
0.951	0.72
1.379	0.79
1.5395	0.82



$$z = wx + b$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma(z) \in (0 \quad 1)$$

$$z = 0.535 * x - 0.654$$

Logistic Regression

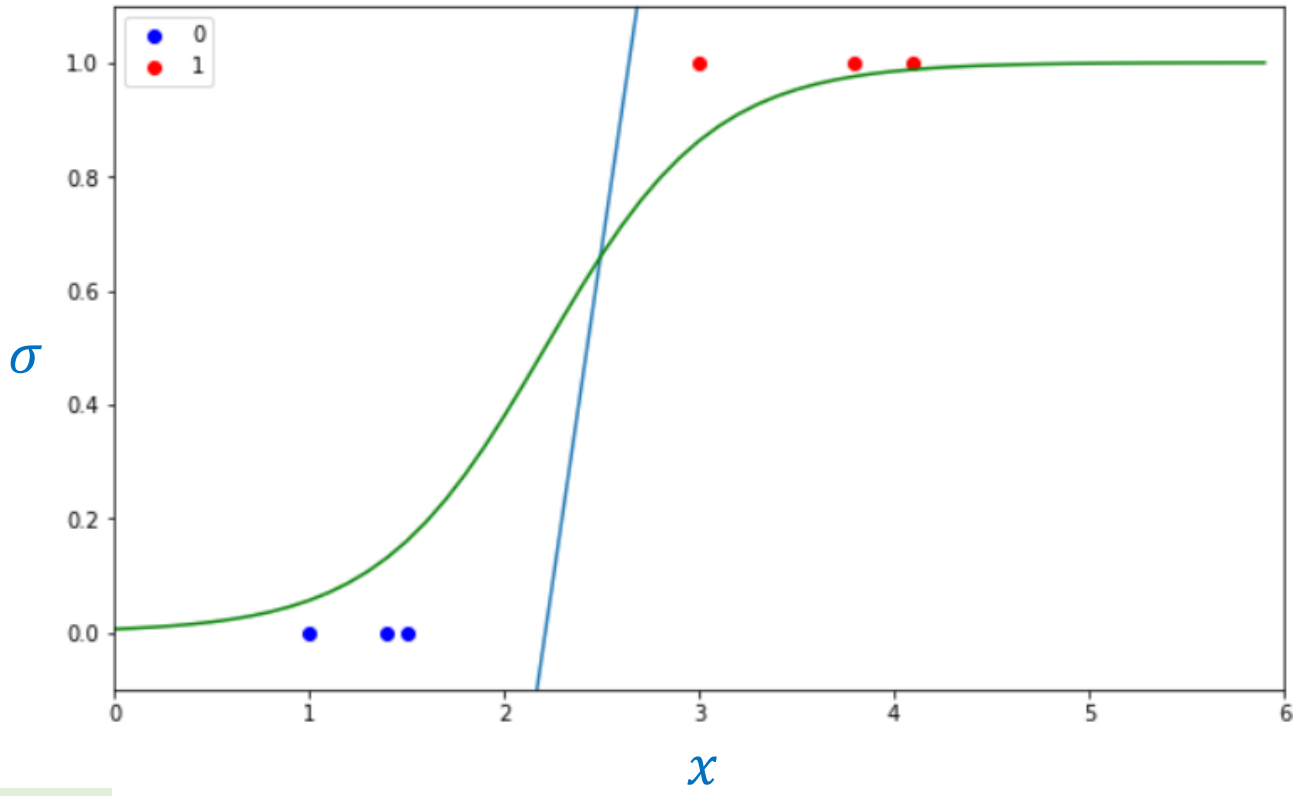
❖ Using the Sigmoid function

Feature	Label
Petal_Length	Category
1.4	0
1	0
1.5	0
3	1
3.8	1
4.1	1

Category 0

Category 1

z	$\sigma(z)$
-1.89	0.1309
-2.82	0.0559
-1.65	0.1598
1.837	0.8625
3.701	0.9759
4.401	0.9878



$$z = wx + b$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma(z) \in (0 \quad 1)$$

$$z = 2.331 * x - 5.156$$

How to Compute Loss?

Feature	Label
Petal_Length	Category
1.4	0
1	0
1.5	0
3	1
3.8	1
4.1	1

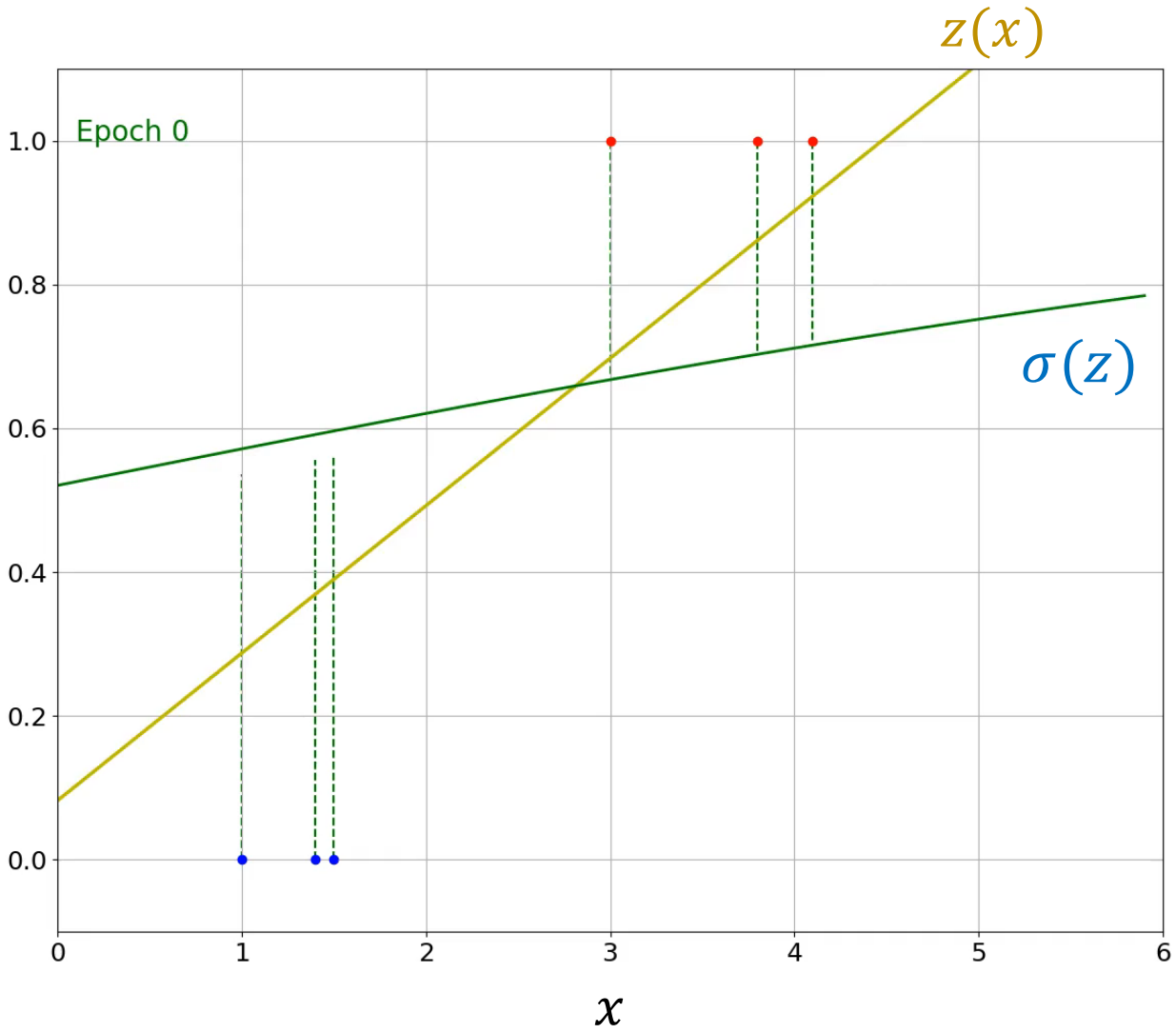
Category 0

Category 1

$$z = wx + b$$

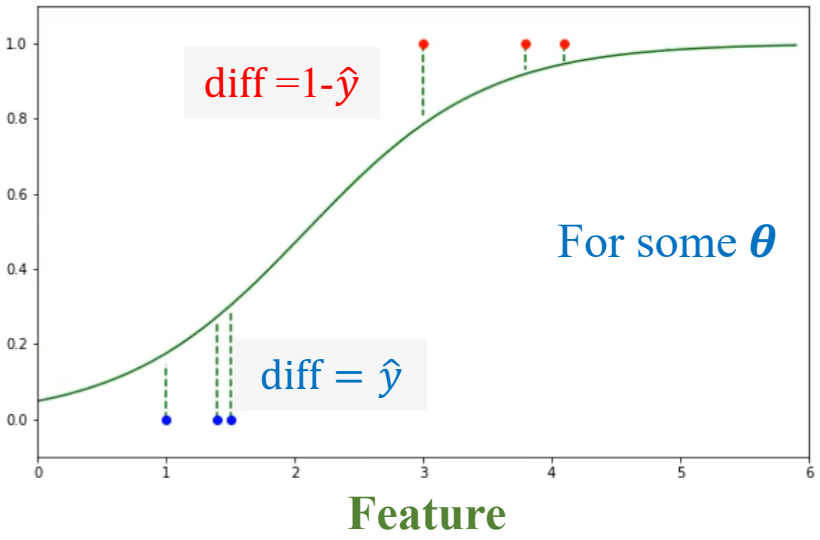
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma(z) \in (0 \quad 1)$$

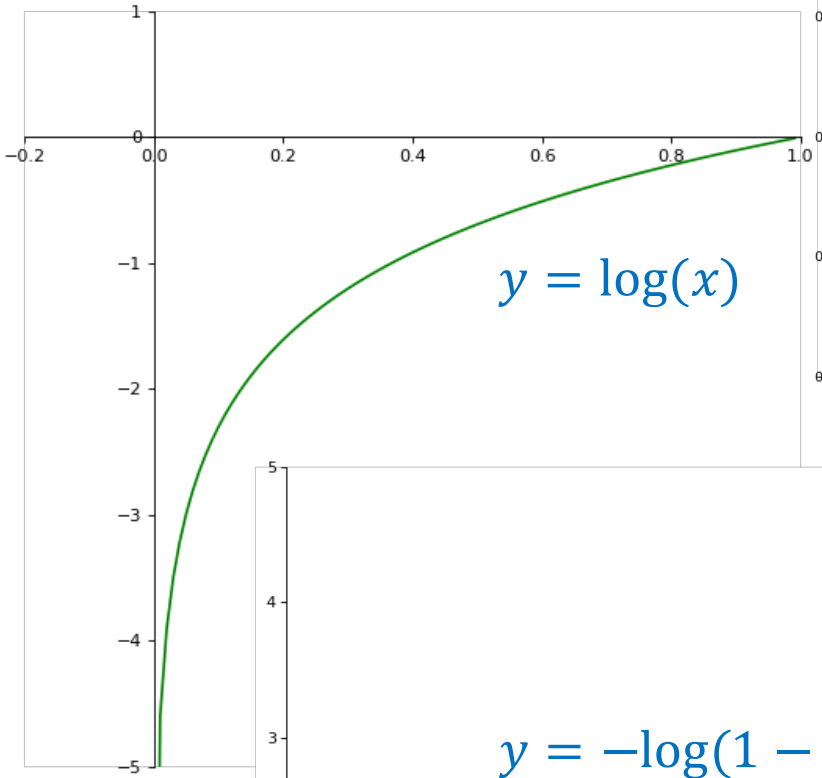


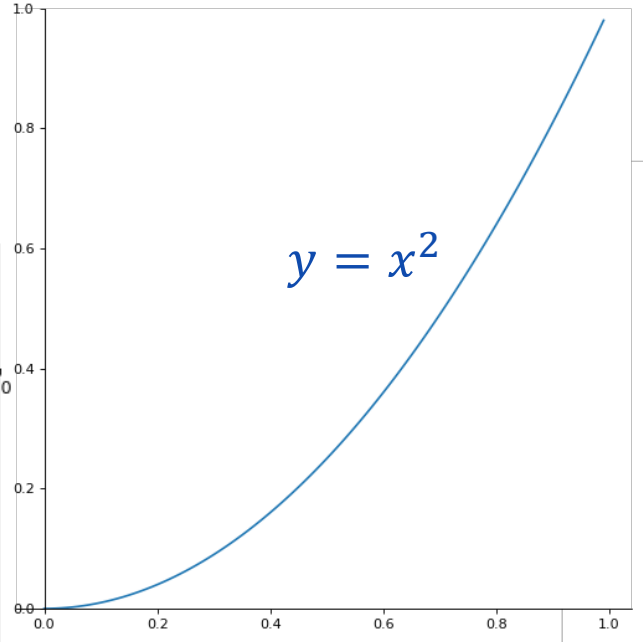
How to evaluate the performance of a model?

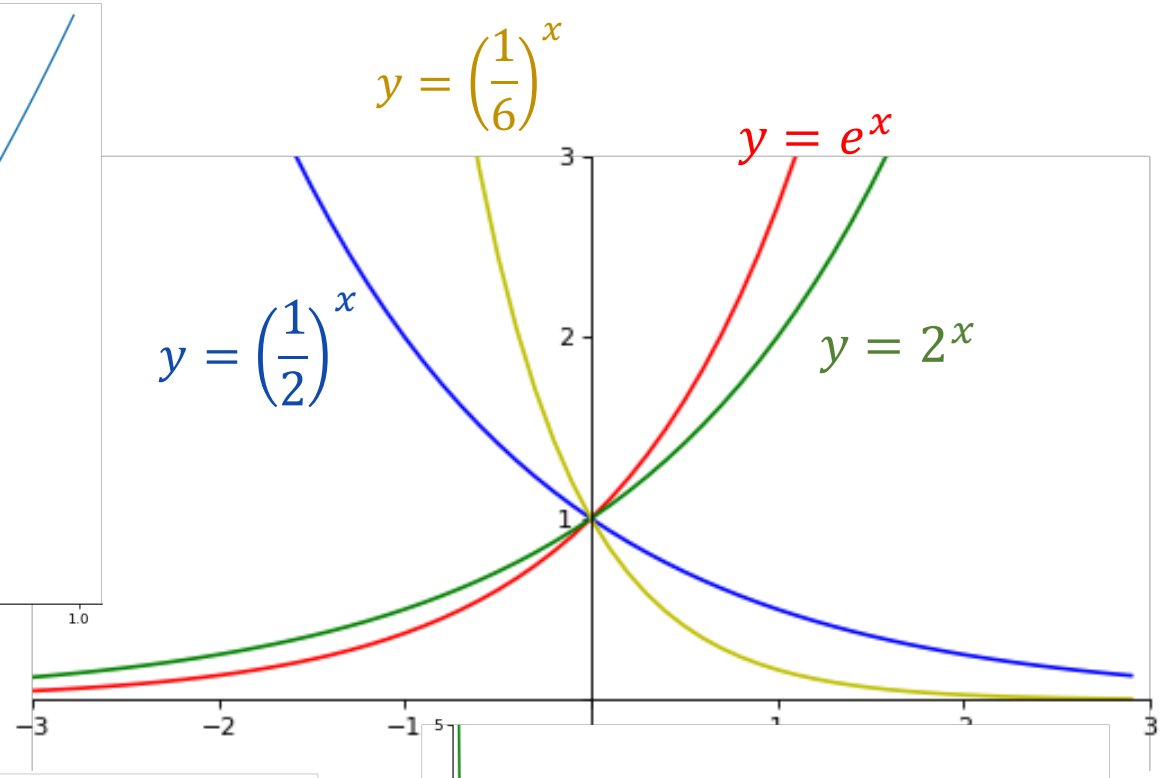
Discussion

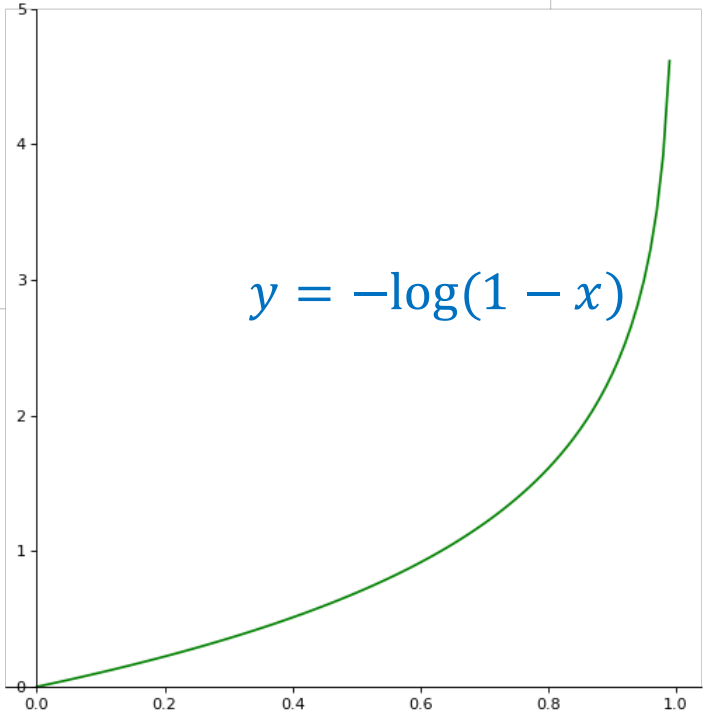

diff =1-$\hat{y}$
For some $\boldsymbol{\theta}$
diff = $\hat{y}$
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5
6
Feature

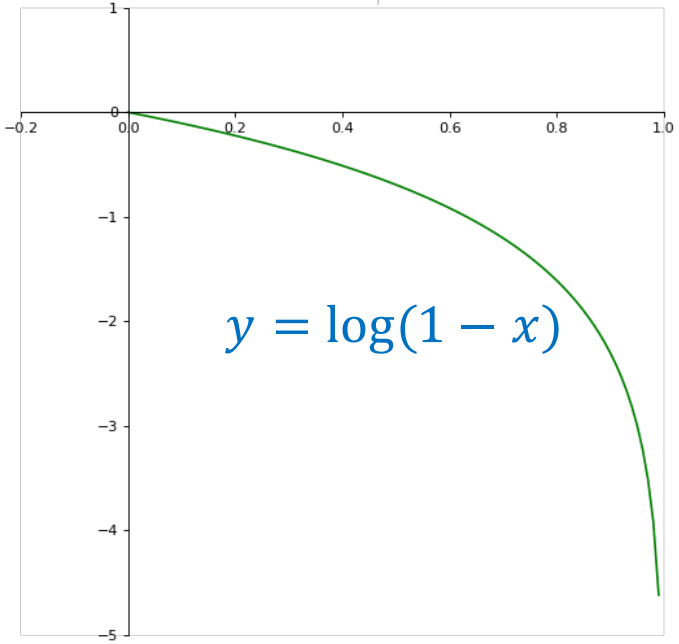
❖ Suggested Functions

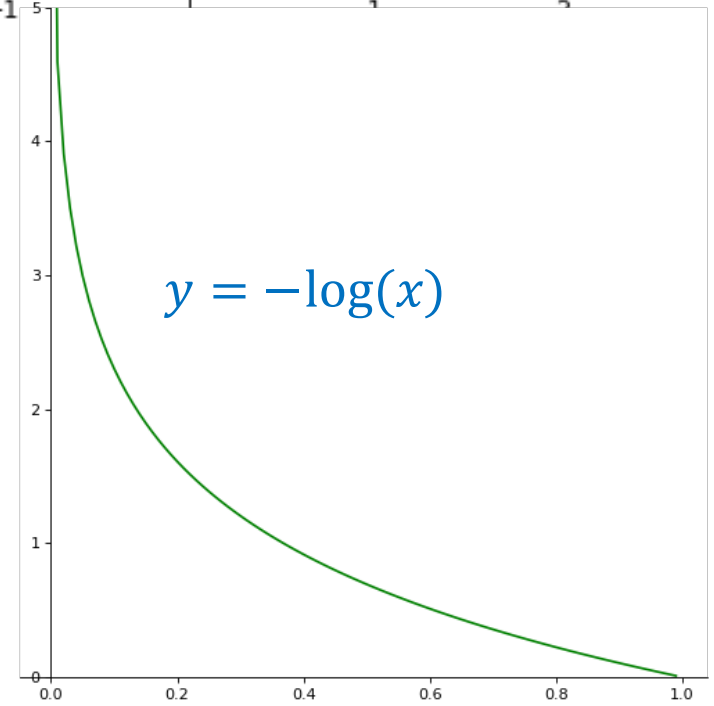












Loss Function

❖ Loss function

Feature	Label
Petal_Length	Category
1.4	0
1	0
1.5	0
3	1
3.8	1
4.1	1

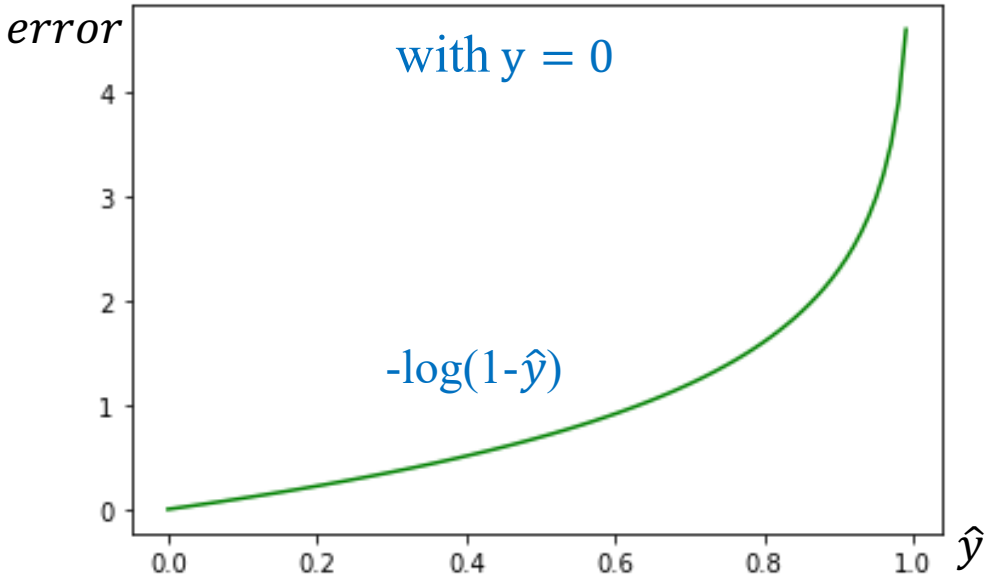
Category 0

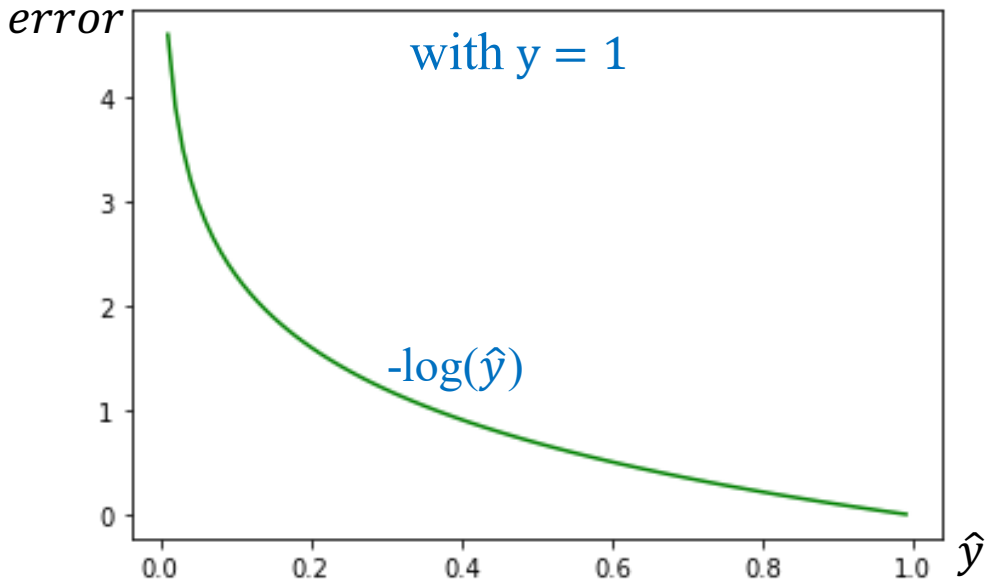
Category 1

$$z = wx + b$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma(z) \in (0 \quad 1)$$





Cơ số e

if y = 1

$$L(\hat{y}) = -\log(\hat{y})$$

if y = 0

$$L(\hat{y}) = -\log(1 - \hat{y})$$

How to remove if?

Loss Function

❖ Loss function

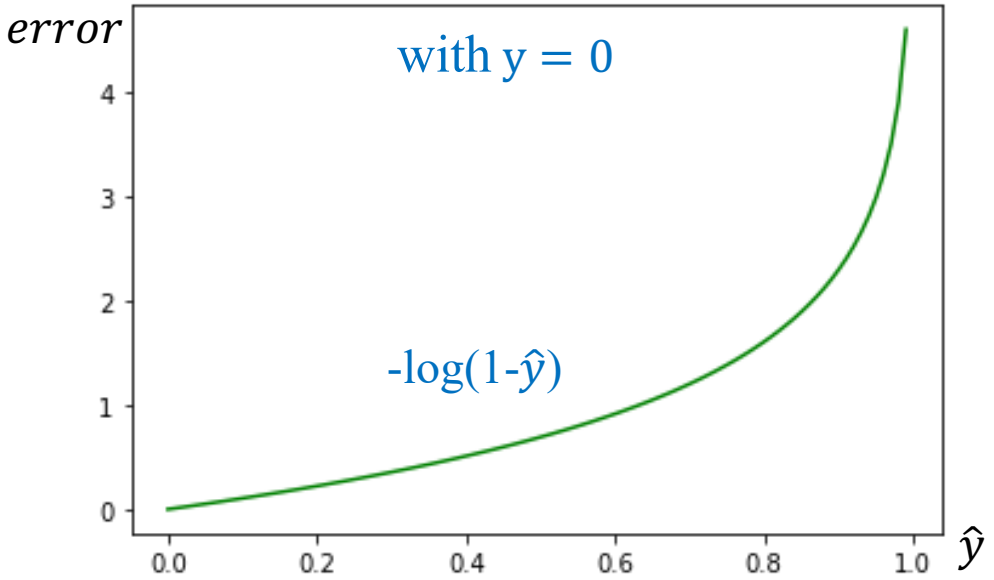
Feature	Output	Label
Input	Output	Label
…	0.3	0
…	0.8	0
…	0.7	0
…	0.4	0
…	0.6	1
…	0.8	1
…	0.9	1
…	0.2	1

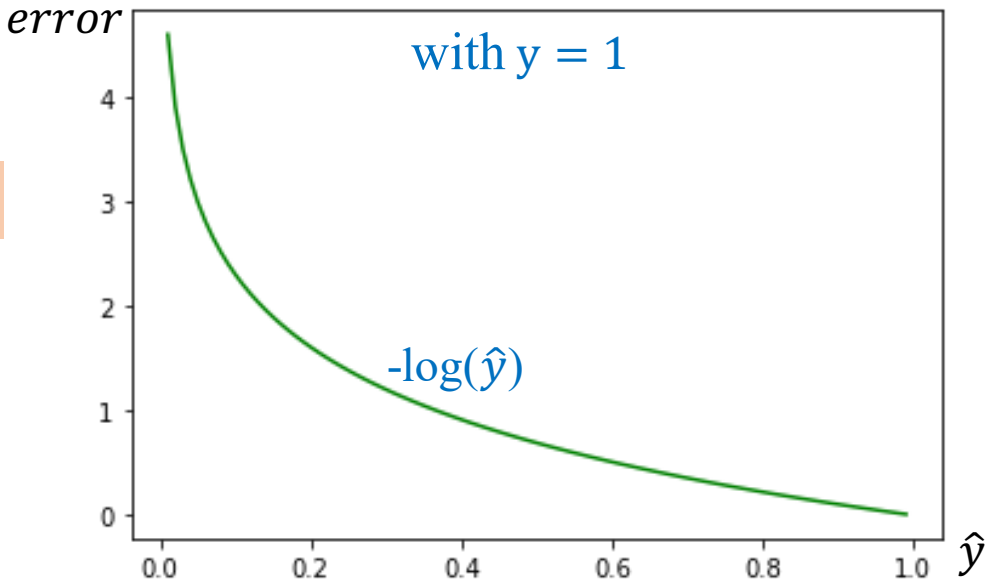
if y = 0

$$L(\hat{y}) = -\log(1-\hat{y})$$

if y = 1

$$L(\hat{y}) = -\log(\hat{y})$$





one sample

Cơ số e

Binary cross-entropy

$$L(y,\hat{y}) = -y\log\hat{y} - (1-y)\log(1-\hat{y})$$

Loss Function

❖ Construct loss

Model and Loss

$$z = wx + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\mathrm{L}(\hat{y}, y) = -y\log(\hat{y}) - (1 - y)\log(1 - \hat{y})$$

Derivative

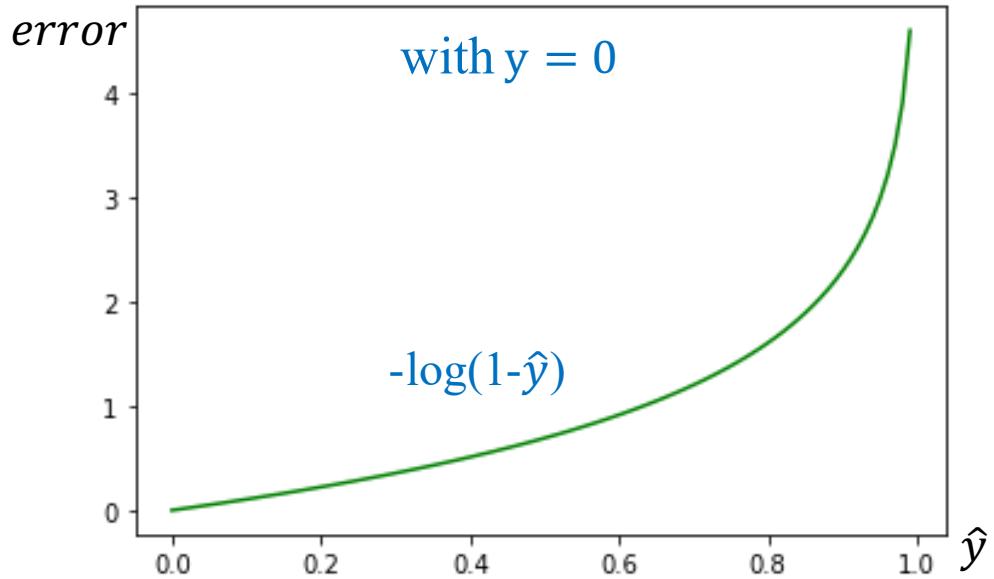
$$\frac{\partial L}{\partial \theta_i} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial \theta_i}$$

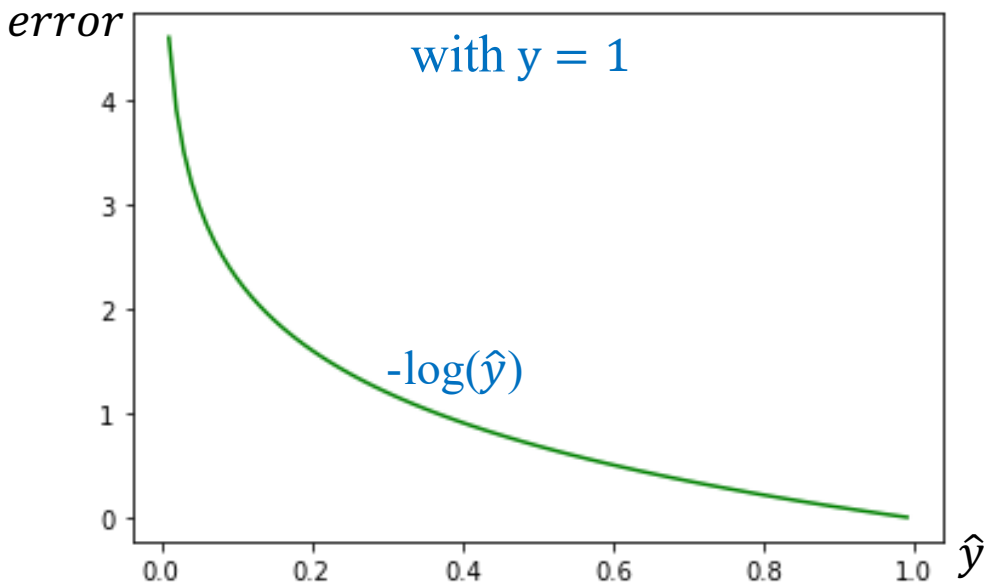
$$\frac{\partial L}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{1 - y}{1 - \hat{y}} = \frac{\hat{y} - y}{\hat{y}(1 - \hat{y})}$$

$$\frac{\partial \hat{y}}{\partial z} = \hat{y}(1 - \hat{y})$$

$$\frac{\partial z}{\partial \theta_i} = x_i$$

$$\frac{\partial L}{\partial \theta_i} = x_i(\hat{y} - y)$$





Logistic Regression-Stochastic

1) Pick a sample (x, y) from training data

2) Compute output $\hat{y}$

$$z = wx + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

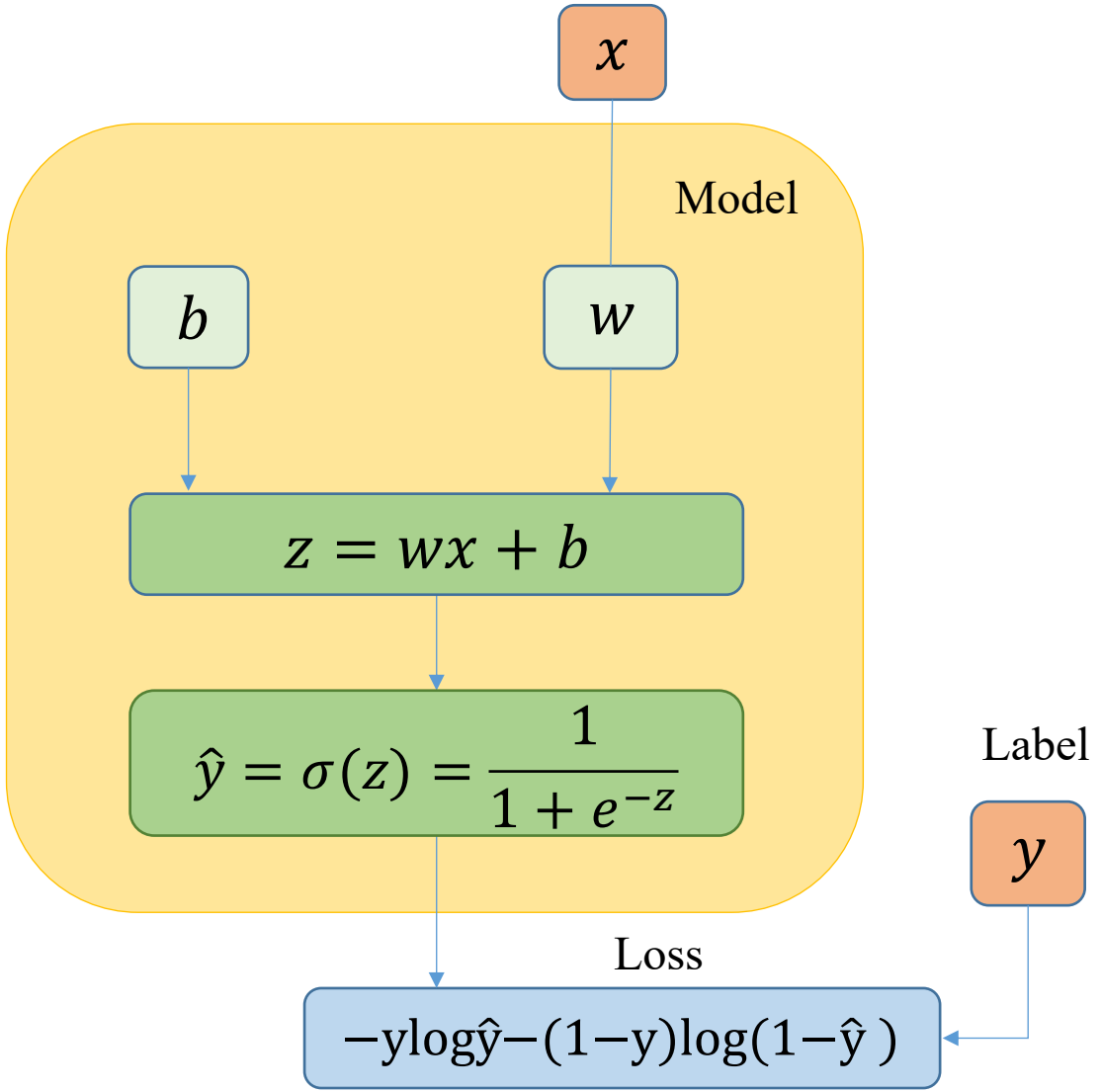
$$L(\hat{y}, y) = (-y\log\hat{y} - (1-y)\log(1-\hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y) \qquad \frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w = w - \eta\frac{\partial L}{\partial w} \qquad b = b - \eta\frac{\partial L}{\partial b}$$



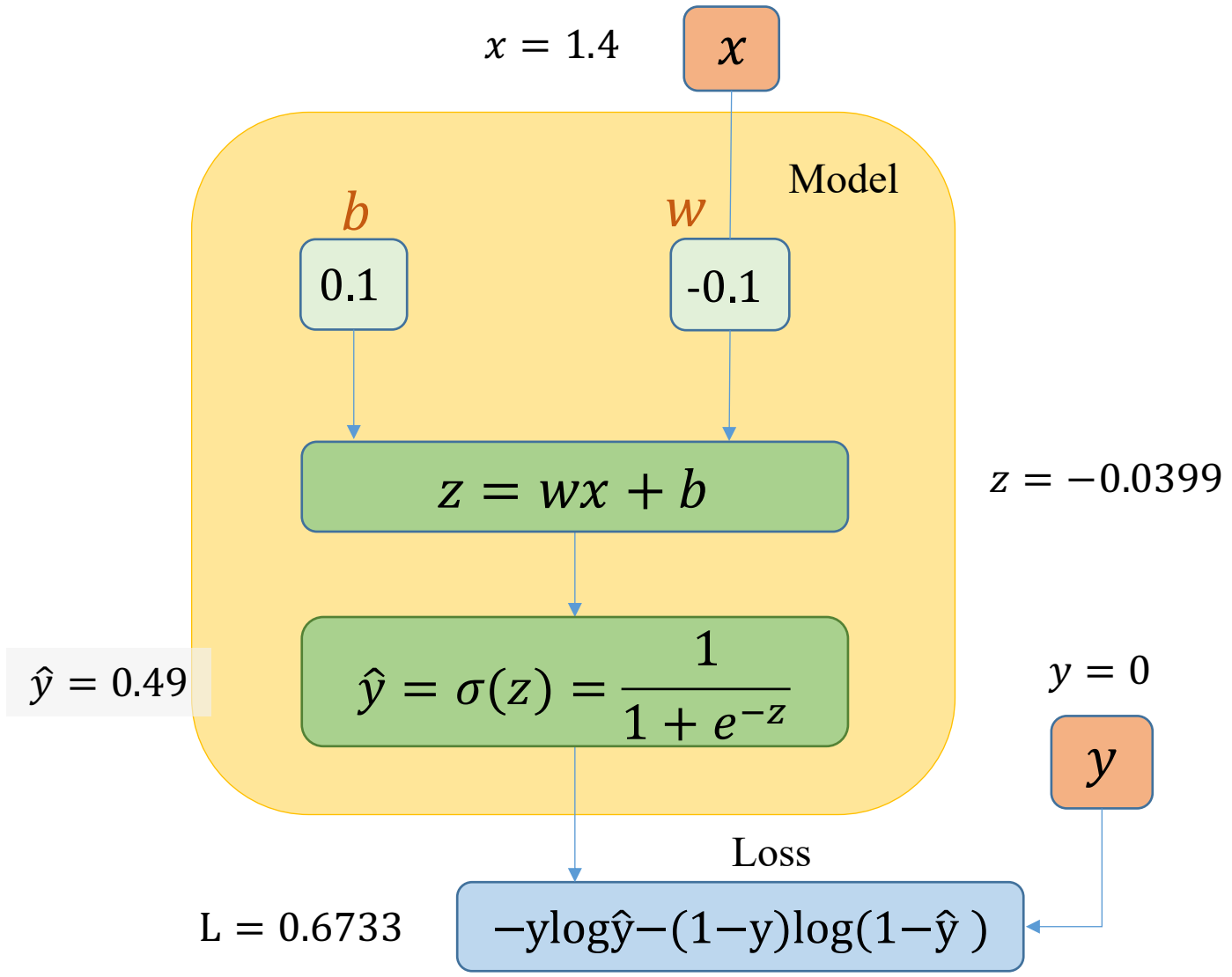
Logistic Regression-Stochastic

❖ Example

Dataset

Petal_Length	Label
1.4	0
1.5	0
3	1
4.1	1

$x = 1.4$ $y = 0$



Logistic Regression-Stochastic

❖ Example

$\eta = 0.01$

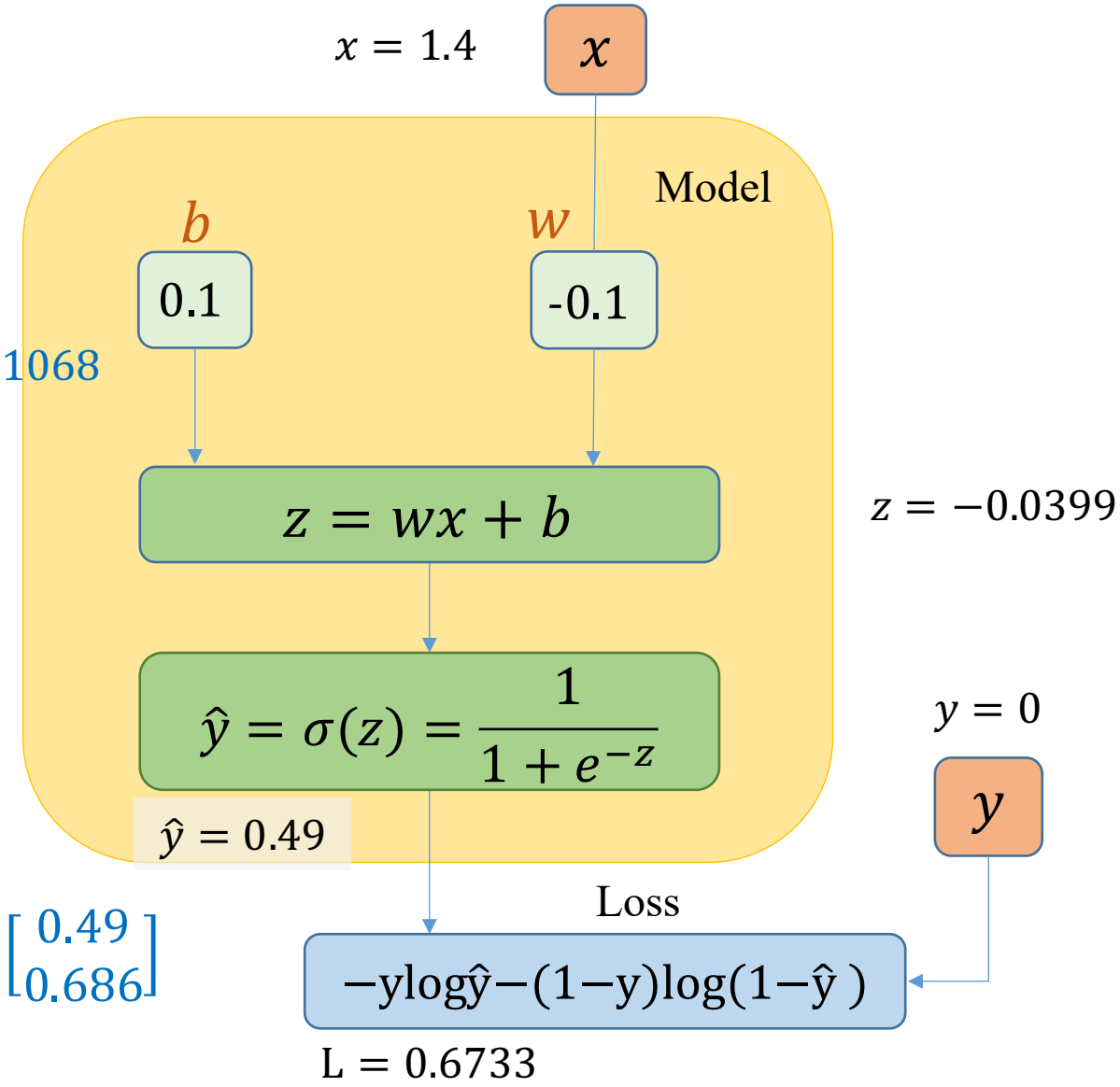
Dataset

Petal_Length	Label
1.4	0
1.5	0
3	1
4.1	1

$x = 1.4$ $y = 0$

$b = 0.1 - \eta 0.49 = 0.095$

$w = -0.1 - \eta 0.686 = -0.1068$



$$\begin{bmatrix} L'_b \\ L'_w \end{bmatrix} = \begin{bmatrix} 1 * 0.49 \\ 1.4 * 0.49 \end{bmatrix} = \begin{bmatrix} 0.49 \\ 0.686 \end{bmatrix}$$

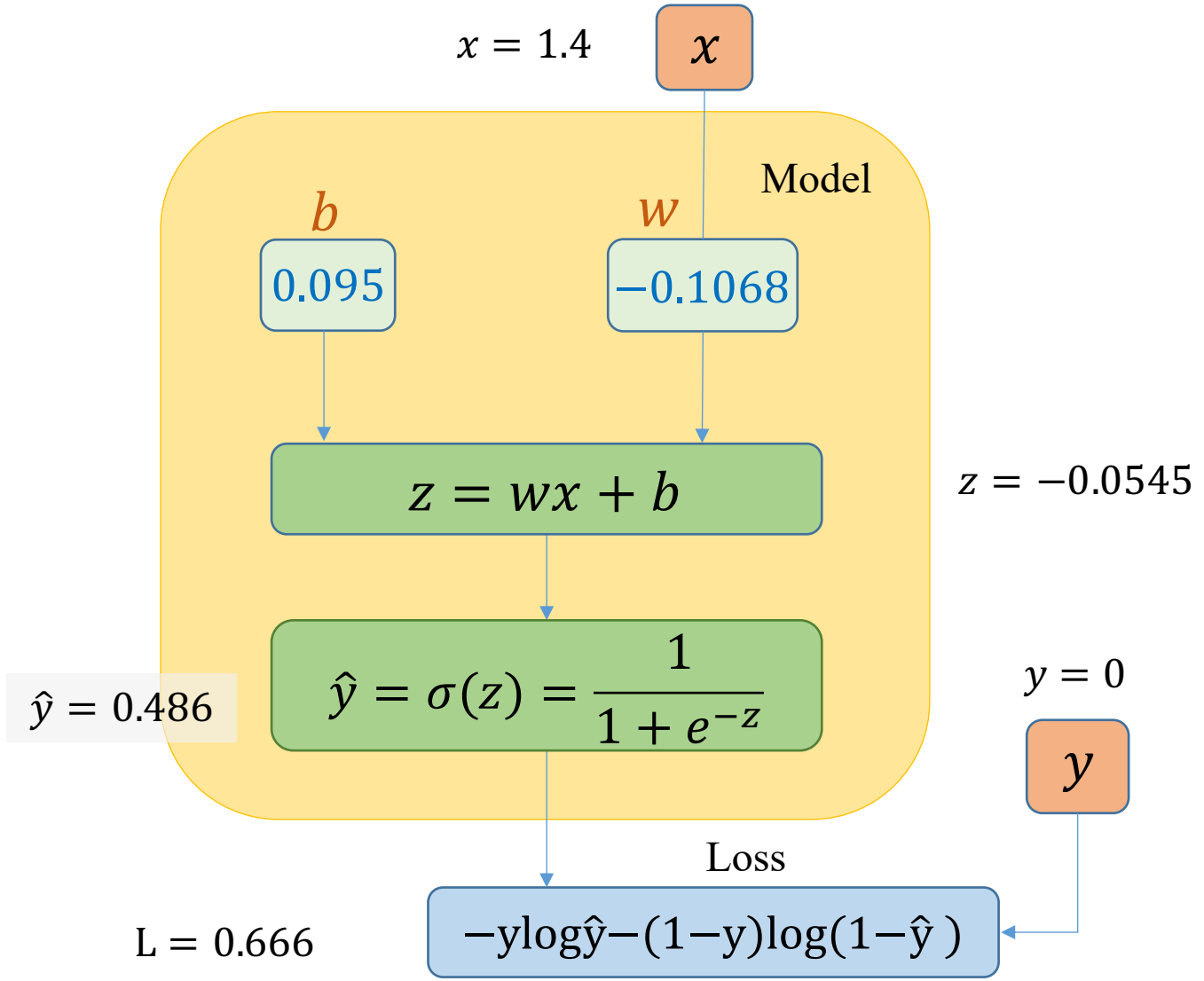
Logistic Regression-Stochastic

❖ Example

Dataset

Petal_Length	Label
1.4	0
1.5	0
3	1
4.1	1

$x = 1.4$ $y = 0$



previous L = 0.6733

L = 0.666

Implementation - One Sample

Feature	Label
Petal_Length	Label
1.4	0
1.5	0
3	1
4.1	1

1) Pick **a sample** (x, y) from training data

2) Compute the output $\hat{y}$

$$z = wx + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y\log\hat{y} - (1-y)\log(1-\hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y) \qquad \frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w = w - \eta\frac{\partial L}{\partial w} \qquad b = b - \eta\frac{\partial L}{\partial b}$$

```python
def sigmoid_function(z):
    return 1 / (1 + math.exp(-z))

def predict(x, w, b):
    z = w*x + b
    y_hat = sigmoid_function(z)

    return y_hat

def loss_function(y_hat, y):
    return -y*math.log(y_hat) - (1 - y)*math.log(1 - y_hat)

def compute_gradient(x, y_hat, y):
    dw = x*(y_hat - y)
    db = (y_hat - y)

    return dw, db

def update(w, b, dw, db, lr):
    w = w - lr*dw
    b = b - lr*db

    return w, b
```

if #features changes, which functions are affected?

demo

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

1) Pick **a sample** (x, y) from training data

2) Compute the output $\hat{y}$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y\log\hat{y} - (1-y)\log(1-\hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w_i} = x_i(\hat{y} - y) \qquad \frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w_i = w_i - \eta \frac{\partial L}{\partial w_i} \qquad b = b - \eta \frac{\partial L}{\partial b}$$

```python
def sigmoid_function(z):
    return 1 / (1 + np.exp(-z))

def predict(x1, x2, b, w1, w2):
    z = x1*w1 + x2*w2 + b
    y_hat = sigmoid_function(z)

    return y_hat

def loss_function(y_hat, y):
    return -y*np.log(y_hat) - (1 - y)*np.log(1 - y_hat)

def compute_gradient(x1, x2, y_hat, y):
    db  = (y_hat - y)
    dw1 = x1*(y_hat - y)
    dw2 = x2*(y_hat - y)

    return (db, dw1, dw2)

def update(b, w1, w2, lr, db, dw1, dw2):
    b  = b - lr*db
    w1 = w1 - lr*dw1
    w2 = w2 - lr*dw2

    return (b, w1, w2)
```

How to solve the problem?

Vectorization

Feature	Label
area	price
6.7	9.1
4.6	5.9
3.5	4.6
5.5	6.7
x	y

1) Pick **a sample** (x, y) from training data

2) Compute the output $\hat{y}$

Traditional

$$z = wx + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y\log\hat{y} - (1-y)\log(1-\hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y) \qquad \frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w = w - \eta\frac{\partial L}{\partial w} \qquad b = b - \eta\frac{\partial L}{\partial b}$$

η is learning rate

$$z = wx + b \qquad \boldsymbol{x} = \begin{bmatrix} 1 \\ x \end{bmatrix} \qquad \boldsymbol{\theta} = \begin{bmatrix} b \\ w \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} b \\ w \end{bmatrix} \rightarrow \boldsymbol{\theta}^T = [b \quad w]$$

$$z = wx + b1 = [b \quad w]\begin{bmatrix} 1 \\ x \end{bmatrix} = \boldsymbol{\theta}^T\boldsymbol{x}$$

dot product

Vectorization

1) Pick **a sample** (x, y) from training data

2) Compute the output $\hat{y}$

Traditional

$$z = wx + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y\log\hat{y} - (1-y)\log(1-\hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y) \qquad \frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w = w - \eta \frac{\partial L}{\partial w} \qquad b = b - \eta \frac{\partial L}{\partial b}$$

η is learning rate

$$z = wx + b \qquad \boldsymbol{x} = \begin{bmatrix} 1 \\ x \end{bmatrix} \qquad \boldsymbol{\theta} = \begin{bmatrix} b \\ w \end{bmatrix}$$

$$z = \boldsymbol{\theta}^T \boldsymbol{x} \qquad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$L(\hat{y}, y) = (-y\log\hat{y} - (1-y)\log(1-\hat{y}))$$

numbers

What will we do?

Vectorization

Traditional

1) Pick **a sample** (x, y) from training data

2) Compute the output $\hat{y}$

$$z = wx + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y\log\hat{y} - (1-y)\log(1-\hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y) \qquad \frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w = w - \eta\frac{\partial L}{\partial w} \qquad b = b - \eta\frac{\partial L}{\partial b}$$

$$z = wx + b \qquad \boldsymbol{x} = \begin{bmatrix} 1 \\ x \end{bmatrix} \qquad \boldsymbol{\theta} = \begin{bmatrix} b \\ w \end{bmatrix}$$

$$\begin{cases} \dfrac{\partial L}{\partial b} = (\hat{y} - y) = (\hat{y} - y)\times 1 \\ \dfrac{\partial L}{\partial w} = x(\hat{y} - y) = (\hat{y} - y)\times x \end{cases}$$

$$\begin{bmatrix} (\hat{y} - y)\times 1 \\ (\hat{y} - y)\times x \end{bmatrix} = (\hat{y} - y)\begin{bmatrix} 1 \\ x \end{bmatrix} = (\hat{y} - y)\boldsymbol{x} = \begin{bmatrix} \dfrac{\partial L}{\partial b} \\ \dfrac{\partial L}{\partial w} \end{bmatrix} = \nabla_{\boldsymbol{\theta}} L \qquad \rightarrow \qquad \nabla_{\boldsymbol{\theta}} L = \boldsymbol{x}(\hat{y} - y)$$

common factor

Vectorization

1) Pick **a sample** (x, y) from training data

2) Compute the output $\hat{y}$

Traditional

$$z = wx + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y\log\hat{y} - (1-y)\log(1-\hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y) \qquad \frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w = w - \eta\frac{\partial L}{\partial w} \qquad b = b - \eta\frac{\partial L}{\partial b}$$

η is learning rate

$$z = \boldsymbol{\theta}^T \boldsymbol{x}$$

$$\boldsymbol{x} = \begin{bmatrix} 1 \\ x \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} b \\ w \end{bmatrix}$$

$$\nabla_{\boldsymbol{\theta}} L = \begin{bmatrix} \frac{\partial L}{\partial b} \\ \frac{\partial L}{\partial w} \end{bmatrix}$$

$$\begin{cases} b = b - \eta\frac{\partial L}{\partial b} \\ w = w - \eta\frac{\partial L}{\partial w} \end{cases}$$

$$\boldsymbol{\theta} \quad \boldsymbol{\theta} \quad \nabla_{\boldsymbol{\theta}} L$$

$$\rightarrow \quad \boldsymbol{\theta} = \boldsymbol{\theta} - \eta\nabla_{\boldsymbol{\theta}} L$$

Vectorization

Traditional

1) Pick a sample (x, y) from training data

2) Compute the output $\hat{y}$

$$z = wx + b \qquad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y\log\hat{y} - (1-y)\log(1-\hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y) \qquad \frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w = w - \eta\frac{\partial L}{\partial w} \qquad b = b - \eta\frac{\partial L}{\partial b}$$

η is learning rate

Vectorized

1) Pick a sample $(\mathbf{x}, y)$ from training data

2) Compute output $\hat{y}$

$$z = \boldsymbol{\theta}^T\boldsymbol{x} = \boldsymbol{x}^T\boldsymbol{\theta} \qquad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y\log\hat{y} - (1-y)\log(1-\hat{y}))$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \boldsymbol{x}(\hat{y} - y)$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta\nabla_{\boldsymbol{\theta}} L$$

η is learning rate

Vectorization

❖ Implementation (using Numpy)

1) Pick a sample $(\boldsymbol{x}, y)$ from training data

↓

2) Compute output $\hat{y}$

↓ $z = \boldsymbol{\theta}^T \boldsymbol{x} = \boldsymbol{x}^T \boldsymbol{\theta}$ $\quad \hat{y} = \sigma(z) = \dfrac{1}{1 + e^{-z}}$

3) Compute loss

↓ $L(\hat{y}, y) = (-y\log\hat{y} - (1-y)\log(1-\hat{y}))$

4) Compute derivative

↓ $\nabla_{\boldsymbol{\theta}} L = \boldsymbol{x}(\hat{y} - y)$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$$

η is learning rate

```python
def sigmoid_function(z):
    return 1 / (1 + np.exp(-z))

def predict(X, theta):
    return sigmoid_function( np.dot(X.T, theta) )

def loss_function(y_hat, y):
    return -y*np.log(y_hat) - (1 - y)*np.log(1 - y_hat)

def compute_gradient(X, y_hat, y):
    return X*(y_hat - y)

def update(theta, lr, gradient):
    return theta - lr*gradient
```

```python
# compute output
y_hat = predict(X, theta)

# compute loss
loss = loss_function(y_hat, y)

# compute mean of gradient
gradient = compute_gradient(X, y_hat, y)

# update
theta = update(theta, lr, gradient)
```

```
# Given X and y
```

Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

1) Pick a sample $(\boldsymbol{x}, y)$ from training data

2) Compute output $\hat{y}$

$$z = \boldsymbol{\theta}^T \boldsymbol{x} = \boldsymbol{x}^T \boldsymbol{\theta} \qquad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y\log\hat{y} - (1-y)\log(1-\hat{y}))$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \boldsymbol{x}(\hat{y} - y)$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$$

1

$$\boldsymbol{x} = \begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1.4 \\ 0.2 \end{bmatrix}$$

Given $\boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$

$\eta = 0.01$

Input $\boldsymbol{x}$

Model

$\boldsymbol{\theta} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$

$\hat{y} = \sigma(\boldsymbol{\theta}^T \boldsymbol{x}) = 0.6856$

Label

$y = 0$

3 Loss

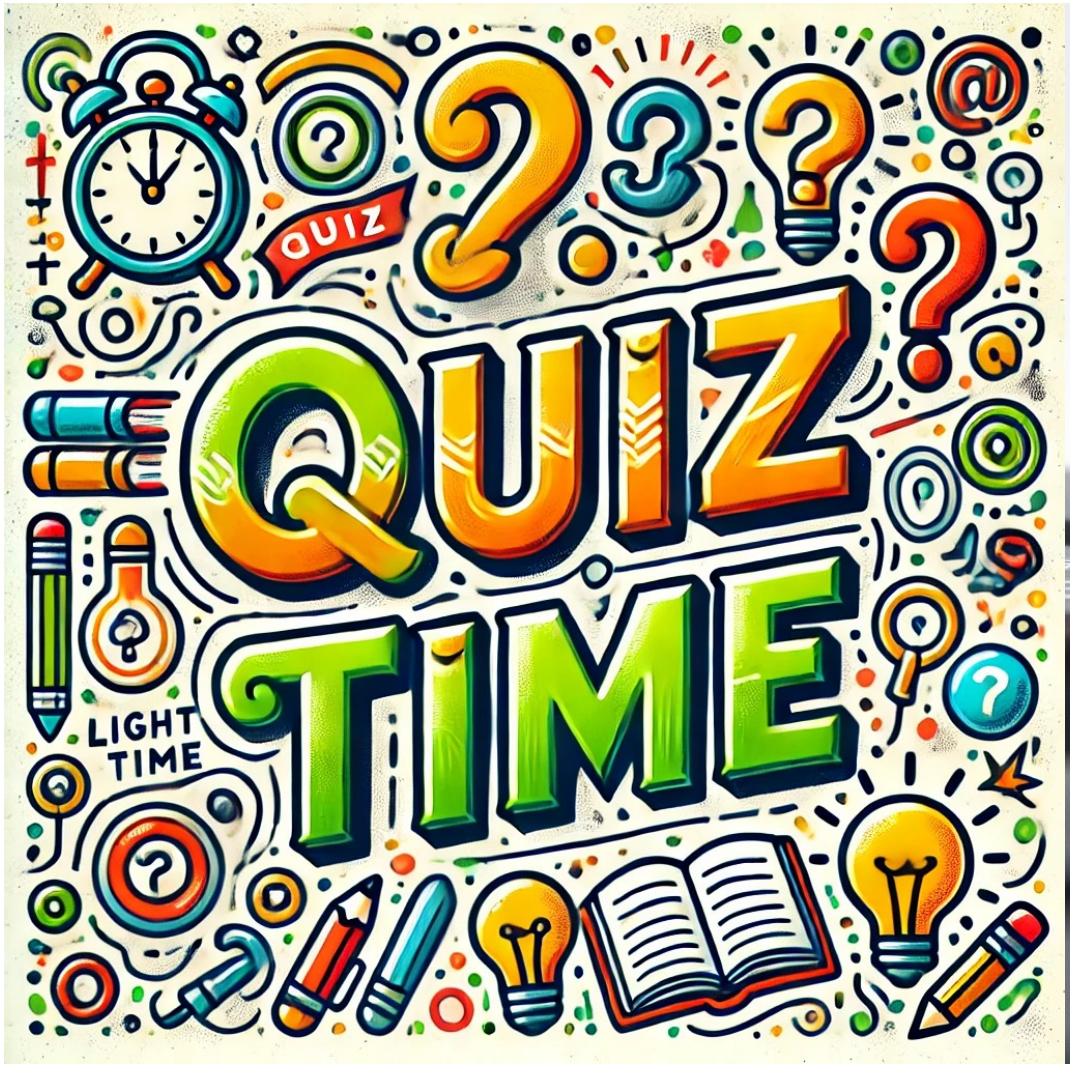
$L = 1.1573$

4

$$\nabla_{\boldsymbol{\theta}} L = \boldsymbol{x}(\hat{y} - y) = \begin{bmatrix} 1 \\ 1.4 \\ 0.2 \end{bmatrix} [0.6856] = \begin{bmatrix} 0.6856 \\ 0.9599 \\ 0.1371 \end{bmatrix} = \begin{bmatrix} L'_b \\ L'_{w_1} \\ L'_{w_2} \end{bmatrix}$$

5

$$\boldsymbol{\theta} - \eta L'_{\boldsymbol{\theta}} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix} - \eta \begin{bmatrix} 0.6856 \\ 0.9599 \\ 0.1371 \end{bmatrix} = \begin{bmatrix} 0.093 \\ 0.499 \\ -0.101 \end{bmatrix}$$


QUIZ
QUIZ
TIME
LIGHT
TIME


BREAK
TIME!!!

Outline

SECTION 1

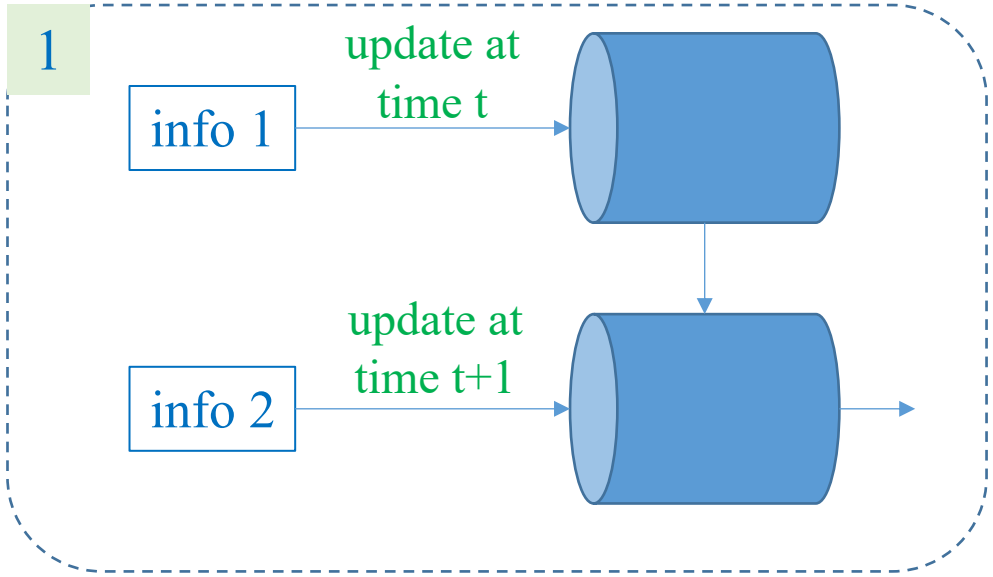
Logistic Regression

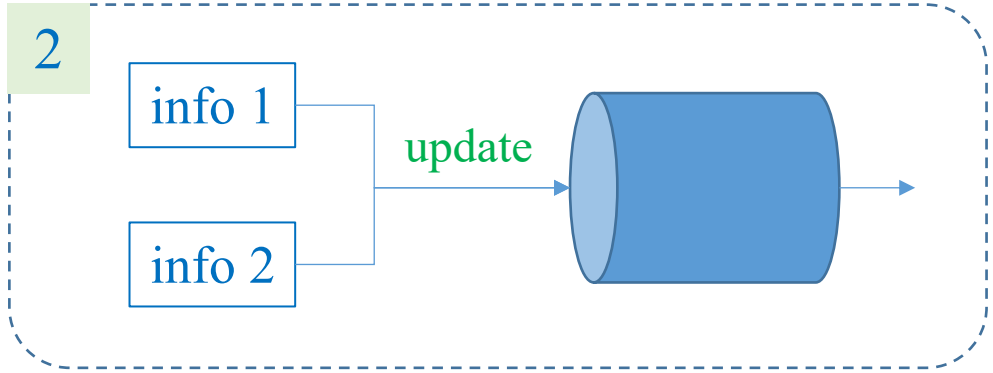
SECTION 2

Using Multi-samples

SECTION 3

Discussion





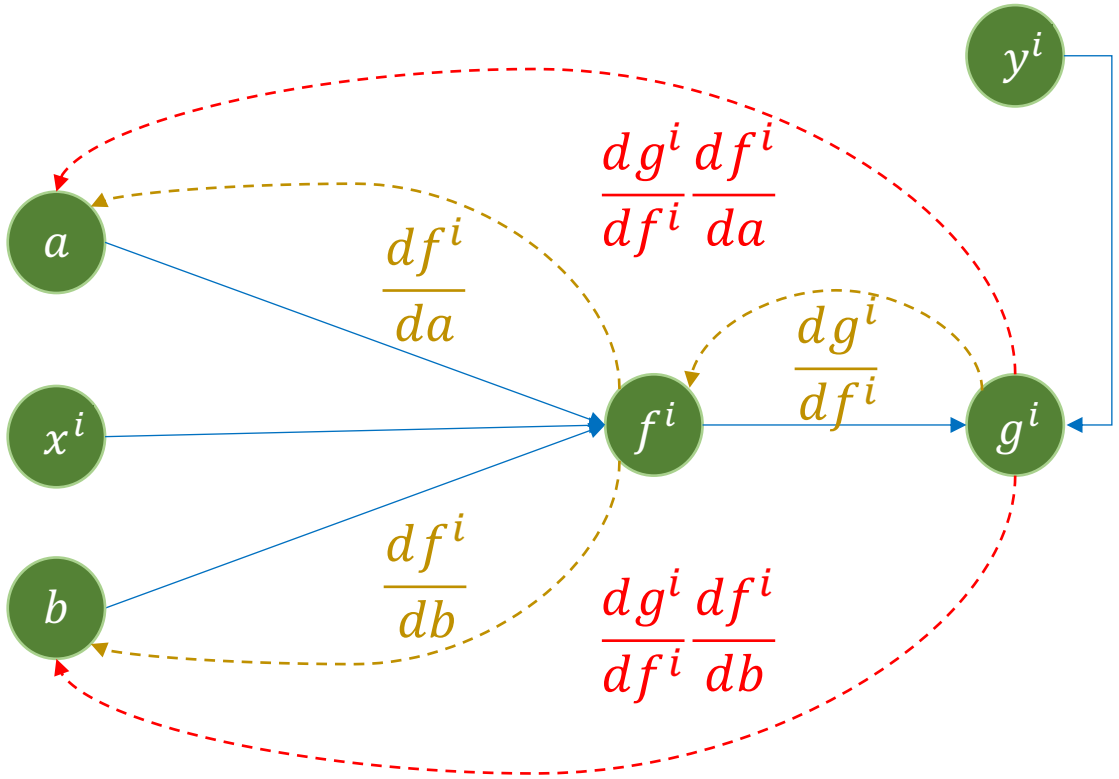
Optimization for One+ Samples

❖ Equations for partial gradients

$$f(x^i) = ax^i + b \qquad (x^1{=}1, y^1{=}5)$$

$$g(f^i) = (f^i - y^i)^2 \qquad (x^2{=}2, y^2{=}7)$$

illustration



$$\frac{df}{da} = x \qquad \frac{df}{db} = 1$$

$$\frac{dg}{df} = 2(f - y)$$

$$\frac{dg}{da} = \frac{dg}{df}\frac{df}{da} = 2x(f - y)$$

$$\frac{dg}{db} = \frac{dg}{df}\frac{df}{db} = 2(f - y)$$

During looking for optimal a and b, at a given time, a and b have concrete values

❖ Optimization for a composite function

Find a and b so that g(f(x)) is minimum

$f(x^i) = ax^i + b \qquad (x^1{=}1, y^1{=}5)$

$g(f^i) = (f^i - y^i)^2 \qquad (x^2{=}2, y^2{=}7)$

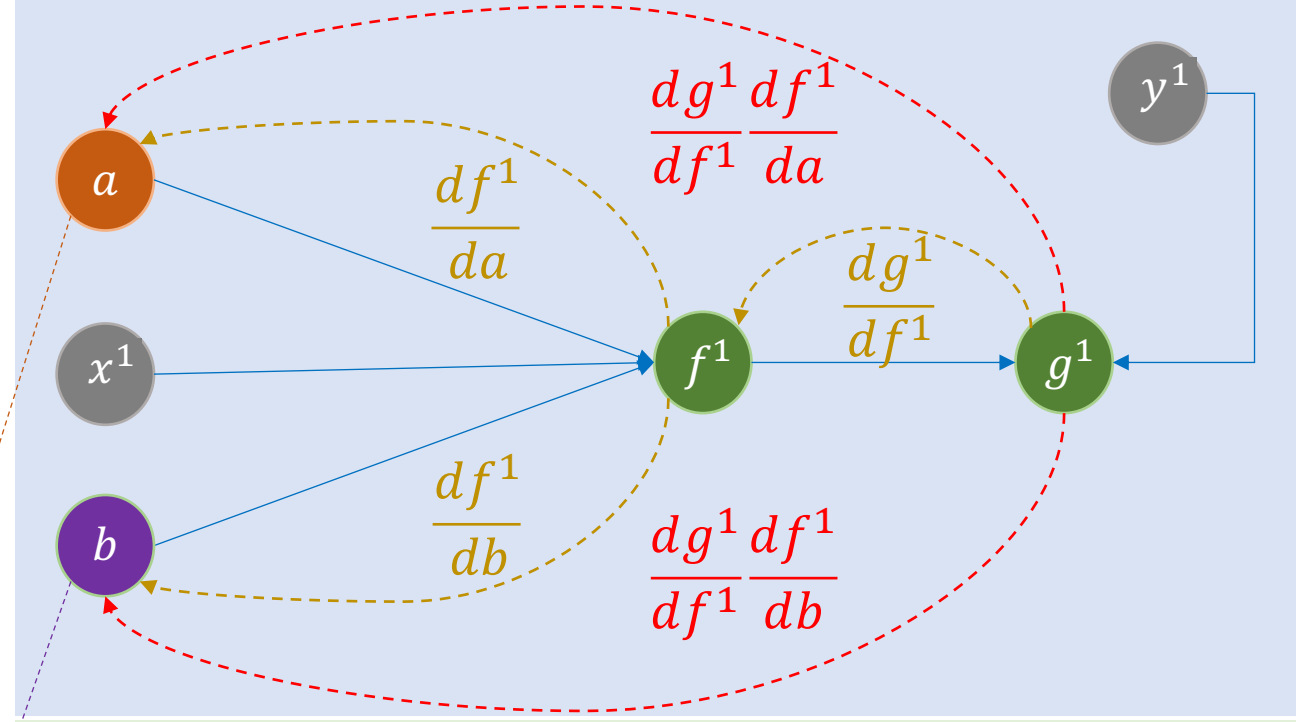
Partial derivative functions

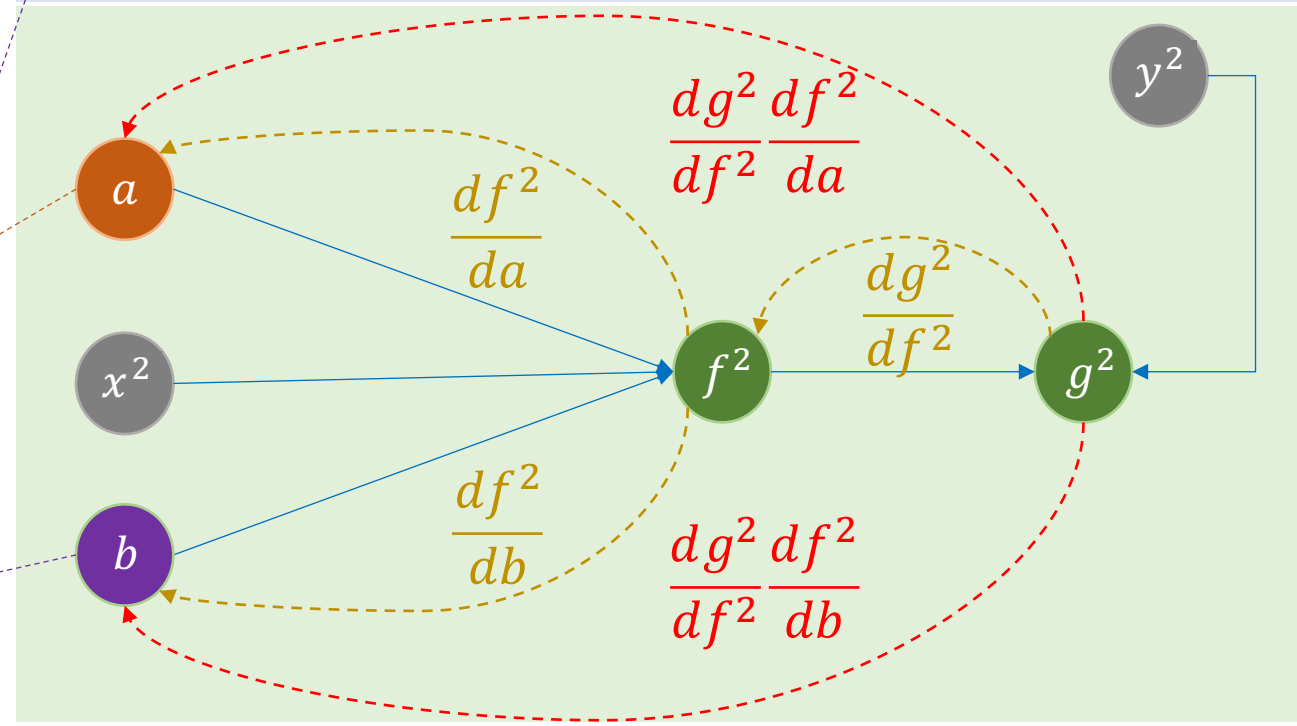
$$\frac{dg}{da} = \frac{dg}{df}\frac{df}{da} = 2x(f-y)$$

$$\frac{dg}{db} = \frac{dg}{df}\frac{df}{db} = 2(f-y)$$

$$\sum_i \frac{dg^i}{da} = \frac{dg^1}{df^1}\frac{df^1}{da} + \frac{dg^2}{df^2}\frac{df^2}{da}$$

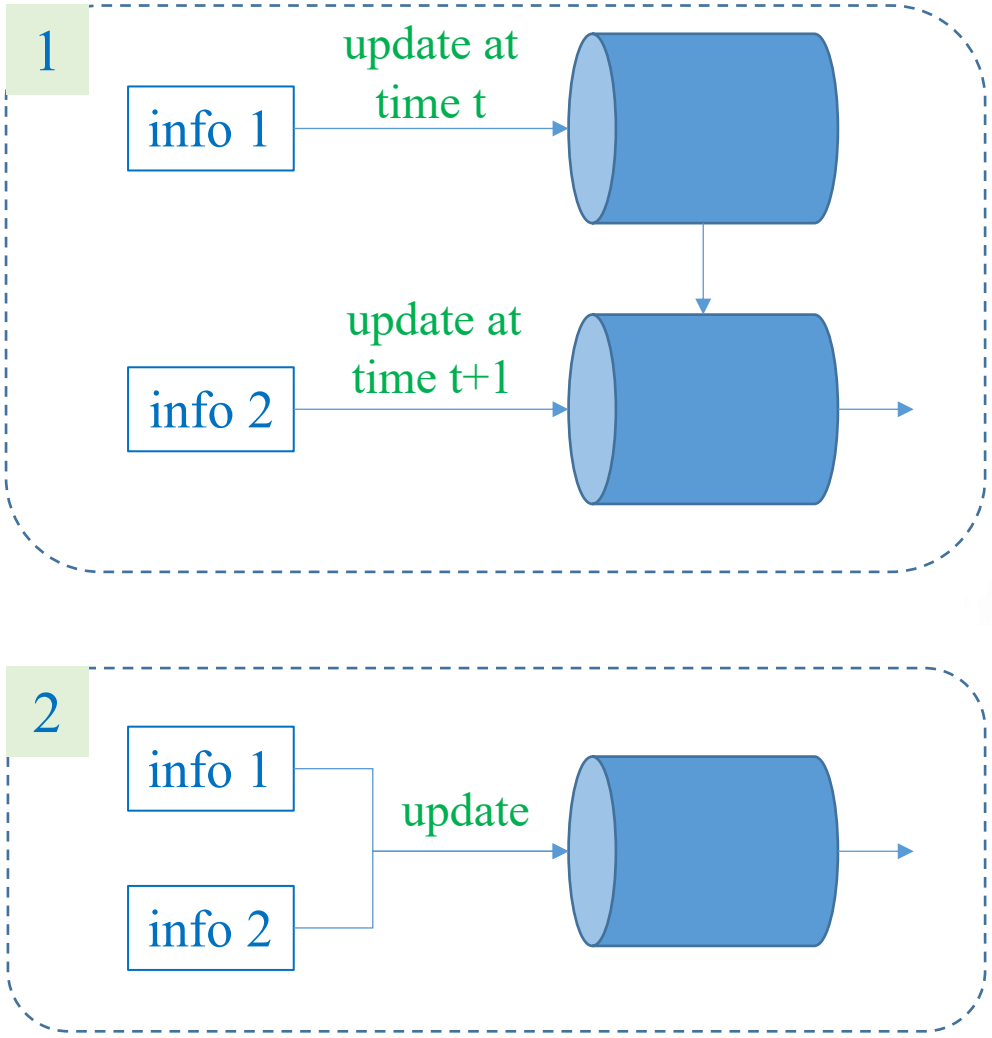
$$\sum_i \frac{dg_i}{db} = \frac{dg^1}{df^1}\frac{df^1}{db} + \frac{dg^2}{df^2}\frac{df^2}{db}$$





Optimization

❖ How to use gradient information



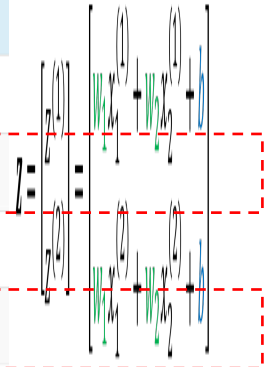


Logistic Regression (m-samples)

❖ Construct formulas

Dataset

Petal_Length	Petal_Width
1.4	0.2
1.5	0.2
3	1.1
4.1	1.3



$$y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3 \end{bmatrix}$$

$$\theta = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$$

2) Compute output $\hat{y}$

$$X = \begin{bmatrix} 1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3 \end{bmatrix} = \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} \end{bmatrix}$$

$$\theta = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$$

$$z = \begin{bmatrix} z^{(1)} \\ z^{(2)} \end{bmatrix} = \begin{bmatrix} w_1 x_1^{(1)} + w_2 x_2^{(1)} + b \\ w_1 x_1^{(2)} + w_2 x_2^{(2)} + b \end{bmatrix}$$

$$= \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} \end{bmatrix} \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix} = X\theta = \begin{bmatrix} 0.83 \\ 2.02 \end{bmatrix}$$

Logistic Regression (m-samples)

❖ Construct formulas

Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$\boldsymbol{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\mathrm{X} = \begin{bmatrix} 1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$$

2) Compute output $\hat{y}$

$$\mathrm{X} = \begin{bmatrix} 1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3 \end{bmatrix} = \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$$

$$\boldsymbol{z} = \mathrm{X}\boldsymbol{\theta} = \begin{bmatrix} 0.83 \\ 2.02 \end{bmatrix}$$

$$\hat{\boldsymbol{y}} = \sigma(\boldsymbol{z}) = \begin{bmatrix} \hat{y}^{(1)} \\ \hat{y}^{(2)} \end{bmatrix} = \begin{bmatrix} \dfrac{1}{1+e^{-z^{(1)}}} \\ \dfrac{1}{1+e^{-z^{(2)}}} \end{bmatrix} = \frac{1}{1+e^{-\boldsymbol{z}}} = \begin{bmatrix} 0.69 \\ 0.88 \end{bmatrix}$$

Linear Regression (m-samples)

❖ Construct formulas

Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$\boldsymbol{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\mathrm{X} = \begin{bmatrix} 1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$$

3) Compute loss

$$L(\hat{\mathbf{y}}, \mathbf{y}) = \frac{1}{\mathrm{m}}\left(-\mathbf{y}^{\mathrm{T}}\log\hat{\mathbf{y}} - (1-\mathbf{y})^{\mathrm{T}}\log(1-\hat{\mathbf{y}})\right)$$

$$L(\hat{\mathbf{y}}, \mathbf{y}) = \frac{L^{(1)}\left(\hat{y}^{(1)}, y^{(1)}\right) + L^{(2)}\left(\hat{y}^{(2)}, y^{(2)}\right)}{\mathrm{m}}$$

$$L^{(1)}\left(\hat{y}^{(1)}, y^{(1)}\right) = -y^{(1)}\log\hat{y}^{(1)} - (1-y^{(1)})\log(1-\hat{y}^{(1)})$$

$$+$$

$$L^{(2)}\left(\hat{y}^{(2)}, y^{(2)}\right) = -y^{(2)}\log\hat{y}^{(2)} - (1-y^{(2)})\log(1-\hat{y}^{(2)})$$

$\mathbf{y}^{\mathrm{T}}\log\hat{\mathbf{y}}$ $\quad$ $(1-\mathbf{y})^{\mathrm{T}}\log(1-\hat{\mathbf{y}})$

4) Compute derivative

sample 1

$$\frac{\partial L^{(1)}}{\partial b} = (\hat{y}^{(1)} - y^{(1)})$$

$$\frac{\partial L^{(1)}}{\partial w_1} = x_1^{(1)}(\hat{y}^{(1)} - y^{(1)})$$

$$\frac{\partial L^{(1)}}{\partial w_2} = x_2^{(1)}(\hat{y}^{(1)} - y^{(1)})$$

sample 2

$$\frac{\partial L^{(2)}}{\partial b} = (\hat{y}^{(2)} - y^{(2)})$$

$$\frac{\partial L^{(2)}}{\partial w_1} = x_1^{(2)}(\hat{y}^{(2)} - y^{(2)})$$

$$\frac{\partial L^{(2)}}{\partial w_2} = x_2^{(2)}(\hat{y}^{(2)} - y^{(2)})$$

$$\frac{\partial L}{\partial b} = \frac{\frac{\partial L^{(1)}}{\partial b} + \frac{\partial L^{(2)}}{\partial b}}{m} = \frac{(\hat{y}^{(1)} - y^{(1)}) + (\hat{y}^{(2)} - y^{(2)})}{m}$$

$$= \frac{1}{m}\left[1 * (\hat{y}^{(1)} - y^{(1)}) + 1 * (\hat{y}^{(2)} - y^{(2)})\right]$$

$$= \frac{1}{m}\begin{bmatrix} x_0^{(1)} & x_0^{(2)} \end{bmatrix}\begin{bmatrix} \hat{y}^{(1)} - y^{(1)} \\ \hat{y}^{(2)} - y^{(2)} \end{bmatrix}$$

$x_0^{(1)} = 1$

$x_0^{(2)} = 1$

$$\frac{\partial L}{\partial w_1} = \frac{\frac{\partial L^{(1)}}{\partial w_1} + \frac{\partial L^{(2)}}{\partial w_1}}{m} = \frac{x_1^{(1)}(\hat{y}^{(1)} - y^{(1)}) + x_1^{(2)}(\hat{y}^{(2)} - y^{(2)})}{m}$$

$$= \frac{1}{m}\begin{bmatrix} x_1^{(1)} & x_1^{(2)} \end{bmatrix}\begin{bmatrix} \hat{y}^{(1)} - y^{(1)} \\ \hat{y}^{(2)} - y^{(2)} \end{bmatrix}$$

$$\frac{\partial L}{\partial w_2} = \frac{\frac{\partial L^{(1)}}{\partial w_2} + \frac{\partial L^{(2)}}{\partial w_2}}{m} = \frac{x_2^{(1)}(\hat{y}^{(1)} - y^{(1)}) + x_2^{(2)}(\hat{y}^{(2)} - y^{(2)})}{m}$$

$$= \frac{1}{m}\begin{bmatrix} x_2^{(1)} & x_2^{(2)} \end{bmatrix}\begin{bmatrix} \hat{y}^{(1)} - y^{(1)} \\ \hat{y}^{(2)} - y^{(2)} \end{bmatrix}$$

4) Compute derivative

sample 1

$$\frac{\partial L^{(1)}}{\partial b} = (\hat{y}^{(1)} - y^{(1)})$$

$$\frac{\partial L^{(1)}}{\partial w_1} = x_1^{(1)}(\hat{y}^{(1)} - y^{(1)})$$

$$\frac{\partial L^{(1)}}{\partial w_2} = x_2^{(1)}(\hat{y}^{(1)} - y^{(1)})$$

sample 2

$$\frac{\partial L^{(2)}}{\partial b} = (\hat{y}^{(2)} - y^{(2)})$$

$$\frac{\partial L^{(2)}}{\partial w_1} = x_1^{(2)}(\hat{y}^{(2)} - y^{(2)})$$

$$\frac{\partial L^{(2)}}{\partial w_2} = x_2^{(2)}(\hat{y}^{(2)} - y^{(2)})$$

$$\frac{\partial L}{\partial b} = \frac{1}{m}\begin{bmatrix} x_0^{(1)} & x_0^{(2)} \end{bmatrix}\begin{bmatrix} \hat{y}^{(1)} - y^{(1)} \\ \hat{y}^{(2)} - y^{(2)} \end{bmatrix}$$

$$\frac{\partial L}{\partial w_1} = \frac{1}{m}\begin{bmatrix} x_1^{(1)} & x_1^{(2)} \end{bmatrix}\begin{bmatrix} \hat{y}^{(1)} - y^{(1)} \\ \hat{y}^{(2)} - y^{(2)} \end{bmatrix}$$

$$\frac{\partial L}{\partial w_2} = \frac{1}{m}\begin{bmatrix} x_2^{(1)} & x_2^{(2)} \end{bmatrix}\begin{bmatrix} \hat{y}^{(1)} - y^{(1)} \\ \hat{y}^{(2)} - y^{(2)} \end{bmatrix}$$

$$\nabla_{\boldsymbol{\theta}} \boldsymbol{L} = \begin{bmatrix} \frac{\partial L}{\partial b} \\ \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \end{bmatrix} = \frac{1}{m}\begin{bmatrix} x_0^{(1)} & x_0^{(2)} \\ x_1^{(1)} & x_1^{(2)} \\ x_2^{(1)} & x_2^{(2)} \end{bmatrix}\begin{bmatrix} \hat{y}^{(1)} - y^{(1)} \\ \hat{y}^{(2)} - y^{(2)} \end{bmatrix}$$

$\left(\boldsymbol{x}^{(1)}\right)^T$ $\left(\boldsymbol{x}^{(2)}\right)^T$ $\hat{\mathbf{y}}$ $\mathbf{y}$

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{m}\boldsymbol{x}^T(\hat{\mathbf{y}} - \boldsymbol{y})$$

5) Update parameters

Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$\mathrm{X} = \begin{bmatrix} 1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3 \end{bmatrix}$$

$$\boldsymbol{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$$

$$\nabla_{\boldsymbol{\theta}} \boldsymbol{L} = \begin{bmatrix} \frac{\partial L}{\partial b} \\ \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \end{bmatrix} = \frac{1}{m} \mathrm{X}^T (\hat{\boldsymbol{y}} - \boldsymbol{y})$$

$$\begin{bmatrix} b \\ w_1 \\ w_2 \\ \boldsymbol{\theta} \end{bmatrix} = \begin{bmatrix} b \\ w_1 \\ w_2 \\ \boldsymbol{\theta} \end{bmatrix} - \eta \begin{bmatrix} \frac{\partial L}{\partial b} \\ \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \\ \nabla_{\boldsymbol{\theta}} \boldsymbol{L} \end{bmatrix}$$

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L$$

Logistic Regression - Minibatch

1) Pick m samples from training data

2) Compute output $\hat{\boldsymbol{y}}$

$$\boldsymbol{z} = \mathrm{X}\boldsymbol{\theta}$$

$$\hat{\boldsymbol{y}} = \sigma(\boldsymbol{z}) = \frac{1}{1+e^{-\boldsymbol{z}}}$$

3) Compute loss

$$L(\hat{\mathbf{y}}, \boldsymbol{y}) = \frac{1}{\mathrm{m}}\left(-\mathbf{y}^{\mathrm{T}}\log\hat{\mathbf{y}} - (\mathbf{1}-\mathbf{y})^{\mathrm{T}}\log(\mathbf{1}-\hat{\mathbf{y}})\right)$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{\mathrm{m}}\mathrm{X}^{\mathrm{T}}(\hat{\mathbf{y}} - \boldsymbol{y})$$

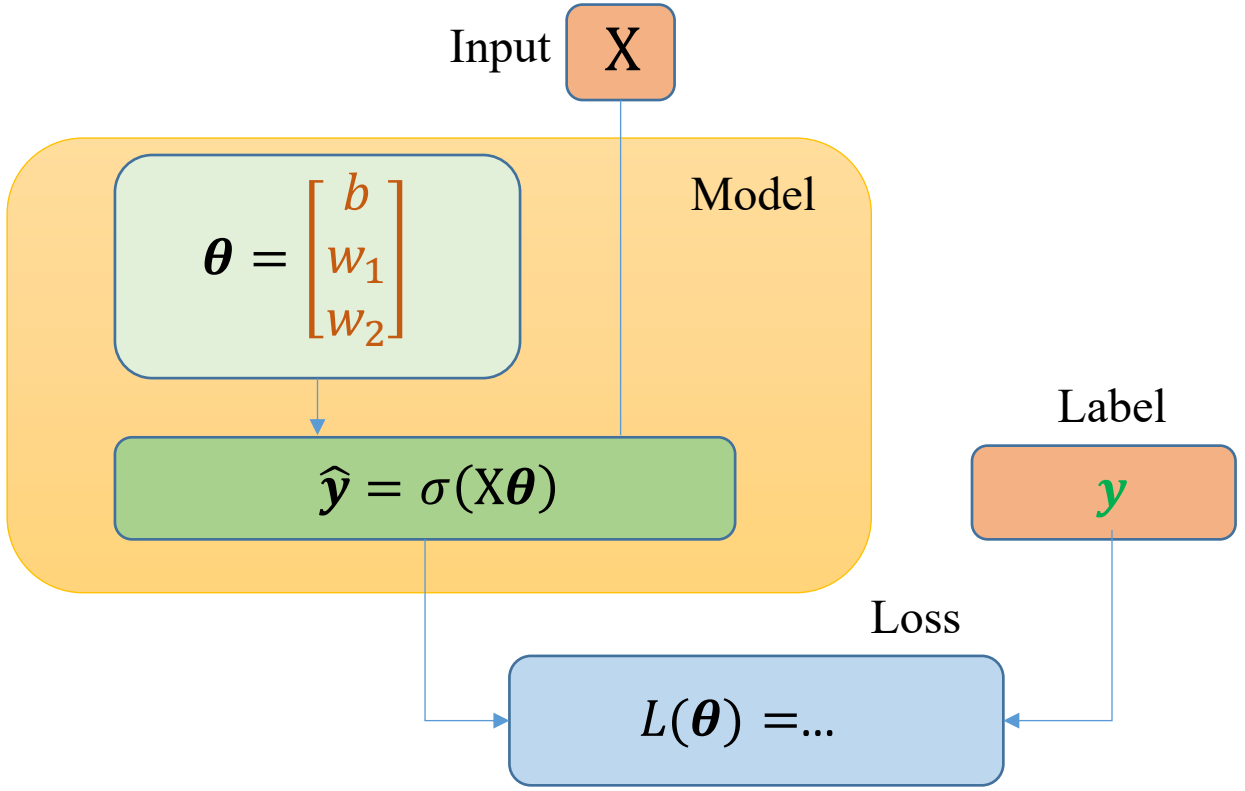
5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta\nabla_{\boldsymbol{\theta}} L$$

η is learning rate

Mini-batch m=2

$$\mathrm{X} = \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} \end{bmatrix} \qquad \boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix}$$



Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

1) Pick m samples from training data

2) Compute output $\hat{\boldsymbol{y}}$

Mini-batch m=2

$$\boldsymbol{z} = \mathrm{X}\boldsymbol{\theta}$$

$$\hat{\boldsymbol{y}} = \sigma(\boldsymbol{z}) = \frac{1}{1+e^{-\boldsymbol{z}}}$$

3) Compute loss

$$L(\hat{\mathbf{y}}, \boldsymbol{y}) = \frac{1}{\mathrm{m}}\left(-\mathbf{y}^{\mathrm{T}}\log\hat{\mathbf{y}} - (\mathbf{1}-\mathbf{y})^{\mathrm{T}}\log(\mathbf{1}-\hat{\mathbf{y}})\right)$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{\mathrm{m}}\mathrm{X}^{\mathrm{T}}(\hat{\mathbf{y}} - \boldsymbol{y})$$

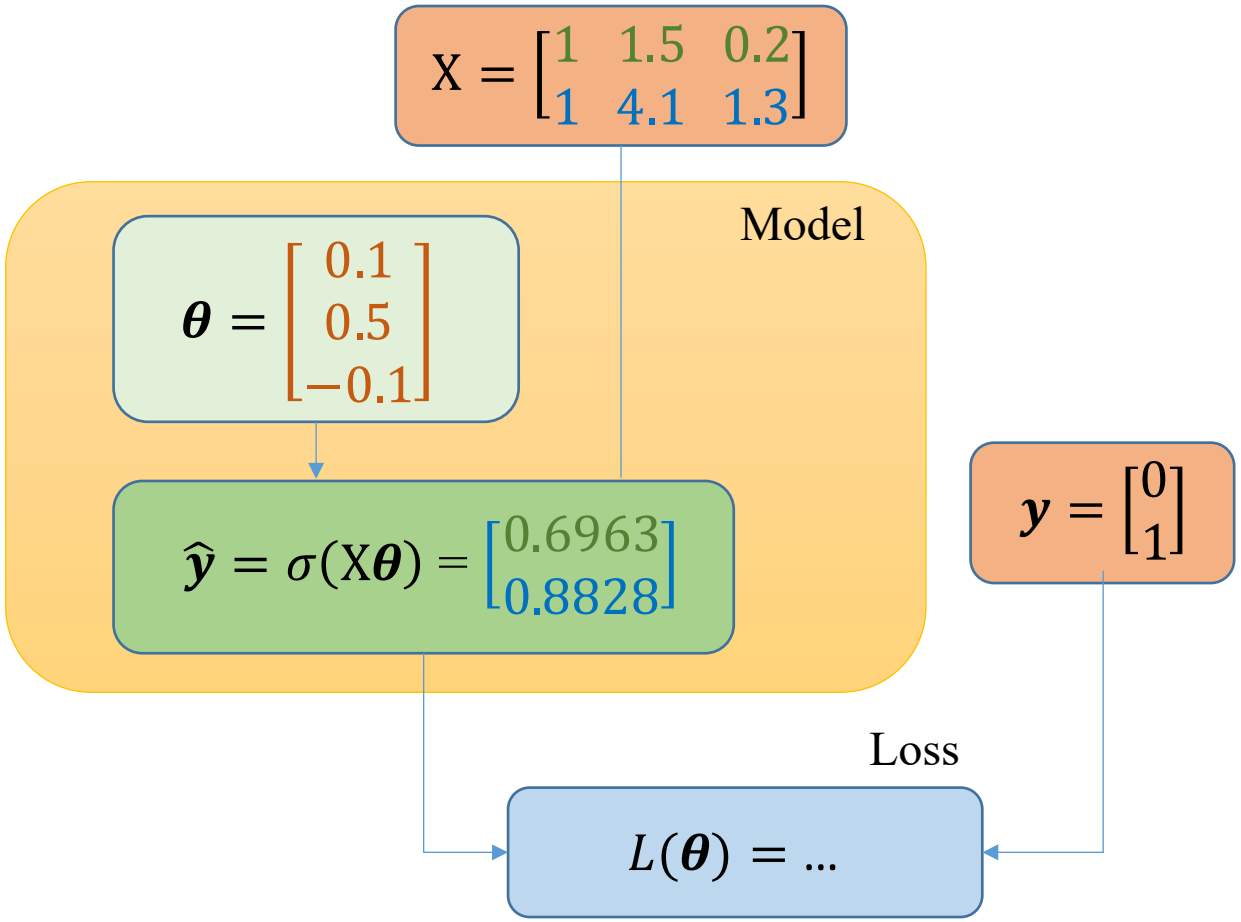
5) Update parameters

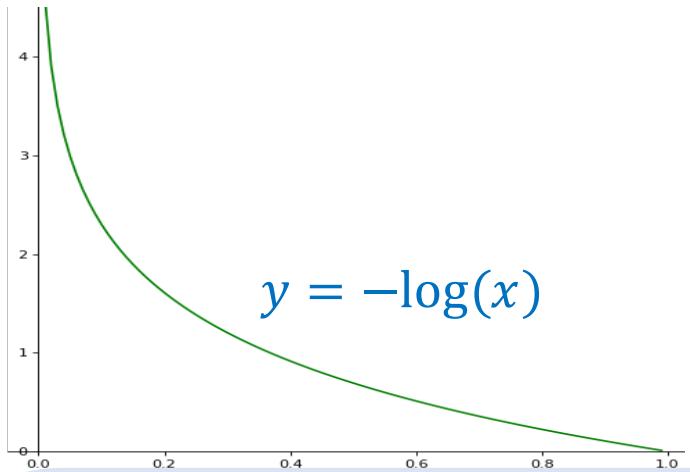
$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta\nabla_{\boldsymbol{\theta}} L$$

2

$$\hat{\boldsymbol{y}} = \sigma(\mathrm{X}\boldsymbol{\theta}) = \sigma\left(\begin{bmatrix}1 & 1.5 & 0.2\\ 1 & 4.1 & 1.3\end{bmatrix}\begin{bmatrix}0.1\\ 0.5\\ -0.1\end{bmatrix}\right)$$

$$= \sigma\left(\begin{bmatrix}0.83\\ 2.02\end{bmatrix}\right) = \begin{bmatrix}0.6963\\ 0.8828\end{bmatrix}$$





3

$$L(\boldsymbol{\theta}) = \frac{1}{\mathrm{m}}\left(-[0 \quad 1]\begin{bmatrix}\log 0.6963 \\ \log 0.8828\end{bmatrix} - [1 \quad 0]\begin{bmatrix}\log(1-0.6963) \\ \log(1-0.8828)\end{bmatrix}\right)$$

$$= \frac{1}{\mathrm{m}}(-\log 0.8828 - \log(1-0.6963))$$

$$= \frac{0.1246 + 1.1917}{\mathrm{m}} = 0.65815$$

1) Pick m samples from training data

2) Compute output $\hat{\boldsymbol{y}}$

Mini-batch m=2

$$\boldsymbol{z} = \mathrm{X}\boldsymbol{\theta}$$

$$\hat{\boldsymbol{y}} = \sigma(\boldsymbol{z}) = \frac{1}{1+e^{-\boldsymbol{z}}}$$

3) Compute loss

$$L(\hat{\mathbf{y}}, \boldsymbol{y}) = \frac{1}{\mathrm{m}}\left(-\mathbf{y}^{\mathrm{T}}\log\hat{\mathbf{y}} - (\mathbf{1}-\mathbf{y})^{\mathrm{T}}\log(\mathbf{1}-\hat{\mathbf{y}})\right)$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{\mathrm{m}}\mathrm{X}^{\mathrm{T}}(\hat{\mathbf{y}} - \boldsymbol{y})$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta\nabla_{\boldsymbol{\theta}} L$$

$$\mathrm{X} = \begin{bmatrix}1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3\end{bmatrix}$$

Model

$$\boldsymbol{\theta} = \begin{bmatrix}0.1 \\ 0.5 \\ -0.1\end{bmatrix}$$

$$\hat{\boldsymbol{y}} = \sigma(\mathrm{X}\boldsymbol{\theta}) = \begin{bmatrix}0.6963 \\ 0.8828\end{bmatrix}$$

$$\boldsymbol{y} = \begin{bmatrix}0 \\ 1\end{bmatrix}$$

Loss

$$L(\boldsymbol{\theta}) = 0.65815$$

Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

1) Pick m samples from training data

Mini-batch m=2

2) Compute output $\hat{\boldsymbol{y}}$

$$\boldsymbol{z} = \mathrm{X}\boldsymbol{\theta}$$

$$\hat{\boldsymbol{y}} = \sigma(\boldsymbol{z}) = \frac{1}{1+e^{-\boldsymbol{z}}}$$

3) Compute loss

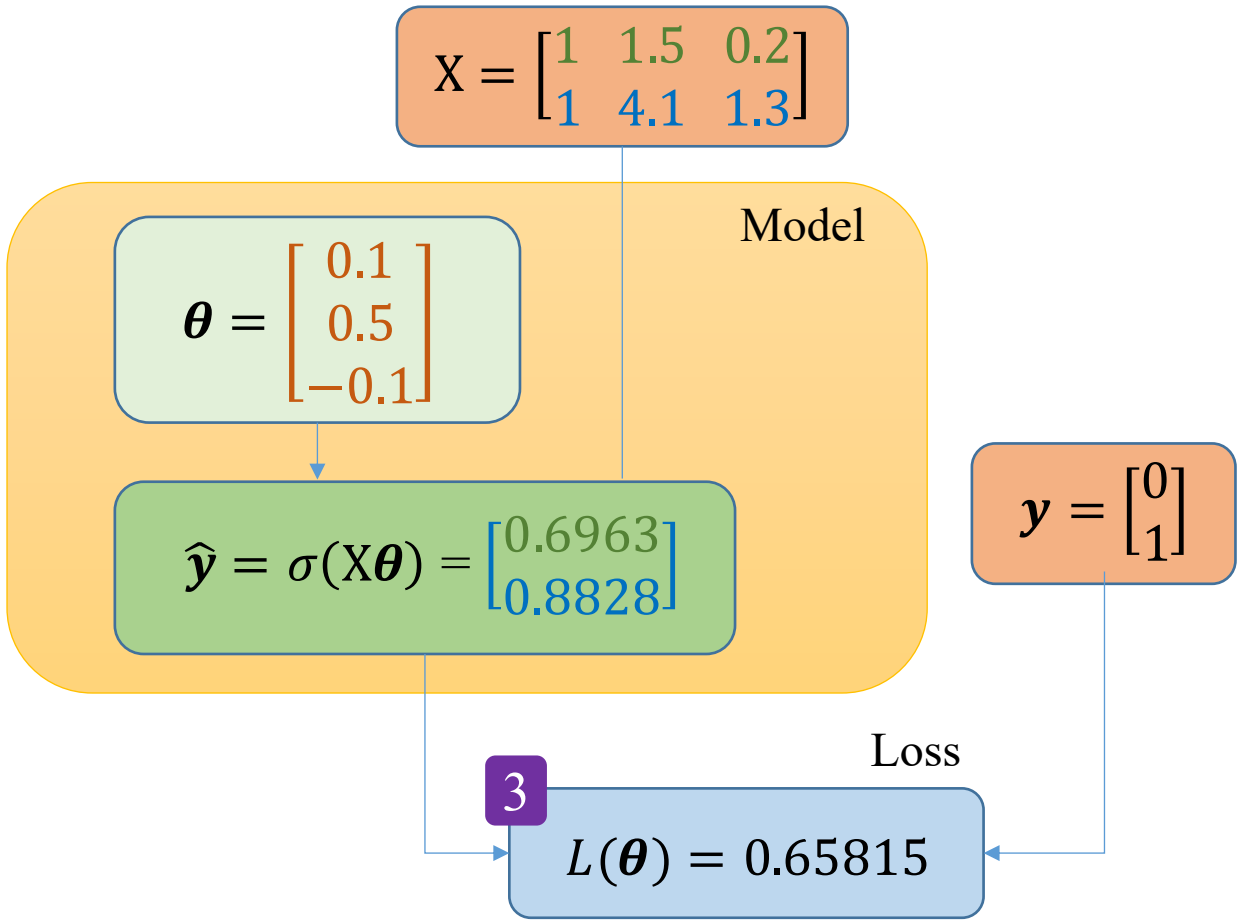
$$L(\hat{\mathbf{y}}, \boldsymbol{y}) = \frac{1}{\mathrm{m}}\left(-\mathbf{y}^{\mathrm{T}}\log\hat{\mathbf{y}} - (\mathbf{1}-\mathbf{y})^{\mathrm{T}}\log(\mathbf{1}-\hat{\mathbf{y}})\right)$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{\mathrm{m}}\mathrm{X}^{\mathrm{T}}(\hat{\mathbf{y}} - \boldsymbol{y})$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta\nabla_{\boldsymbol{\theta}} L$$



4

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{\mathrm{m}}\begin{bmatrix} 1 & 1 \\ 1.5 & 4.1 \\ 0.2 & 1.3 \end{bmatrix}\left(\begin{bmatrix} 0.6963 \\ 0.8828 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$$

$$= \frac{1}{\mathrm{m}}\begin{bmatrix} 1 & 1 \\ 1.5 & 4.1 \\ 0.2 & 1.3 \end{bmatrix}\begin{bmatrix} 0.6963 \\ -0.1171 \end{bmatrix} = \begin{bmatrix} 0.28961 \\ 0.28217 \\ -0.0064 \end{bmatrix}$$

Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

1) Pick m samples from training data

Mini-batch m=2

2) Compute output $\hat{y}$

$$z = X\theta$$

$$\hat{y} = \sigma(z) = \frac{1}{1+e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = \frac{1}{m}\left(-y^{T}\log\hat{y} - (1-y)^{T}\log(1-\hat{y})\right)$$

4) Compute derivative

$$\nabla_{\theta} L = \frac{1}{m} X^{T}(\hat{y} - y)$$

5) Update parameters

$$\theta = \theta - \eta\nabla_{\theta} L$$

$$X = \begin{bmatrix} 1 & 1.5 & 0.2 \\ 1 & 4.1 & 1.3 \end{bmatrix}$$

Model

$$\theta = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}$$

$$\hat{y} = \sigma(X\theta) = \begin{bmatrix} 0.6963 \\ 0.8828 \end{bmatrix}$$

$$y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Loss

3 $L(\theta) = 0.65815$

4 $\nabla_{\theta} L = \begin{bmatrix} 0.28961 \\ 0.28217 \\ -0.0064 \end{bmatrix}$

5 $\theta - \eta L'_{\theta} = \begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix} - \eta \begin{bmatrix} 0.28961 \\ 0.28217 \\ -0.0064 \end{bmatrix} = \begin{bmatrix} 0.0971 \\ 0.4971 \\ -0.099 \end{bmatrix}$

Using all the samples

Logistic Regression - Batch

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

1) Pick all the samples from training data

2) Compute output $\hat{\boldsymbol{y}}$

$$z = \mathrm{X}\boldsymbol{\theta}$$

$$\hat{\boldsymbol{y}} = \sigma(\boldsymbol{z}) = \frac{1}{1+e^{-\boldsymbol{z}}}$$

3) Compute loss

$$L(\hat{\mathbf{y}}, \boldsymbol{y}) = \frac{1}{\mathrm{N}}\left(-\mathbf{y}^{\mathrm{T}}\log\hat{\mathbf{y}} - (\mathbf{1}-\mathbf{y})^{\mathrm{T}}\log(\mathbf{1}-\hat{\mathbf{y}})\right)$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{\mathrm{N}}\mathrm{X}^{\mathrm{T}}(\hat{\mathbf{y}} - \boldsymbol{y})$$

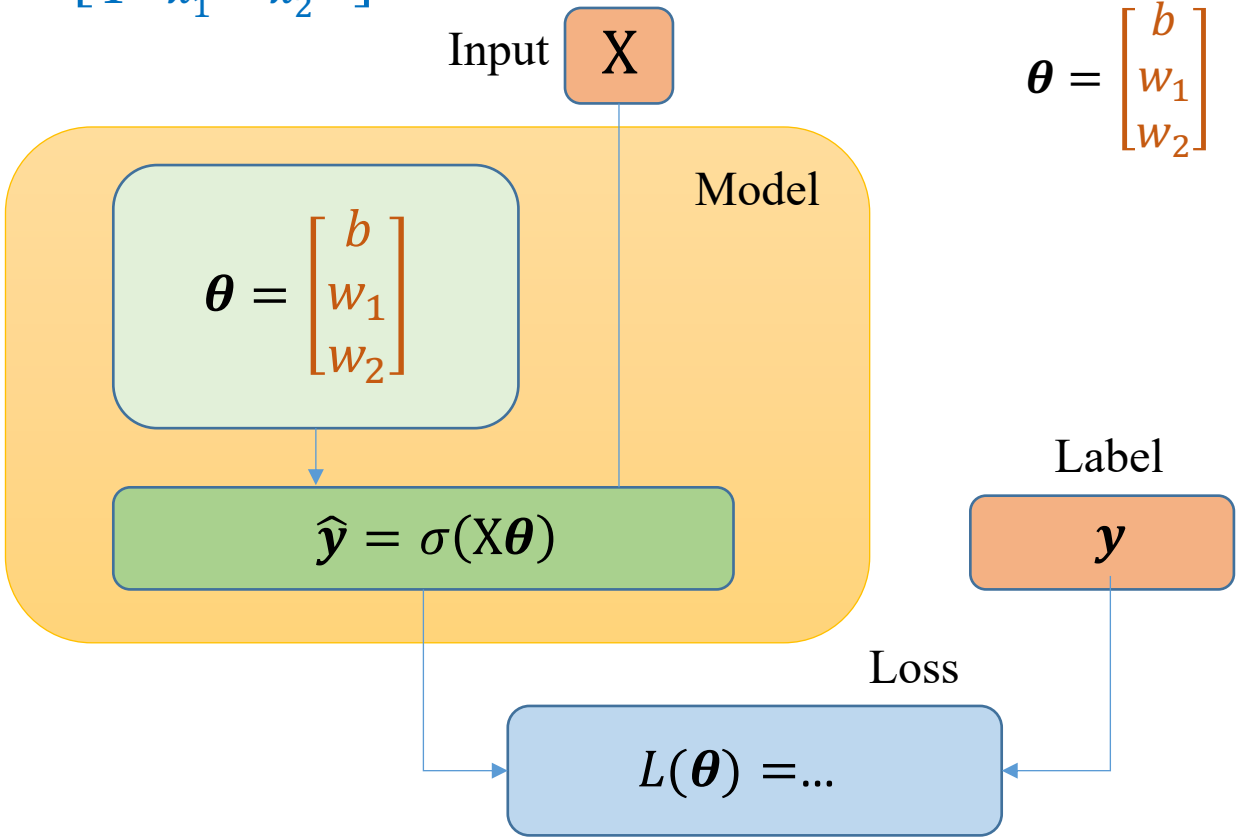
5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta\nabla_{\boldsymbol{\theta}} L$$

η is learning rate

$$\mathrm{X} = \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} \\ 1 & x_1^{(3)} & x_2^{(3)} \\ 1 & x_1^{(4)} & x_2^{(4)} \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix}$$



Logistic Regression - Batch

Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$\mathrm{X} = \begin{bmatrix} 1 & 1.4 & 0.2 \\ 1 & 1.5 & 0.2 \\ 1 & 3.0 & 1.1 \\ 1 & 4.1 & 1.3 \end{bmatrix} \qquad \boldsymbol{y} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \qquad \boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix}$$

1) Pick all the samples from training data

2) Compute output $\hat{\boldsymbol{y}}$

$$z = \mathrm{X}\boldsymbol{\theta}$$

$$\hat{\boldsymbol{y}} = \sigma(\boldsymbol{z}) = \frac{1}{1 + e^{-\boldsymbol{z}}}$$

3) Compute loss

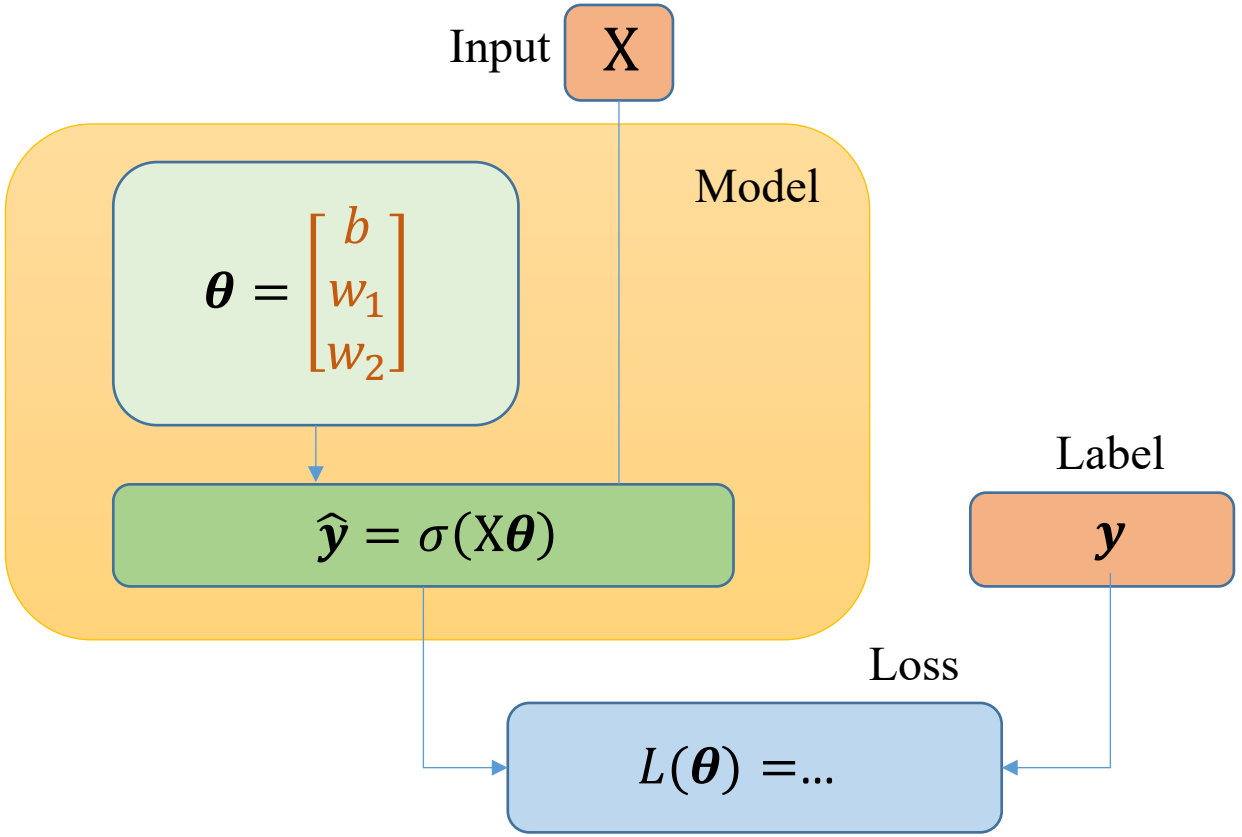
$$L(\hat{\mathbf{y}}, \boldsymbol{y}) = \frac{1}{\mathrm{N}}\left(-\mathbf{y}^{\mathrm{T}}\log\hat{\mathbf{y}} - (\mathbf{1}-\mathbf{y})^{\mathrm{T}}\log(\mathbf{1}-\hat{\mathbf{y}})\right)$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{\mathrm{N}}\mathrm{X}^{\mathrm{T}}(\hat{\mathbf{y}} - \boldsymbol{y})$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta\nabla_{\boldsymbol{\theta}} L$$



Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$\boldsymbol{x} = \begin{bmatrix} 1 & 1.4 & 0.2 \\ 1 & 1.5 & 0.2 \\ 1 & 3.0 & 1.1 \\ 1 & 4.1 & 1.3 \end{bmatrix}$$

1) Pick all the samples from training data

2) Compute output $\hat{\boldsymbol{y}}$

$$z = \mathrm{X}\boldsymbol{\theta}$$

$$\hat{\boldsymbol{y}} = \sigma(\boldsymbol{z}) = \frac{1}{1 + e^{-\boldsymbol{z}}}$$

3) Compute loss

$$L(\hat{\mathbf{y}}, \boldsymbol{y}) = \frac{1}{\mathrm{N}}\left(-\mathbf{y}^{\mathrm{T}}\log\hat{\mathbf{y}} - (\mathbf{1}-\mathbf{y})^{\mathrm{T}}\log(\mathbf{1}-\hat{\mathbf{y}})\right)$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{\mathrm{N}}\mathrm{X}^{\mathrm{T}}(\hat{\mathbf{y}} - \boldsymbol{y})$$

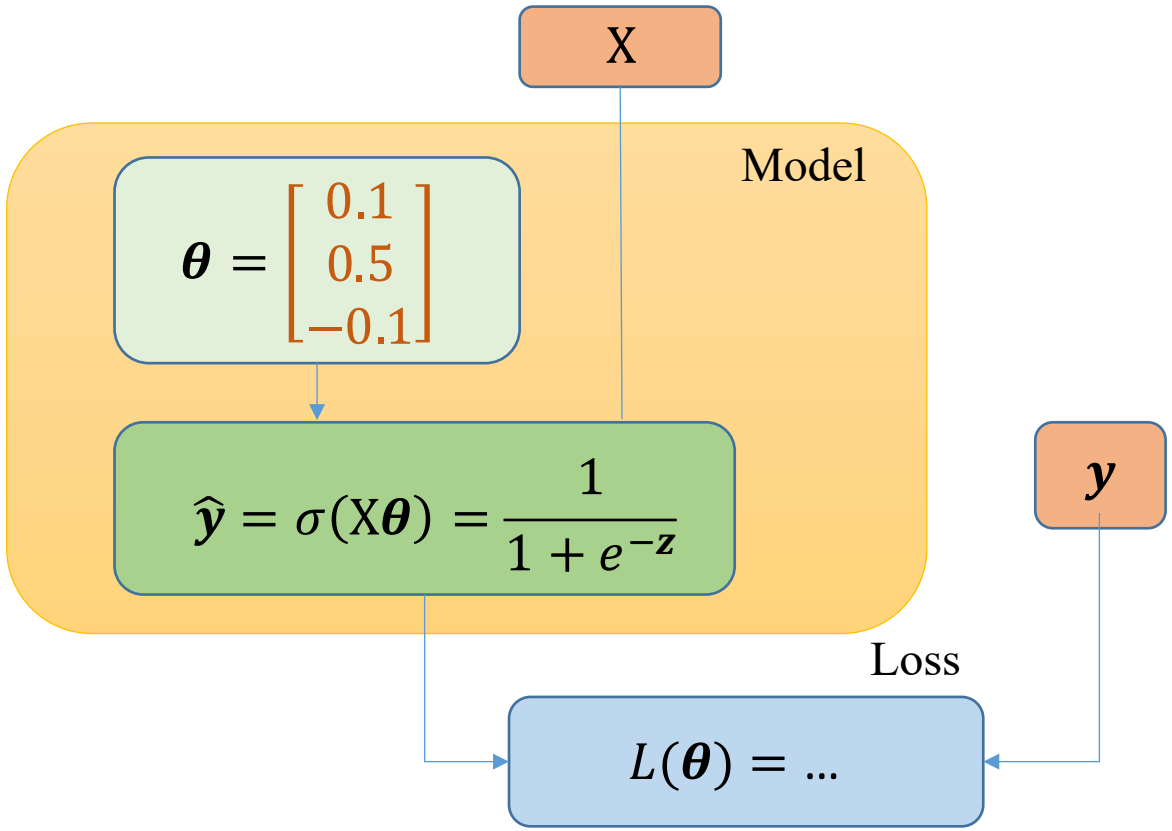
5) Update parameters

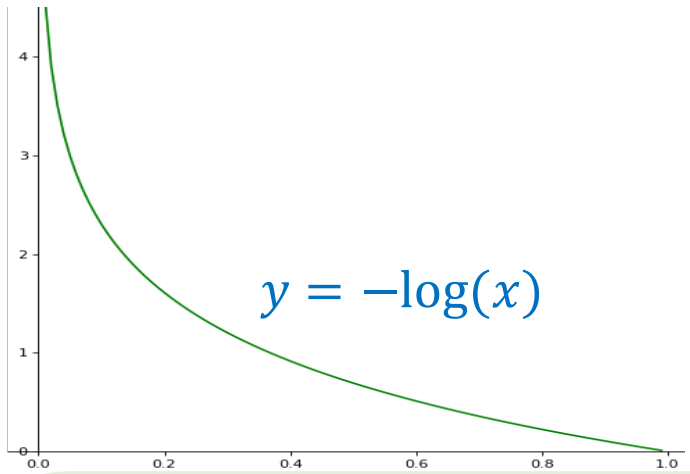
$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta\nabla_{\boldsymbol{\theta}} L$$

2

$$\hat{\boldsymbol{y}} = \sigma(\mathrm{X}\boldsymbol{\theta}) = \sigma\left(\begin{bmatrix} 1 & 1.4 & 0.2 \\ 1 & 1.5 & 0.2 \\ 1 & 3.0 & 1.1 \\ 1 & 4.1 & 1.3 \end{bmatrix}\begin{bmatrix} 0.1 \\ 0.5 \\ -0.1 \end{bmatrix}\right)$$

$$= \sigma\left(\begin{bmatrix} 0.78 \\ 0.83 \\ 1.49 \\ 2.02 \end{bmatrix}\right) = \begin{bmatrix} 0.6856 \\ 0.6963 \\ 0.8160 \\ 0.8828 \end{bmatrix}$$





3

$$L(\boldsymbol{\theta}) = \frac{1}{\mathrm{N}}\left\{-\begin{bmatrix}0\\0\\1\\1\end{bmatrix}^{\mathrm{T}} \log\begin{bmatrix}0.6856\\0.6963\\0.8160\\0.8828\end{bmatrix} - \left(\begin{bmatrix}1\\1\\1\\1\end{bmatrix} - \begin{bmatrix}0\\0\\1\\1\end{bmatrix}\right)^{T} \log\left(1 - \begin{bmatrix}0.6856\\0.6963\\0.8160\\0.8828\end{bmatrix}\right)\right\}$$

1) Pick all the samples from training data

2) Compute output $\hat{\boldsymbol{y}}$

$$z = \mathrm{X}\boldsymbol{\theta}$$

$$\hat{\boldsymbol{y}} = \sigma(\boldsymbol{z}) = \frac{1}{1+e^{-\boldsymbol{z}}}$$

3) Compute loss

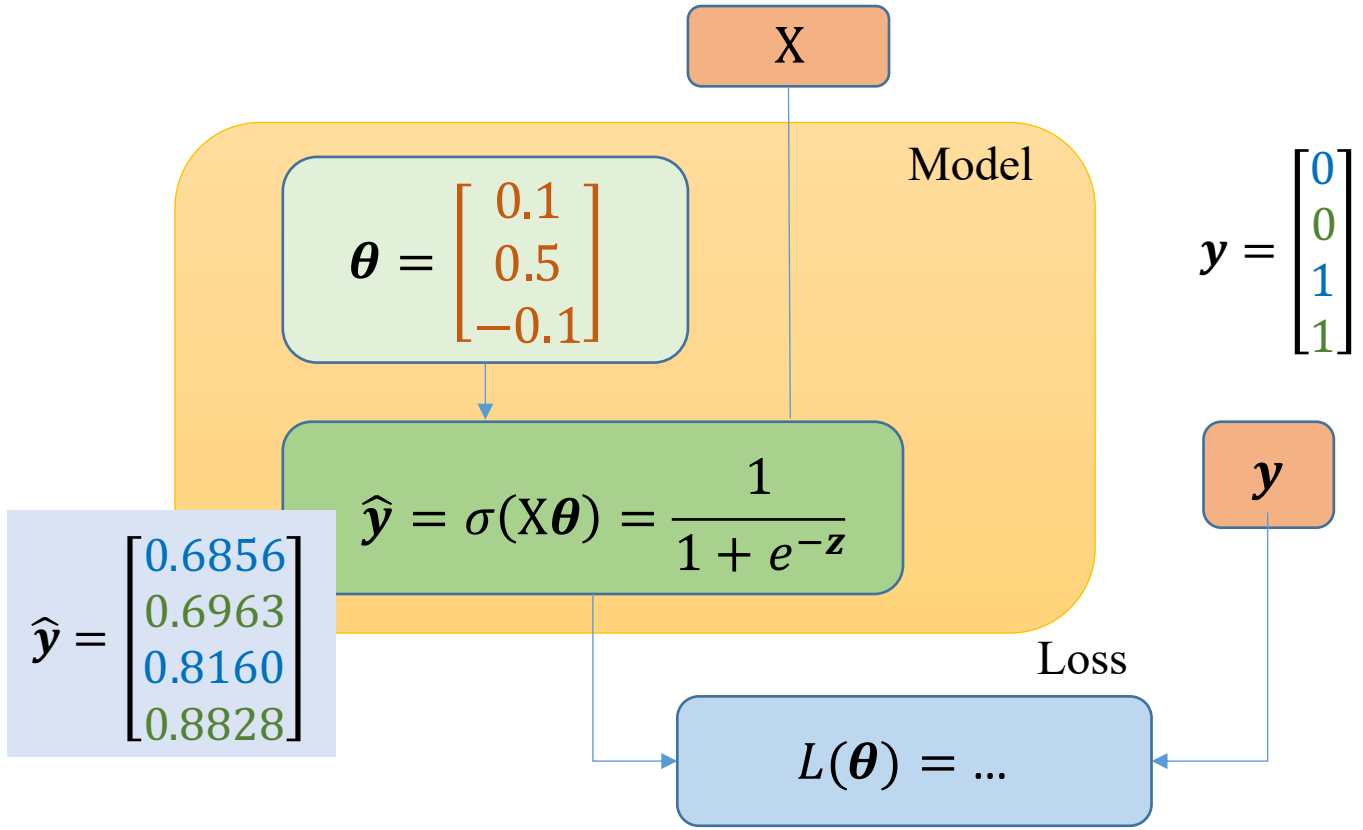
$$L(\hat{\mathbf{y}}, \boldsymbol{y}) = \frac{1}{\mathrm{N}}\left(-\mathbf{y}^{\mathrm{T}}\log\hat{\mathbf{y}} - (\mathbf{1}-\mathbf{y})^{\mathrm{T}}\log(\mathbf{1}-\hat{\mathbf{y}})\right)$$

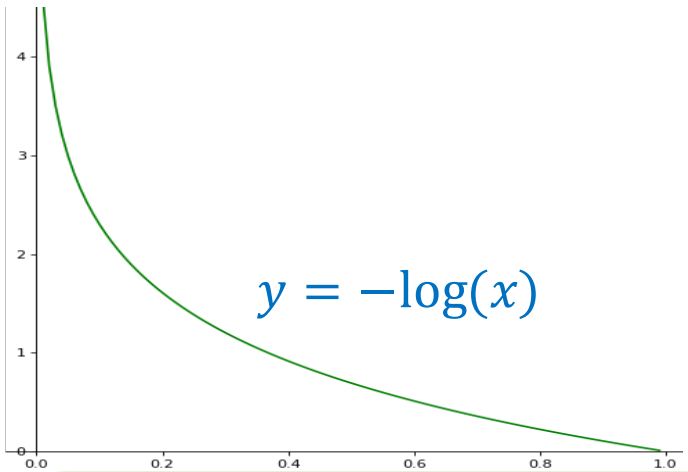
4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{\mathrm{N}}\mathrm{X}^{\mathrm{T}}(\hat{\mathbf{y}} - \boldsymbol{y})$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta\nabla_{\boldsymbol{\theta}} L$$





3

$$L(\boldsymbol{\theta}) = \frac{1}{\mathrm{N}}\left\{-\begin{bmatrix}0\\0\\1\\1\end{bmatrix}^{\mathrm{T}} \log\left(\begin{bmatrix}0.6856\\0.6963\\0.8160\\0.8828\end{bmatrix}\right) - \begin{bmatrix}1\\1\\0\\0\end{bmatrix}^{T} \log\left(\begin{bmatrix}0.3144\\0.3037\\0.1840\\0.1172\end{bmatrix}\right)\right\}$$

$$= \frac{1}{\mathrm{N}}(-\log 0.8160 - \log 0.8828 - \log 0.3144 - \log 0.3037)$$

$$= 0.6691$$

1) Pick all the samples from training data

2) Compute output $\hat{\boldsymbol{y}}$

$$z = \mathrm{X}\boldsymbol{\theta}$$

$$\hat{\boldsymbol{y}} = \sigma(\boldsymbol{z}) = \frac{1}{1+e^{-\boldsymbol{z}}}$$

3) Compute loss

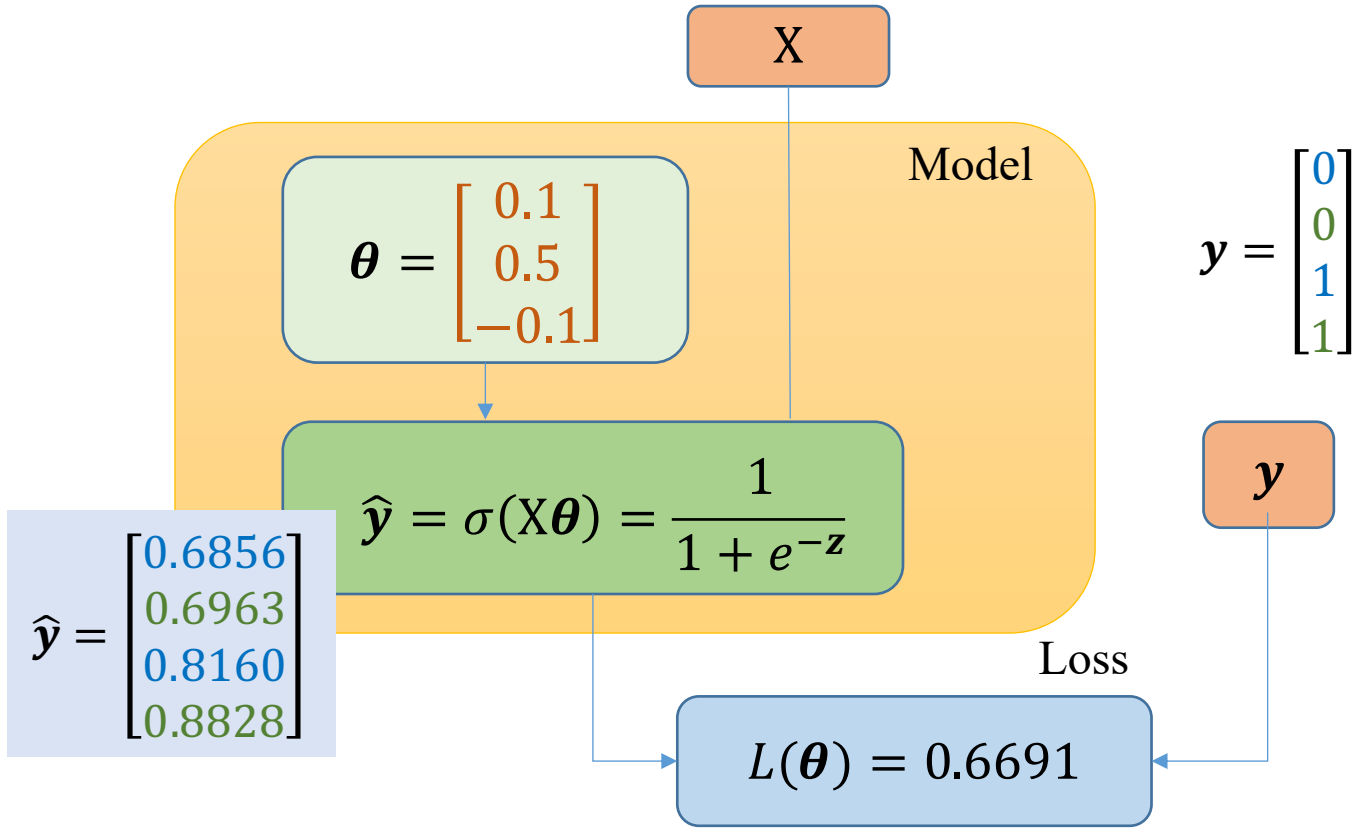
$$L(\hat{\mathbf{y}}, \boldsymbol{y}) = \frac{1}{\mathrm{N}}\left(-\mathbf{y}^{\mathrm{T}}\log\hat{\mathbf{y}} - (\mathbf{1}-\mathbf{y})^{\mathrm{T}}\log(\mathbf{1}-\hat{\mathbf{y}})\right)$$

4) Compute derivative

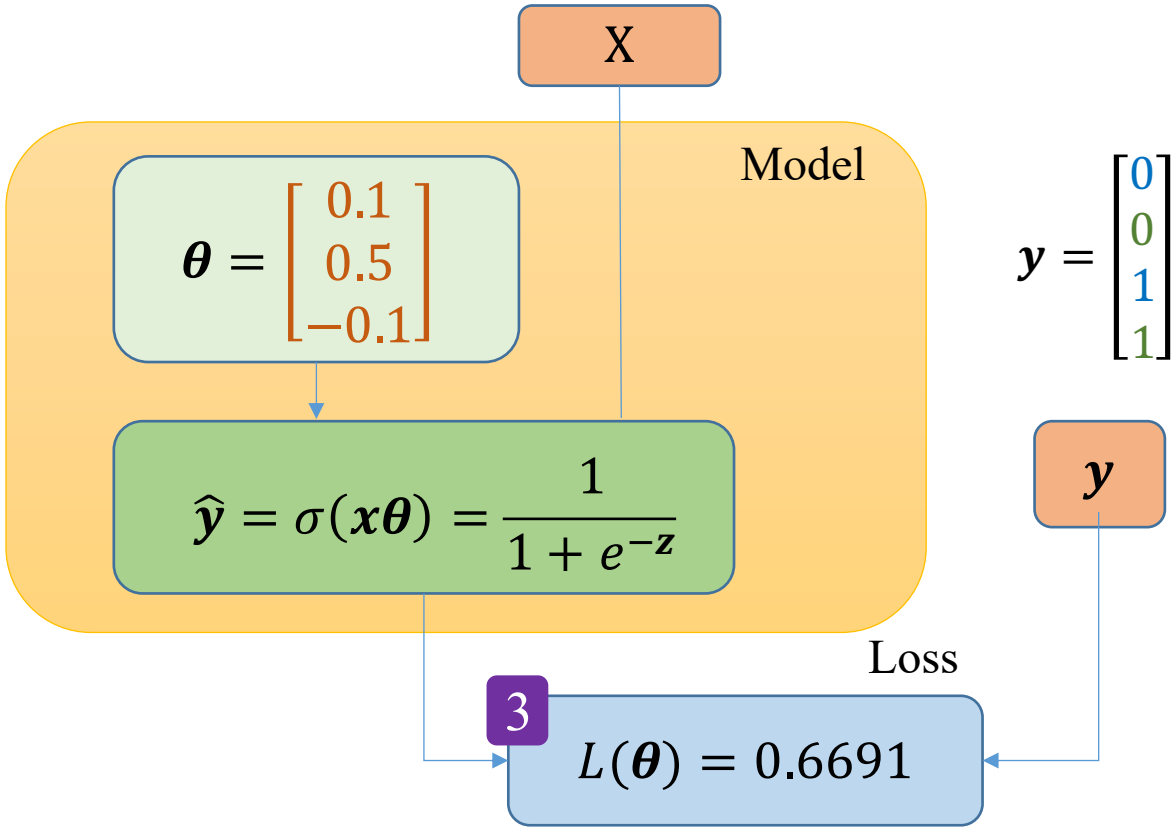
$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{\mathrm{N}}\mathrm{X}^{\mathrm{T}}(\hat{\mathbf{y}} - \boldsymbol{y})$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta\nabla_{\boldsymbol{\theta}} L$$



$$x = \begin{bmatrix} 1 & 1.4 & 0.2 \\ 1 & 1.5 & 0.2 \\ 1 & 3.0 & 1.1 \\ 1 & 4.1 & 1.3 \end{bmatrix} \qquad \hat{y} = \begin{bmatrix} 0.6856 \\ 0.6963 \\ 0.8160 \\ 0.8828 \end{bmatrix}$$



1) Pick all the samples from training data

2) Compute output $\hat{y}$

$$z = \mathrm{X}\boldsymbol{\theta}$$

$$\hat{\boldsymbol{y}} = \sigma(\boldsymbol{z}) = \frac{1}{1+e^{-\boldsymbol{z}}}$$

3) Compute loss

$$L(\hat{\mathbf{y}}, \boldsymbol{y}) = \frac{1}{\mathrm{N}}\left(-\mathbf{y}^{\mathrm{T}}\log\hat{\mathbf{y}} - (\mathbf{1}-\mathbf{y})^{\mathrm{T}}\log(\mathbf{1}-\hat{\mathbf{y}})\right)$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{\mathrm{N}}\mathrm{X}^{\mathrm{T}}(\hat{\mathbf{y}} - \boldsymbol{y})$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta\nabla_{\boldsymbol{\theta}} L$$

4

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{\mathrm{N}}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1.4 & 1.5 & 3.0 & 4.1 \\ 0.2 & 0.2 & 1.1 & 1.3 \end{bmatrix}\left(\begin{bmatrix} 0.6856 \\ 0.6963 \\ 0.8160 \\ 0.8828 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}\right)$$

$$= \frac{1}{\mathrm{N}}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1.4 & 1.5 & 3.0 & 4.1 \\ 0.2 & 0.2 & 1.1 & 1.3 \end{bmatrix}\begin{bmatrix} 0.6856 \\ 0.6963 \\ -0.184 \\ -0.117 \end{bmatrix} = \begin{bmatrix} 0.2702 \\ 0.2431 \\ -0.019 \end{bmatrix}$$

Dataset

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

1) Pick all the samples from training data

2) Compute output $\hat{y}$

$$z = X\theta$$

$$\hat{y} = \sigma(z) = \frac{1}{1+e^{-z}}$$

3) Compute loss

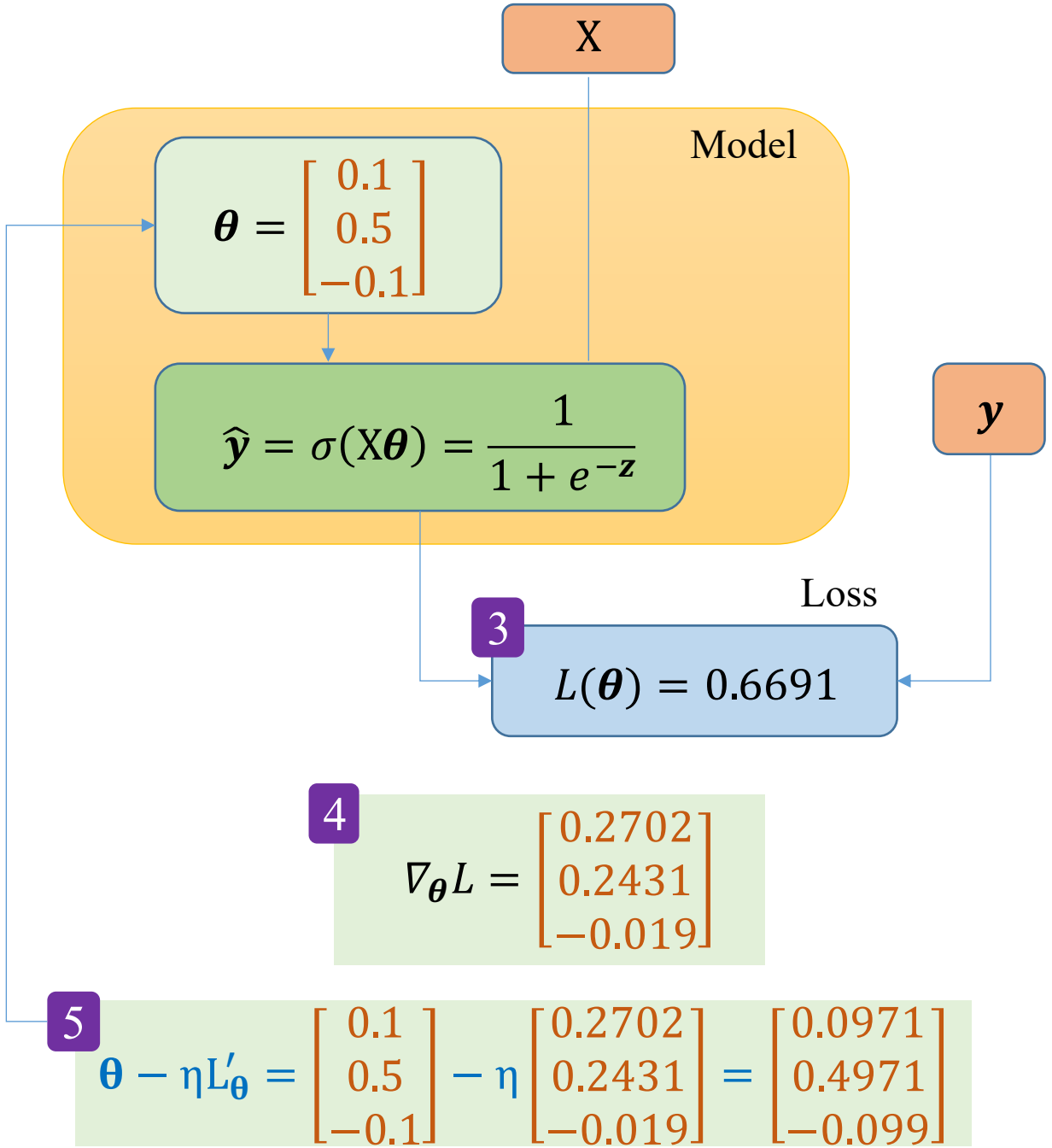
$$L(\hat{y}, y) = \frac{1}{N}\left(-y^{T}\log\hat{y} - (1-y)^{T}\log(1-\hat{y})\right)$$

4) Compute derivative

$$\nabla_{\theta} L = \frac{1}{N} X^{T}(\hat{y} - y)$$

5) Update parameters

$$\theta = \theta - \eta\nabla_{\theta} L$$



Different Setting

1) Pick all the samples from training data

2) Compute output $\hat{\boldsymbol{y}}$

$$\boldsymbol{z} = \mathrm{X}\boldsymbol{\theta} + \boldsymbol{b}$$

$$\hat{\boldsymbol{y}} = \sigma(\boldsymbol{z}) = \frac{1}{1 + e^{-\boldsymbol{z}}}$$

3) Compute loss

$$L(\hat{\mathbf{y}}, \boldsymbol{y}) = \frac{1}{\mathrm{N}}\left(-\mathbf{y}^{\mathrm{T}}\log\hat{\mathbf{y}} - (\mathbf{1}-\mathbf{y})^{\mathrm{T}}\log(\mathbf{1}-\hat{\mathbf{y}})\right)$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \frac{1}{\mathrm{N}}\mathrm{X}^{\mathrm{T}}(\hat{\mathbf{y}} - \boldsymbol{y}) \qquad \mathrm{L'_b} = \frac{1}{\mathrm{N}}\mathbf{1}^{\mathrm{T}}(\hat{\mathbf{y}} - \boldsymbol{y})$$

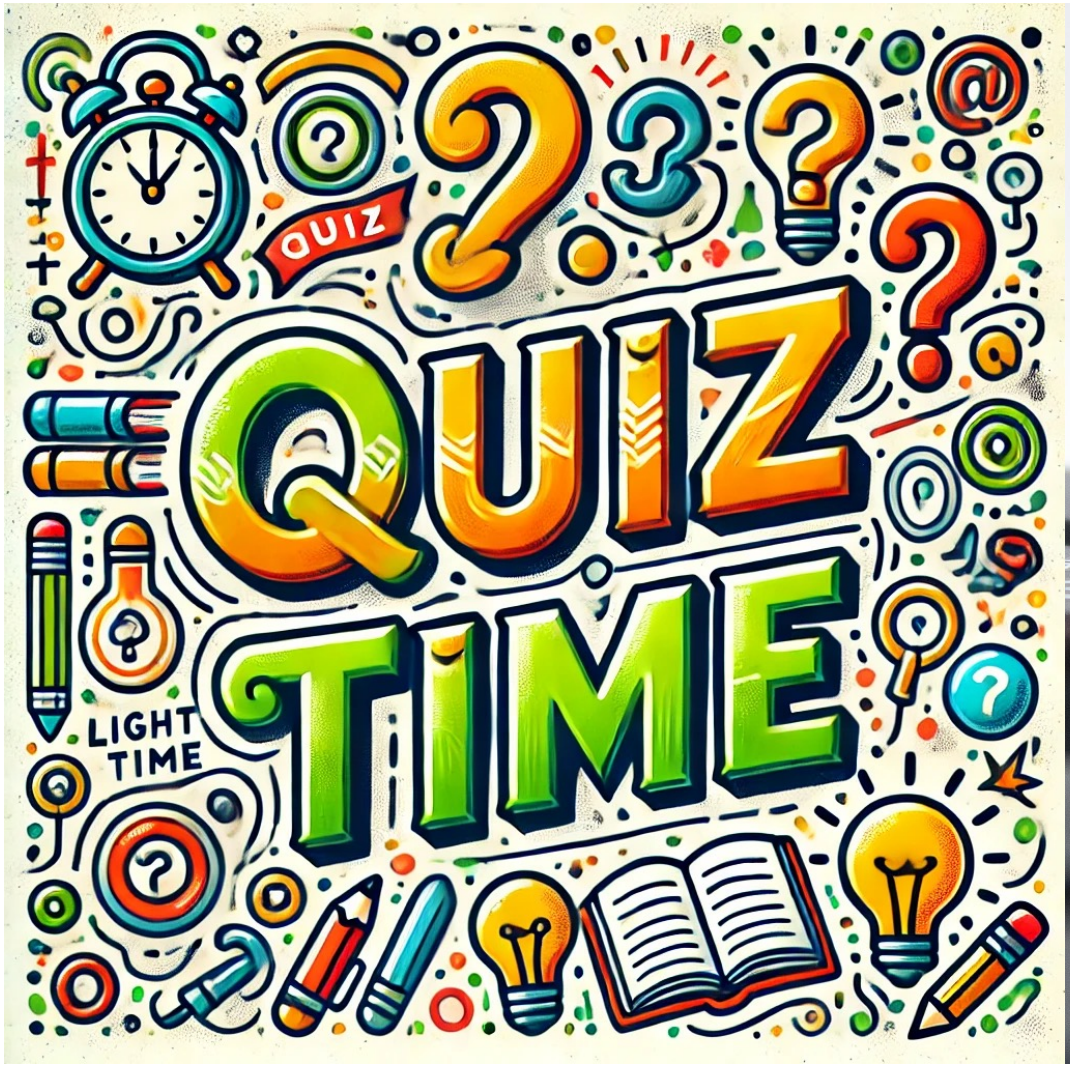
5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta\nabla_{\boldsymbol{\theta}} L \qquad \mathrm{b} = \mathrm{b} - \eta\mathrm{L'_b}$$

Petal_Length	Petal_Width	Label
1.4	0.2	0
1.5	0.2	0
3	1.1	1
4.1	1.3	1

$$\mathrm{X} = \begin{bmatrix} x_1^{(1)} & x_2^{(1)} \\ x_1^{(2)} & x_2^{(2)} \\ x_1^{(3)} & x_2^{(3)} \\ x_1^{(4)} & x_2^{(4)} \end{bmatrix} \qquad \boldsymbol{\theta} = \begin{bmatrix} w_1 \\ w_1 \end{bmatrix}$$

$$\mathrm{b} \qquad \boldsymbol{b} = \begin{bmatrix} \mathrm{b} \\ \mathrm{b} \end{bmatrix}$$


QUIZ
QUIZ
TIME
LIGHT
TIME


BREAK
TIME!!!

Summary

Traditional

1) Pick a sample (x, y) from training data

2) Compute the output $\hat{y}$

$$z = wx + b \qquad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y\log\hat{y} - (1-y)\log(1-\hat{y}))$$

4) Compute derivative

$$\frac{\partial L}{\partial w} = x(\hat{y} - y) \qquad \frac{\partial L}{\partial b} = (\hat{y} - y)$$

5) Update parameters

$$w = w - \eta\frac{\partial L}{\partial w} \qquad b = b - \eta\frac{\partial L}{\partial b}$$

η is learning rate

Vectorized

1) Pick a sample (x, y) from training data

2) Compute output $\hat{y}$

$$z = \boldsymbol{\theta}^T\boldsymbol{x} = \boldsymbol{x}^T\boldsymbol{\theta} \qquad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Compute loss

$$L(\hat{y}, y) = (-y\log\hat{y} - (1-y)\log(1-\hat{y}))$$

4) Compute derivative

$$\nabla_{\boldsymbol{\theta}} L = \boldsymbol{x}(\hat{y} - y)$$

5) Update parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta\nabla_{\boldsymbol{\theta}} L$$

η is learning rate

Outline

SECTION 1

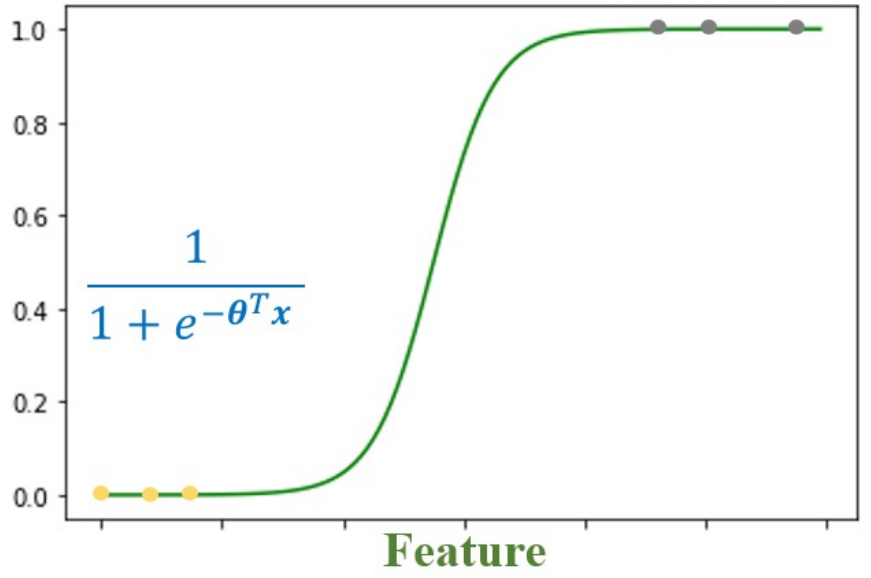
Logistic Regression

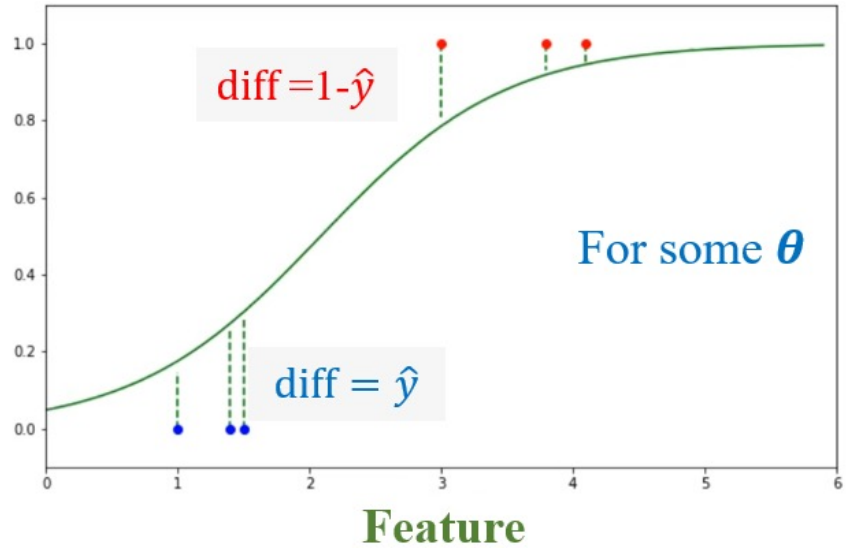
SECTION 2

Using Multi-samples

SECTION 3

Discussion (Optional)





BCE Reconstruction

Rethinking Logistic Regression

❖ What about the loss function?

Feature	Label	
Petal_Length	Category	
1.4	Flower A	Category 0
1	Flower A	
1.5	Flower A	
3	Flower B	Category 1
3.8	Flower B	
4.1	Flower B	

Assign numbers to categories

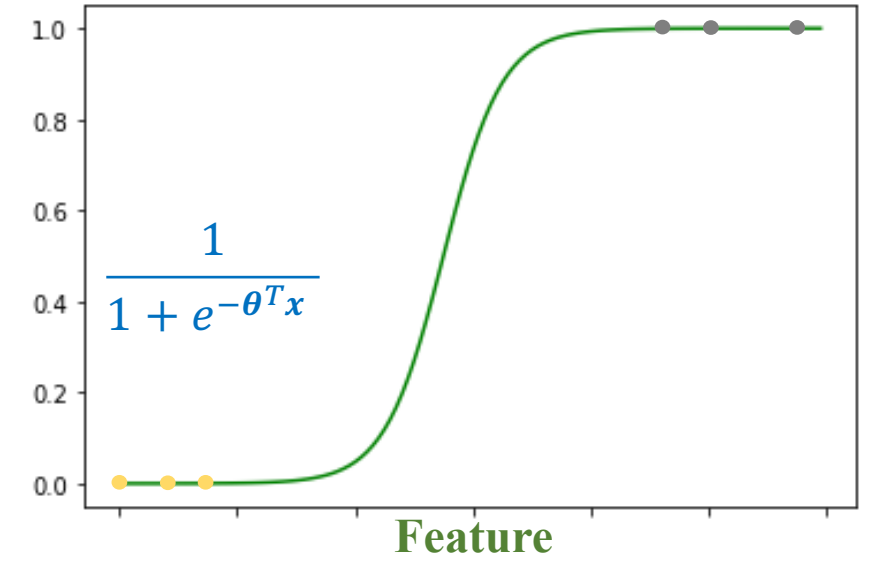
Feature	Label	
Petal_Length	Category	
1.4	0	Category 0
1	0	
1.5	0	
3	1	Category 1
3.8	1	
4.1	1	

Sigmoid function could fit the data

$$z = wx + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\hat{y} \in (0 \quad 1)$$



Error

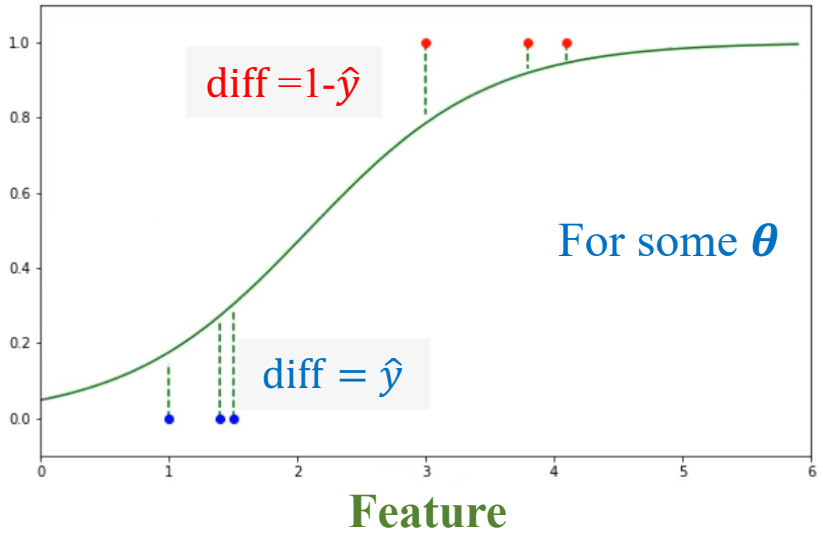
if $y = 1$

$\quad \text{diff} = 1 - \hat{y}$

if $y = 0$

$\quad \text{diff} = \hat{y}$

$$L(\hat{y}) = (\hat{y} - y)^2$$



Logistic Regression-MSE

❖ Construct loss

Model and Loss

$$z = \boldsymbol{\theta}^T \boldsymbol{x} = \boldsymbol{x}^T \boldsymbol{\theta}$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\mathrm{L} = (\hat{y} - y)^2$$

Derivative

$$\frac{\partial L}{\partial \theta_i} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial \theta_i}$$

$$\frac{\partial \hat{y}}{\partial z} = \hat{y}(1 - \hat{y})$$

$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y)$$

$$\frac{\partial z}{\partial \theta_i} = x_i$$

$$\frac{\partial L}{\partial \theta_i} = 2x_i(\hat{y} - y)\hat{y}(1 - \hat{y})$$

Logistic Regression-MSE

❖ Implement

Model and Loss

$$z = \boldsymbol{\theta}^T \boldsymbol{x} = \boldsymbol{x}^T \boldsymbol{\theta}$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\mathrm{L} = (\hat{y} - y)^2$$

$$\frac{\partial L}{\partial \theta_i} = 2x_i(\hat{y} - y)\hat{y}(1 - \hat{y})$$

```python
def sigmoid_function(z):
    return 1 / (1 + np.exp(-z))

def predict(x, theta):
    y_hat = sigmoid_function(np.dot(x, theta))
    return y_hat

def loss_function(y_hat, y):
    return (y_hat-y)**2

def compute_gradient(x, y_hat, y):
    return 2*x*(y_hat - y)*y_hat*(1-y_hat)
```

```python
# predict z
y_hat = predict(xi, theta)

# compute loss
loss = loss_function(y_hat, yi)

# compute gradient and update
gradient = compute_gradient(xi, y_hat, yi)
theta -= lr*gradient
```

Logistic Regression-MSE

❖ **Result**

Done?

Feature	Label	
Petal_Length	Category	
1.4	0	Category 0
1	0	
1.5	0	
3	1	Category 1
3.8	1	
4.1	1	

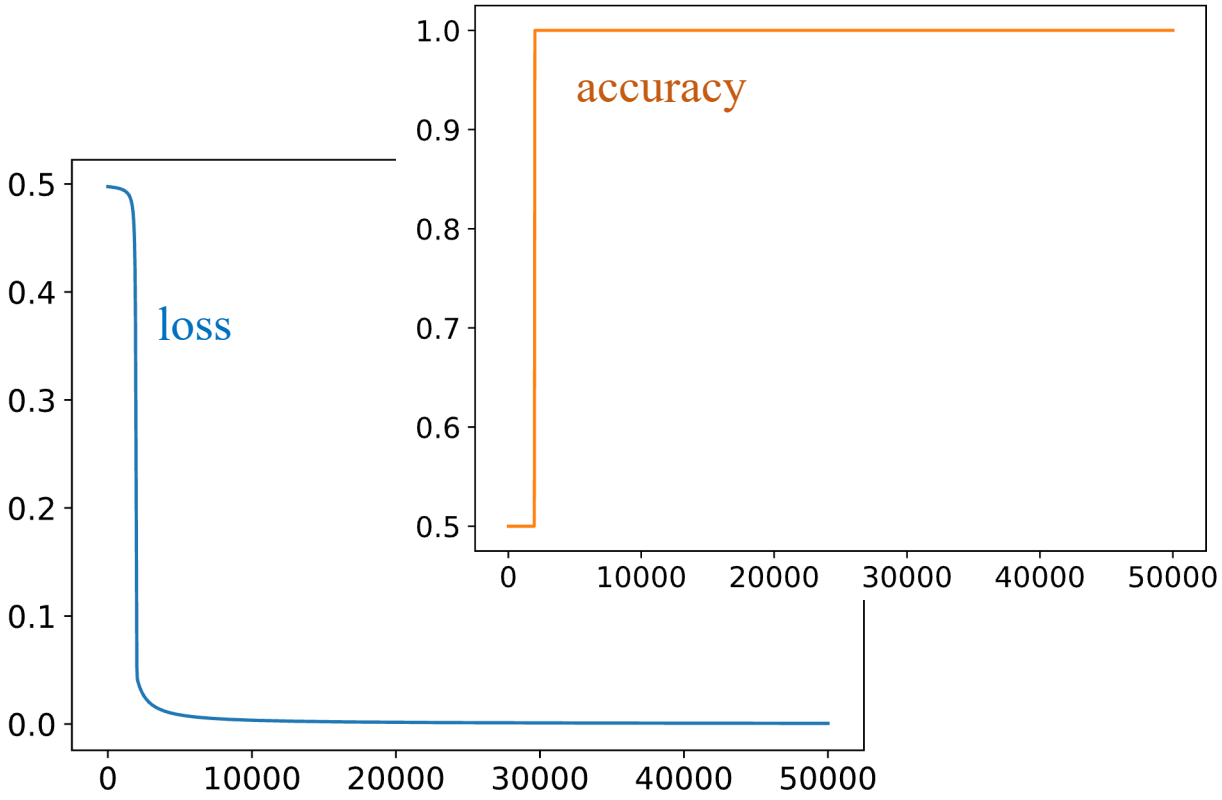
Model and Loss

$$z = \boldsymbol{\theta}^T \boldsymbol{x} = \boldsymbol{x}^T \boldsymbol{\theta}$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\mathrm{L} = (\hat{y} - y)^2$$

$$\frac{\partial L}{\partial \theta_i} = 2x_i(\hat{y} - y)\hat{y}(1 - \hat{y})$$



Discussion

Hessian Matrices

❖ Definition

The Hessian matrix or Hessian is a square matrix of second-order partial derivatives of a scalar-valued function

https://en.wikipedia.org/wiki/Hessian_matrix

Given $f(x, y)$

$f: R^2 \rightarrow R$

$$H_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

Given $f(x, y) = x^2 + 2x^2y + y^3$

$$\frac{\partial f}{\partial x} = 2x + 4xy$$

$$\frac{\partial f}{\partial y} = 2x^2 + 3y^2$$

$$H_f = \begin{bmatrix} 2 + 4y & 4x \\ 4x & 6y \end{bmatrix}$$

Logistic Regression-MSE

❖ Formulae

Model and Loss

$$z = \boldsymbol{\theta}^T \boldsymbol{x} = \boldsymbol{x}^T \boldsymbol{\theta}$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\mathrm{L} = (\hat{y} - y)^2$$

Derivative

$$\frac{\partial L}{\partial \theta_i} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial \theta_i} \qquad \frac{\partial \hat{y}}{\partial z} = \hat{y}(1 - \hat{y})$$

$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y) \qquad \frac{\partial z}{\partial \theta_i} = x_i$$

First-order partial derivatives

$$\frac{\partial L}{\partial \theta_i} = 2x_i(\hat{y} - y)\hat{y}(1 - \hat{y})$$

Mean Squared Error

Model and Loss

$$z = \boldsymbol{\theta}^T \boldsymbol{x} = \boldsymbol{x}^T \boldsymbol{\theta}$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\mathrm{L} = (\hat{y} - y)^2$$

Derivative

$$\frac{\partial L}{\partial \theta_i} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial \theta_i}$$

$$\frac{\partial \hat{y}}{\partial z} = \hat{y}(1 - \hat{y})$$

$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y)$$

$$\frac{\partial z}{\partial \theta_i} = x_i$$

$$\frac{\partial L}{\partial \theta_i} = 2x_i(\hat{y} - y)\hat{y}(1 - \hat{y})$$

$$\frac{\partial L}{\partial \theta_i} = 2x_i(\hat{y} - y)\hat{y}(1 - \hat{y}) = 2x_i\left(-\hat{y}^3 + \hat{y}^2 - y\hat{y} + y\hat{y}^2\right)$$

$$\frac{\partial^2 L}{\partial \theta_i^2} = \frac{\partial}{\partial \theta_i}\left[2x_i(-\hat{y}^3 + \hat{y}^2 - y\hat{y} + y\hat{y}^2)\right]$$

$$= 2x_i\left[-3\hat{y}^2 x_i \hat{y}(1 - \hat{y}) + 2x_i \hat{y}\hat{y}(1 - \hat{y}) - y x_i \hat{y}(1 - \hat{y}) + 2x_i y \hat{y}\hat{y}(1 - \hat{y})\right]$$

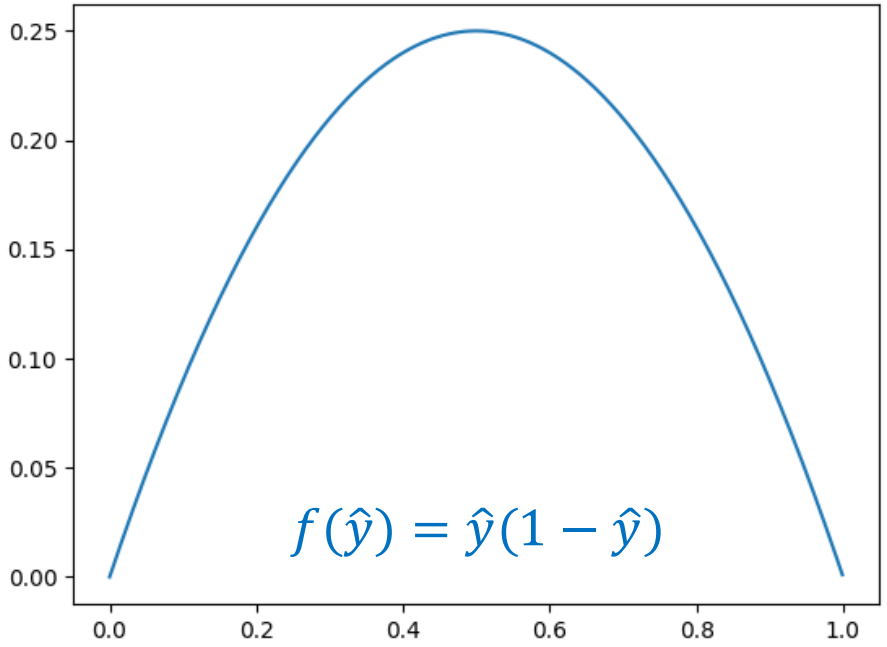
$$= 2x_i^2 \hat{y}(1 - \hat{y})\left[-3\hat{y}^2 + 2\hat{y} - y + 2y\hat{y}\right]$$

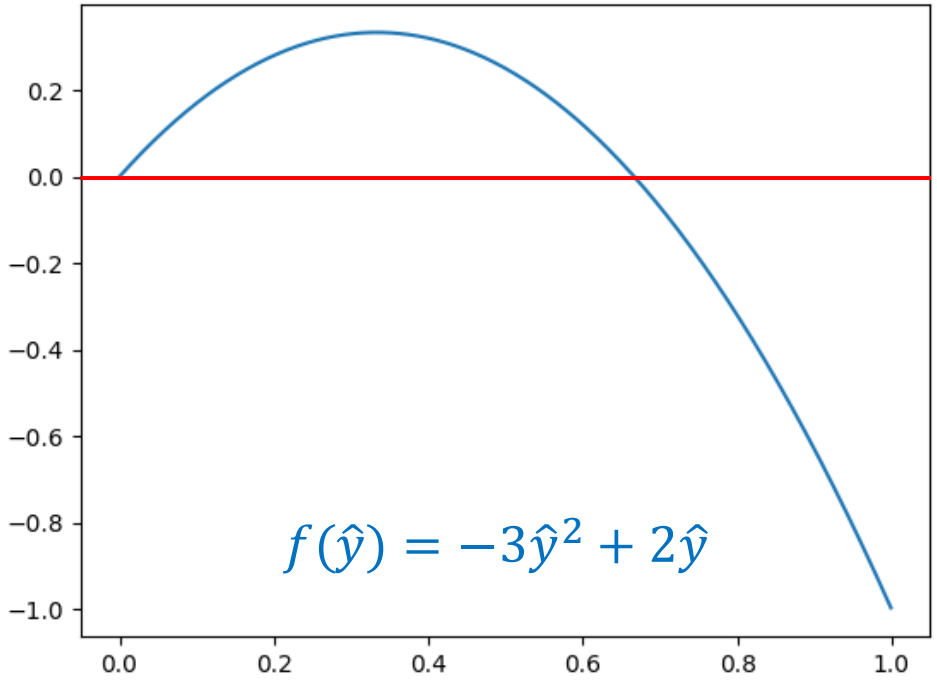
Mean Squared Error

$$\frac{\partial^2 L}{\partial \theta_i^2} = 2x_i^2 \hat{y}(1-\hat{y})[-3\hat{y}^2 + 2\hat{y} - y + 2y\hat{y}]$$

$x_i^2 \geq 0$

$\hat{y}(1-\hat{y}) \in \left[0, \frac{1}{4}\right]$



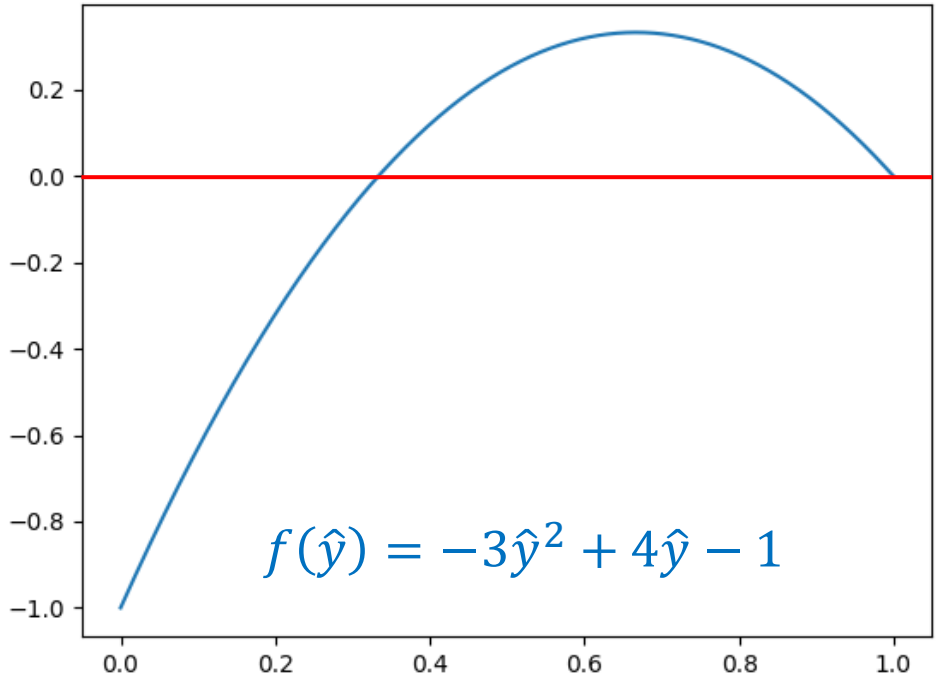


$y = 0$

$f(\hat{y}) = -3\hat{y}^2 + 2\hat{y}$

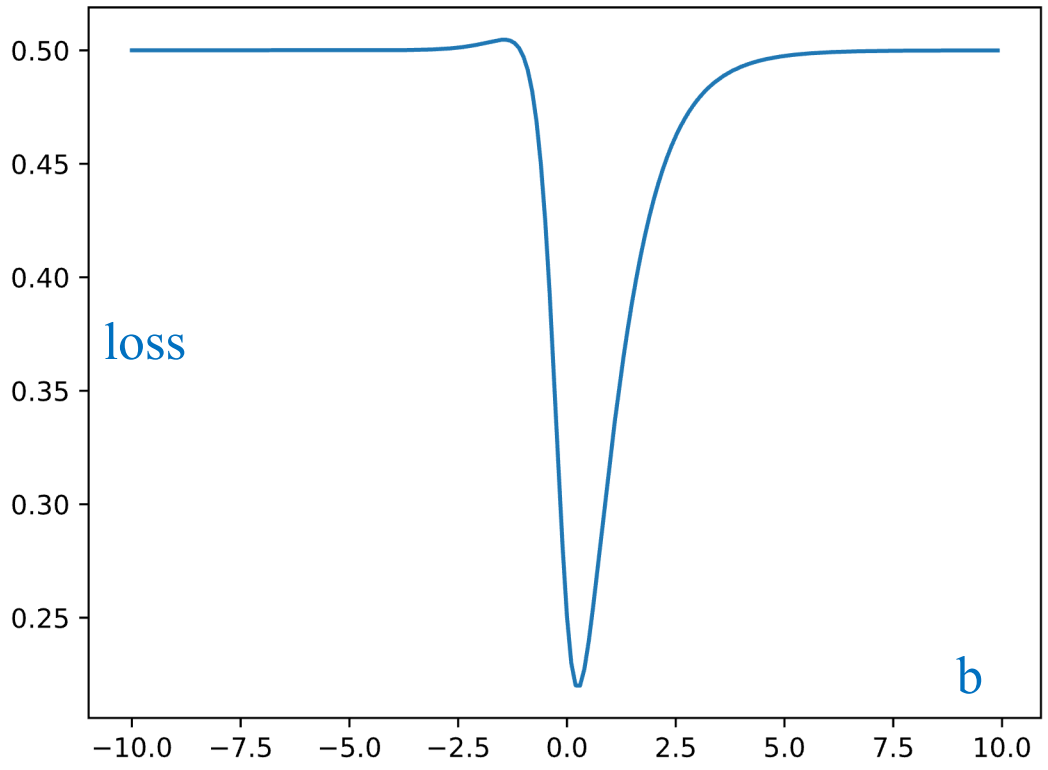
$y = 1$

$f(\hat{y}) = -3\hat{y}^2 + 4\hat{y} - 1$



Mean Squared Error

❖ Visualization



Mean Squared Error

Model and Loss

$$z = \boldsymbol{\theta}^T \boldsymbol{x} = \boldsymbol{x}^T \boldsymbol{\theta}$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\mathrm{L} = (\hat{y} - y)^2$$

$$\frac{\partial L}{\partial \theta_i} = 2x_i(\hat{y} - y)\hat{y}(1 - \hat{y})$$

What about Logistics Regression Using Binary Cross-entropy?

Logistic Regression (BCE)

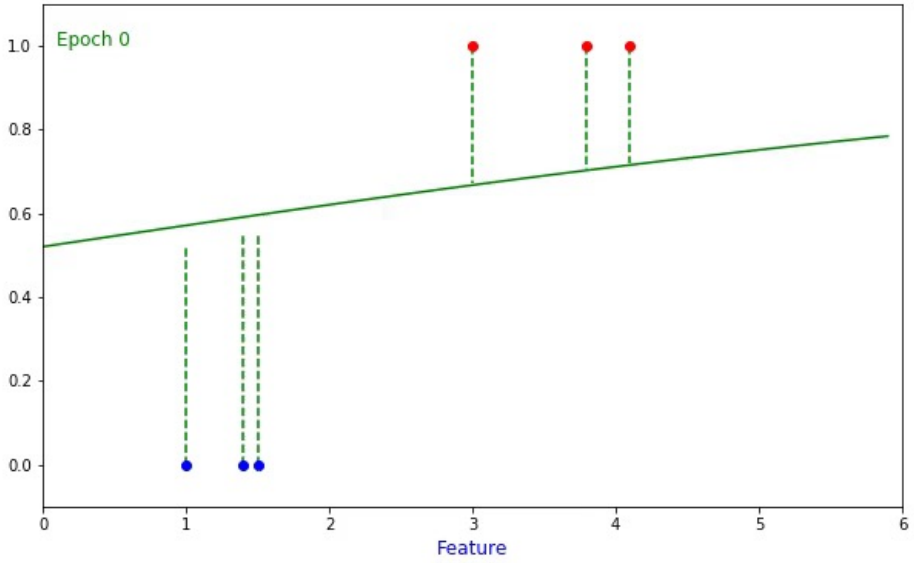
Feature	Label	
Petal_Length	Category	
1.4	0	Category 0
1	0	
1.5	0	
3	1	Category 1
3.8	1	
4.1	1	

$$z = wx + b$$

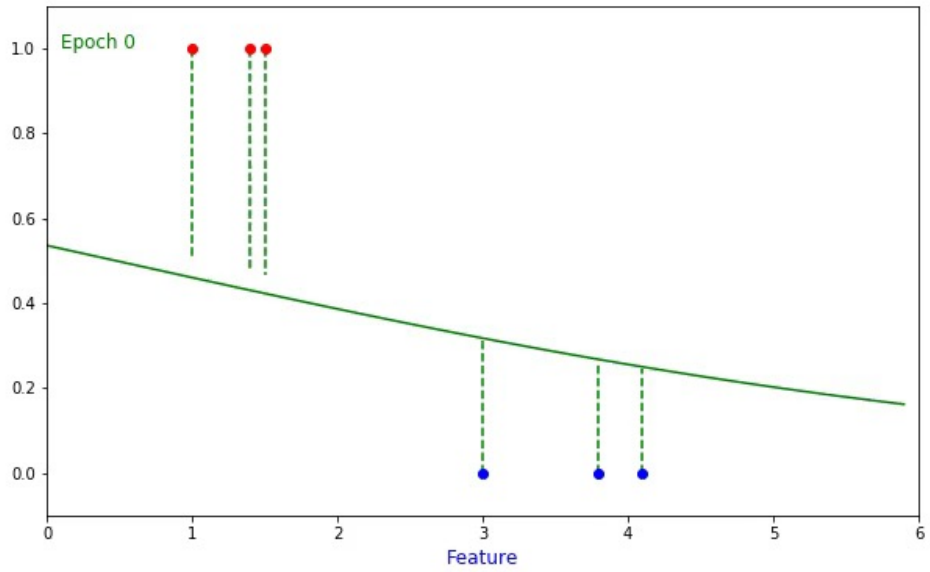
$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Feature	Label	
Petal_Length	Category	
1.4	1	Category 0
1	1	
1.5	1	
3	0	Category 1
3.8	0	
4.1	0	

$\frac{1}{1 + e^{-z}}$



$\frac{1}{1 + e^{-z}}$



Binary Cross-entropy

❖ Convex function

Model and Loss

$$z = \boldsymbol{\theta}^T \boldsymbol{x}$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\mathrm{L} = -y\log(\hat{y}) - (1-y)\log(1-\hat{y})$$

simplified version

$$\frac{\partial L}{\partial \theta_i} = x_i(\hat{y} - y)$$

$$\frac{\partial^2 L}{\partial \theta_i^2} = \frac{\partial}{\partial \theta_i}[x_i(\hat{y} - y)] = x_i^2(\hat{y} - \hat{y}^2) \geq 0$$

$$x_i^2 \geq 0 \qquad \hat{y} - \hat{y}^2 \in \left[0, \frac{1}{4}\right]$$

Derivative

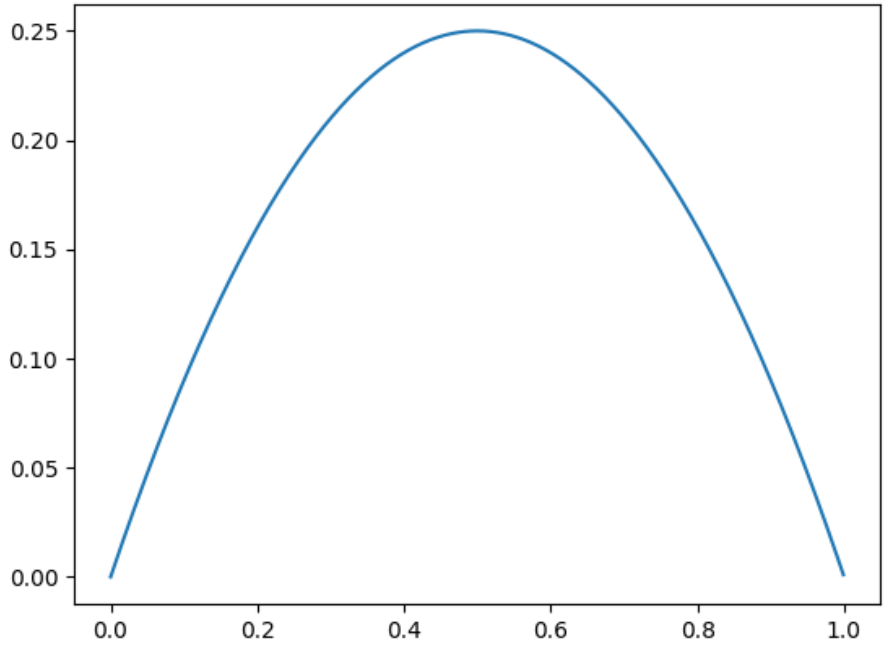
$$\frac{\partial L}{\partial \theta_i} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial \theta_i}$$

$$\frac{\partial L}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}$$

$$\frac{\partial \hat{y}}{\partial z} = \hat{y}(1-\hat{y})$$

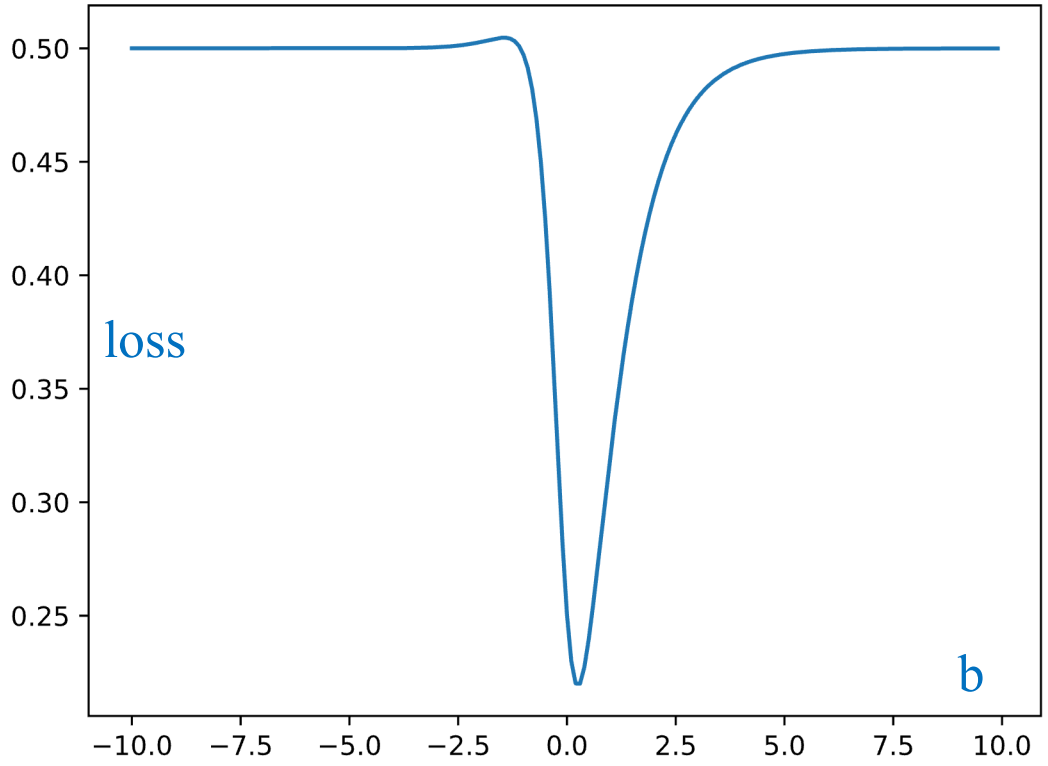
$$\frac{\partial z}{\partial \theta_i} = x_i$$

$$\frac{\partial L}{\partial \theta_i} = x_i(\hat{y} - y)$$

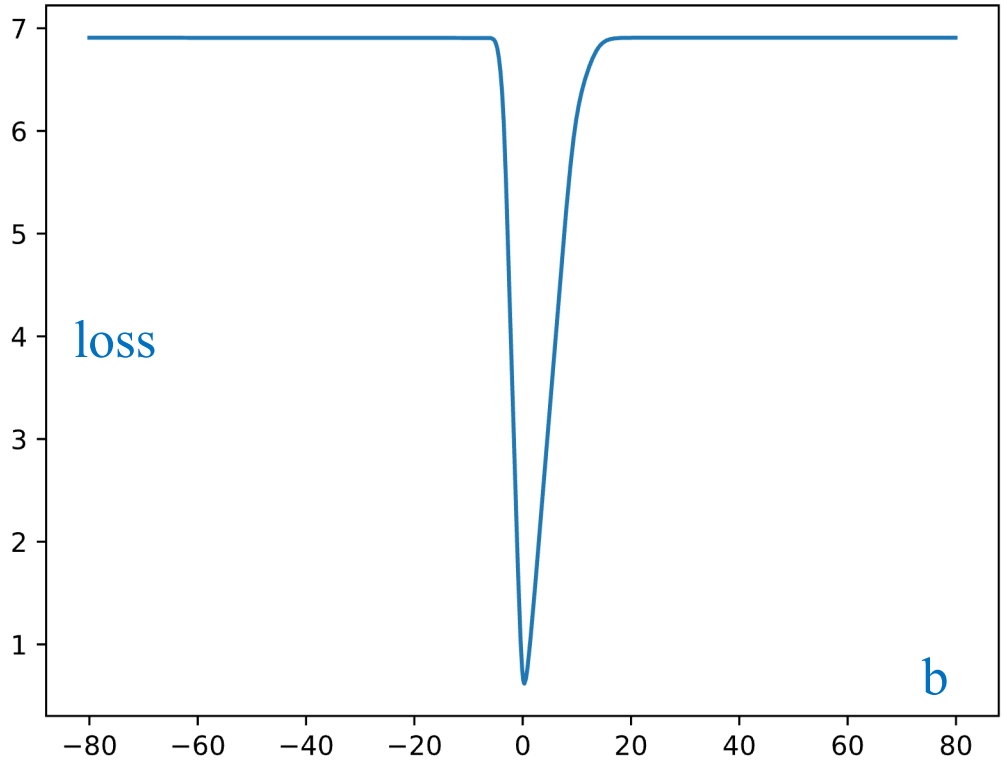


MSE and BCE

❖ Visualization



Mean Squared Error



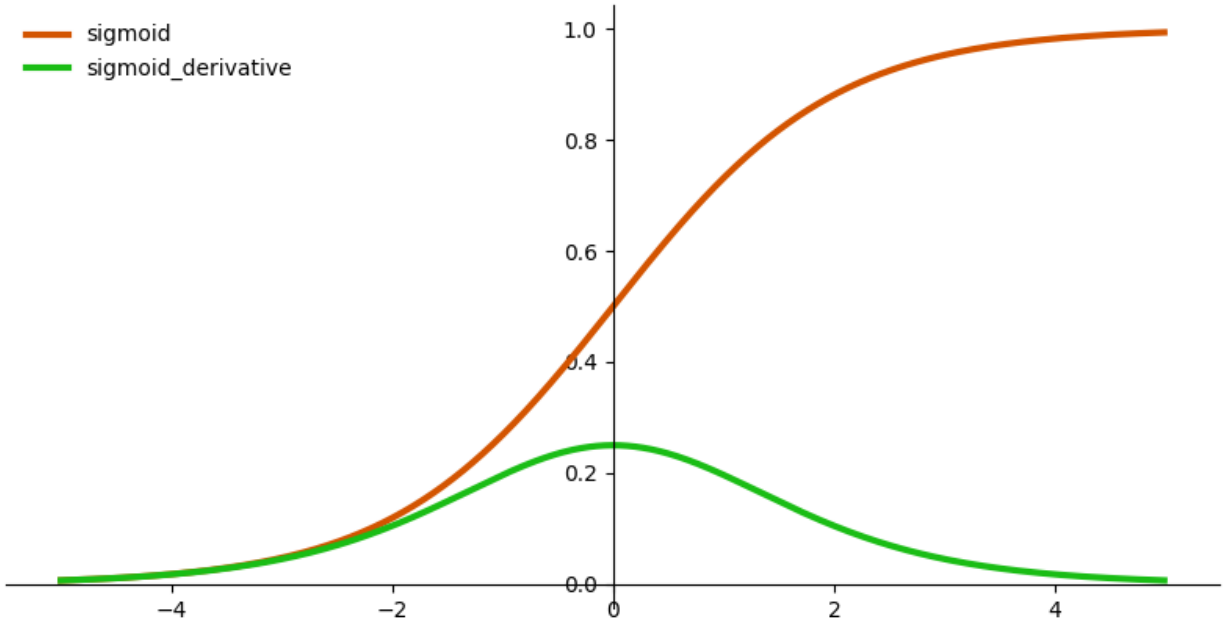
Binary Cross-Entropy

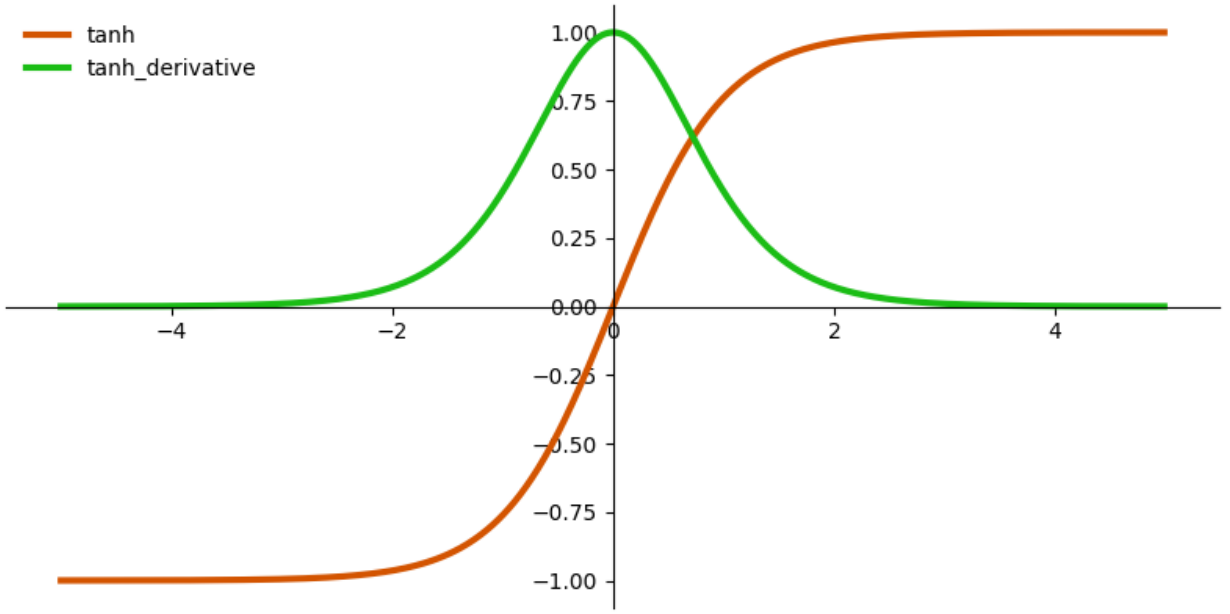
Sigmoid and Tanh Functions

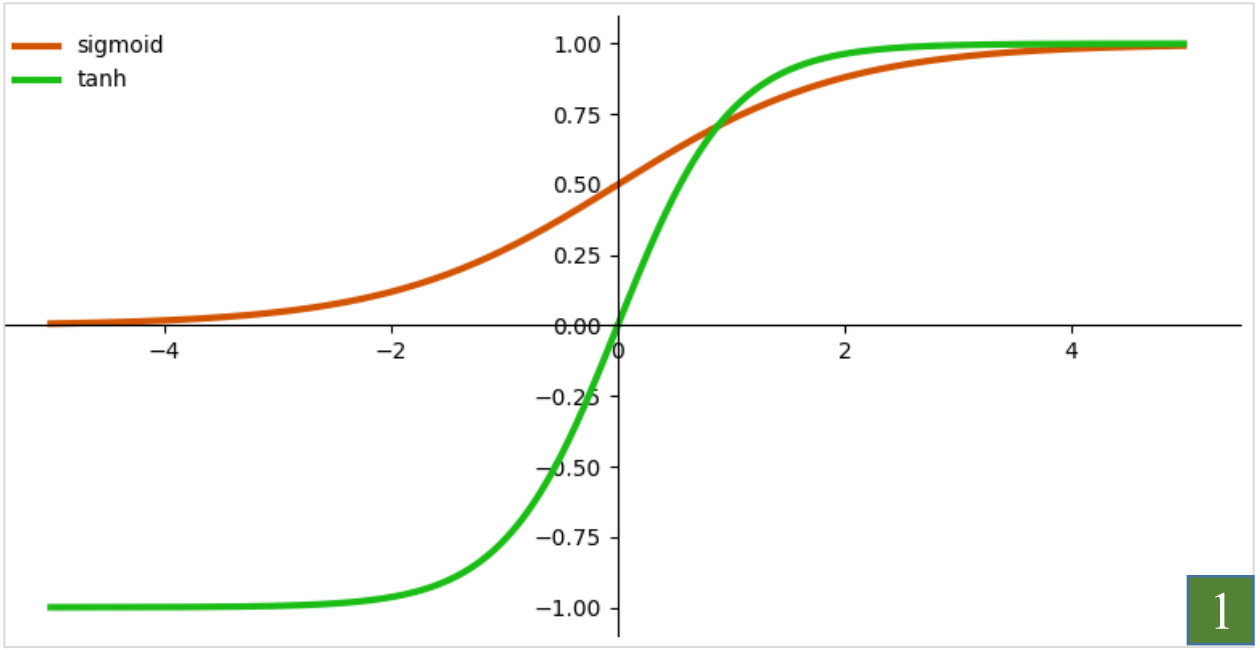
Sigmoid and Tanh Functions

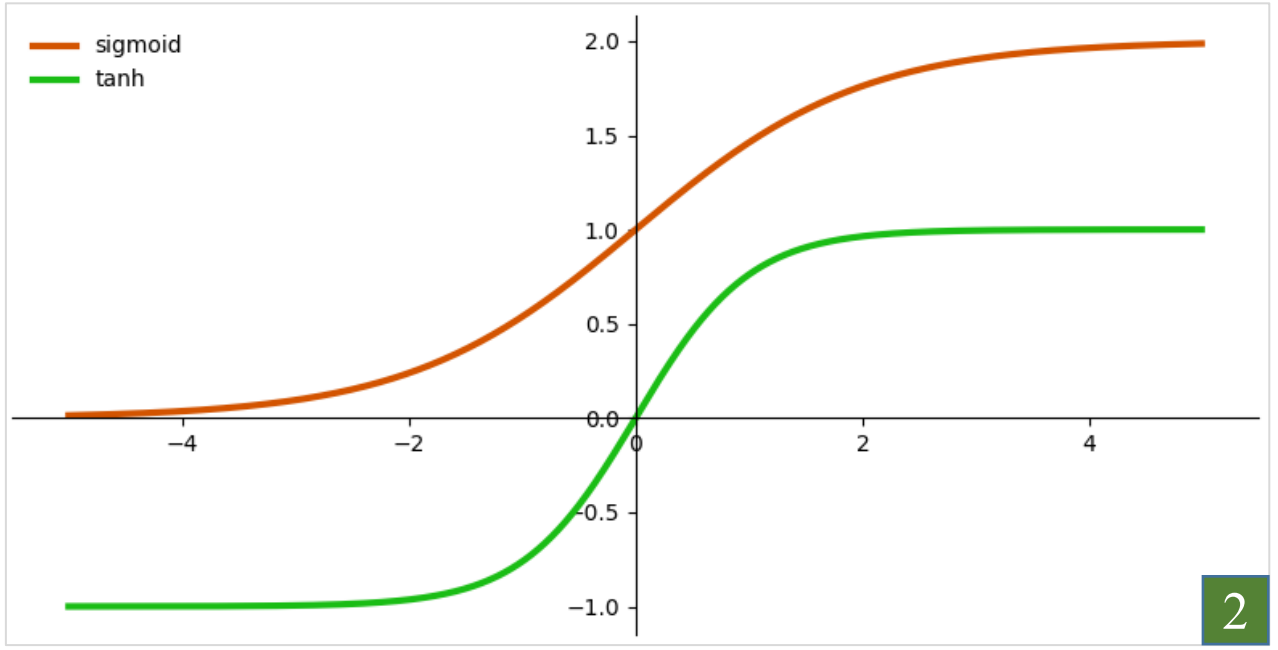
$$\text{sigmoid}(x) = \frac{1}{1 + e^{-x}}$$

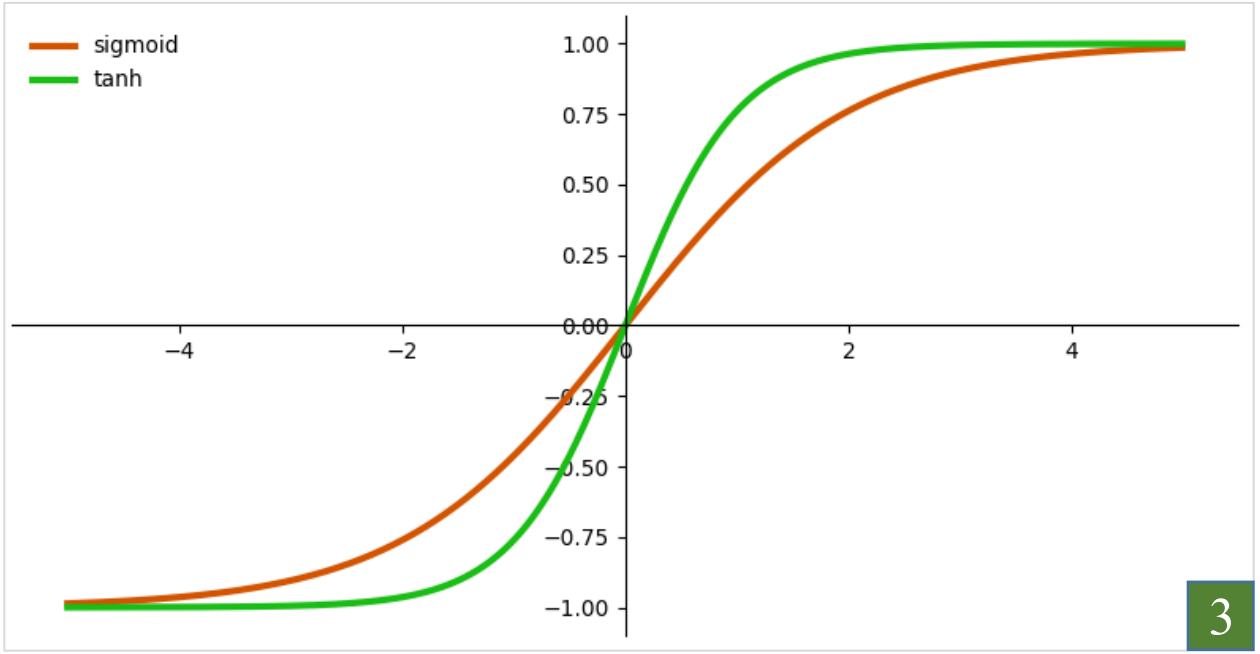
$$\tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

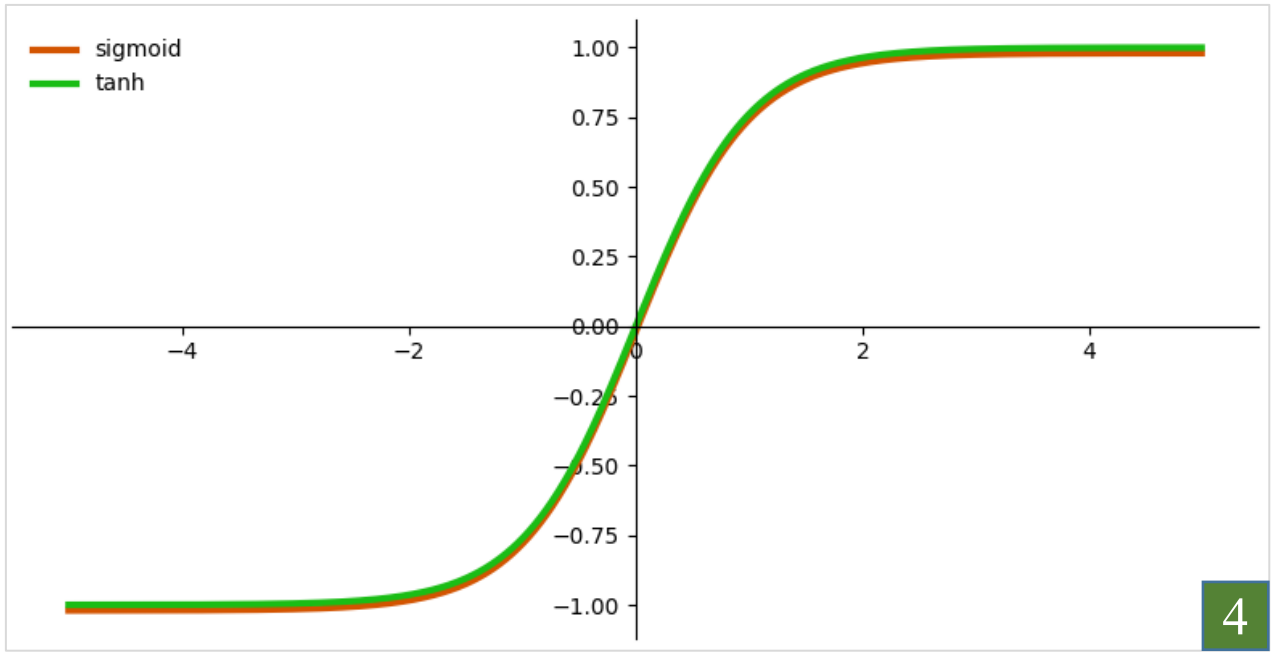





sigmoid
tanh
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−4
−2
0
2
4
1


sigmoid
tanh
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−4
−2
0
2
4
2


sigmoid
tanh
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−4
−2
0
2
4
3


sigmoid
tanh
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−4
−2
0
2
4
4

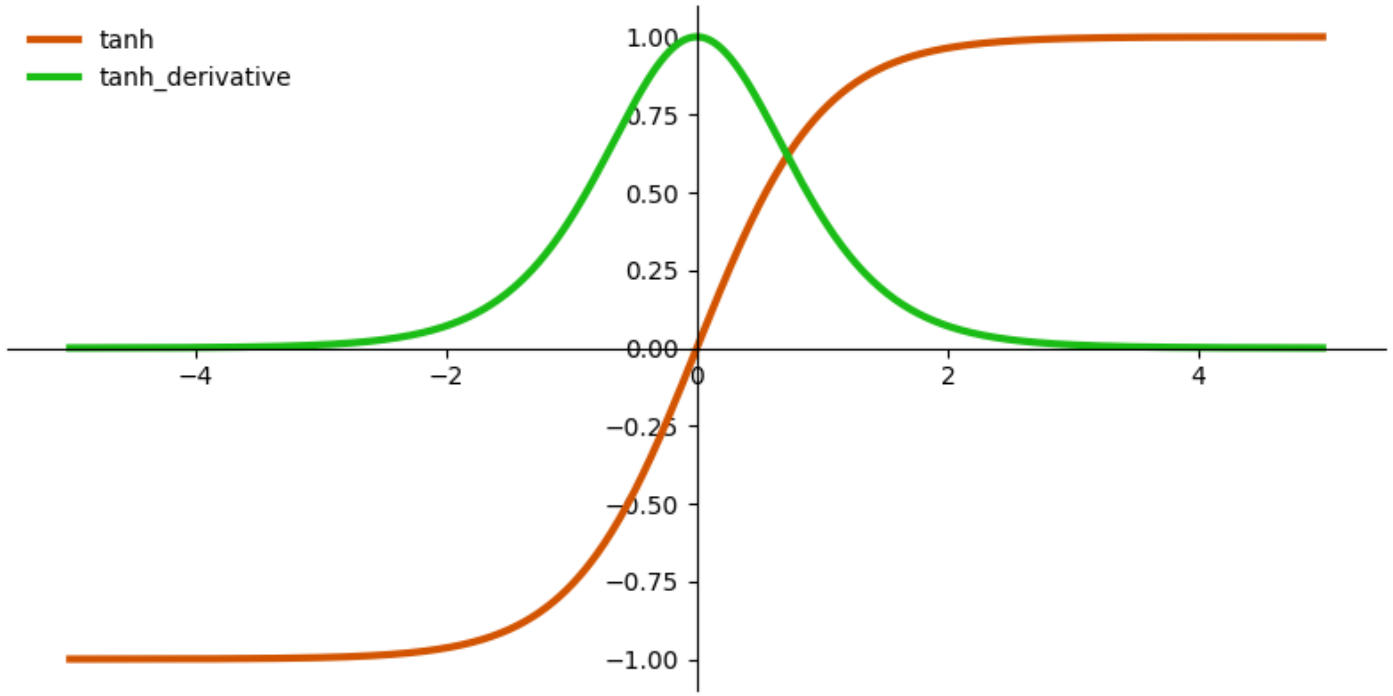
Tanh function

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}$$

$$= 1 - \frac{2}{e^{2x} + 1}$$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1 - e^{-2x}}{1 + e^{-2x}}$$

$$= -\frac{e^{-2x} - 1}{e^{-2x} + 1} = \frac{2}{e^{-2x} + 1} - 1$$



Tanh function

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1} = 1 - \frac{2}{e^{2x} + 1}$$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1 - e^{-2x}}{1 + e^{-2x}} = -\frac{e^{-2x} - 1}{e^{-2x} + 1} = \frac{2}{e^{-2x} + 1} - 1$$

$$tanh'(x) = \left(\frac{e^x - e^{-x}}{e^x + e^{-x}}\right)' = \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2}$$

$$= \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2}$$

$$= 1 - \left(\frac{e^x - e^{-x}}{e^x + e^{-x}}\right)^2 = 1 - tanh^2(x)$$

Tanh function

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1} = 1 - \frac{2}{e^{2x} + 1}$$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1 - e^{-2x}}{1 + e^{-2x}} = -\frac{e^{-2x} - 1}{e^{-2x} + 1} = \frac{2}{e^{-2x} + 1} - 1$$

$$tanh'(x) = \left(\frac{2}{e^{-2x} + 1} - 1\right)' = \frac{4e^{-2x}}{(e^{-2x} + 1)^2} = 4\left(\frac{e^{-2x} + 1 - 1}{(e^{-2x} + 1)^2}\right)$$

$$= 4\left(\frac{1}{e^{-2x} + 1} - \frac{1}{(e^{-2x} + 1)^2}\right) = -\left(\frac{4}{(e^{-2x} + 1)^2} - \frac{4}{e^{-2x} + 1}\right)$$

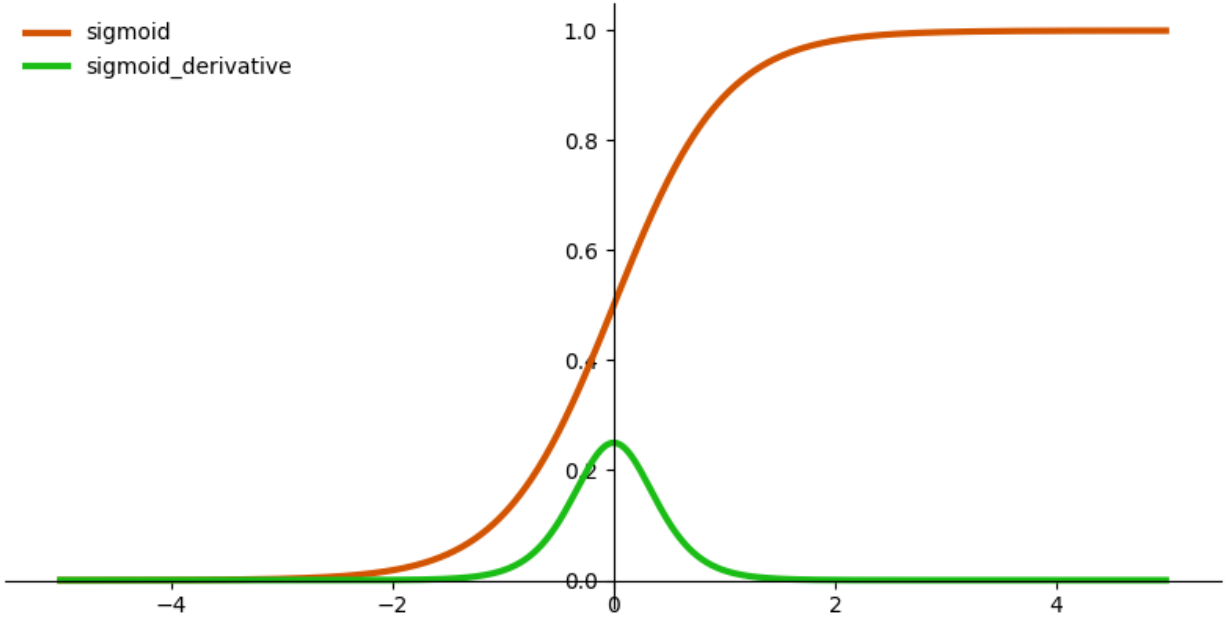
$$= -\left(\frac{4}{(e^{-2x} + 1)^2} - \frac{4}{e^{-2x} + 1} + 1 - 1\right) = 1 - \left(\frac{2}{e^{-2x} + 1} - 1\right)^2 = 1 - tanh^2(x)$$

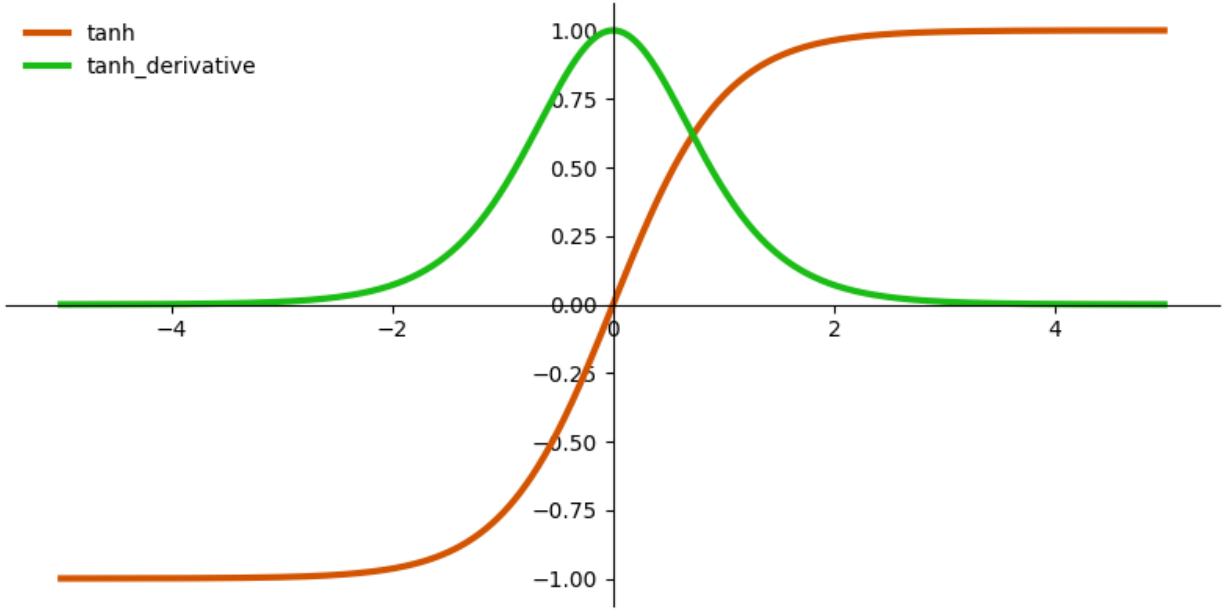
Sigmoid and Tanh Functions

$$\text{sigmoid}(2x) = \frac{1}{1 + e^{-2x}}$$

$$\tanh(x) = 2 \times \frac{1}{1 + e^{-2x}} - 1$$

$$\tanh(x) = 2 \times \text{sigmoid}(2x) - 1$$

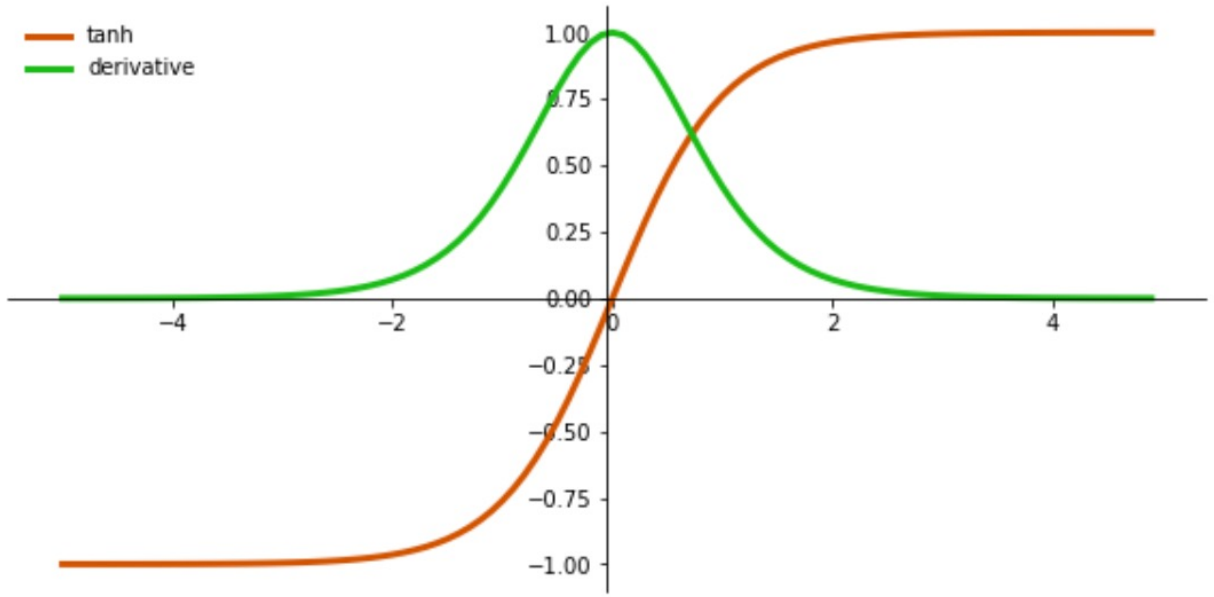




Logistic Regression-Tanh

❖ Construct loss

$$\tanh(x) = \frac{2}{1 + e^{-2x}} - 1$$



Model and Loss

$$z = \boldsymbol{\theta}^T \boldsymbol{x} = \boldsymbol{x}^T \boldsymbol{\theta}$$

$$\hat{y} = tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$$\hat{y}_s = \frac{\hat{y} + 1}{2}$$

$$\mathrm{L} = -y\log(\hat{y}_s) - (1 - y)\log(1 - \hat{y}_s)$$

Derivative

$$\frac{\partial L}{\partial \theta_i} = \frac{\partial L}{\partial \hat{y}_s}\frac{\partial \hat{y}_s}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial z}\frac{\partial z}{\partial \theta_i}$$

$$\frac{\partial L}{\partial \hat{y}_s} = -\frac{y}{\hat{y}_s} + \frac{1 - y}{1 - \hat{y}_s} = \frac{\hat{y}_s - y}{\hat{y}_s(1 - \hat{y}_s)}$$

$$\frac{\partial \hat{y}_s}{\partial \hat{y}} = \frac{1}{2} \qquad \frac{\partial \hat{y}}{\partial z} = 1 - \hat{y}^2 \qquad \frac{\partial z}{\partial \theta_i} = x_i$$

$$\frac{\partial L}{\partial \theta_i} = x_i \frac{(\hat{y}_s - y)(1 - \hat{y}^2)}{2\hat{y}_s(1 - \hat{y}_s)}$$

Logistic Regression-Tanh

❖ Construct loss

$$\tanh(x) = \frac{2}{1+e^{-2x}} - 1$$

Model and Loss

$$z = \boldsymbol{\theta}^T \boldsymbol{x} = \boldsymbol{x}^T \boldsymbol{\theta}$$

$$\hat{y} = tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$$\hat{y}_s = \frac{\hat{y}+1}{2}$$

$$\mathrm{L} = -y\log(\hat{y}_s) - (1-y)\log(1-\hat{y}_s)$$

Derivative

$$\frac{\partial L}{\partial \theta_i} = \frac{\partial L}{\partial \hat{y}_s}\frac{\partial \hat{y}_s}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial z}\frac{\partial z}{\partial \theta_i}$$

$$\frac{\partial L}{\partial \hat{y}_s} = -\frac{y}{\hat{y}_s} + \frac{1-y}{1-\hat{y}_s} = \frac{\hat{y}_s - y}{\hat{y}_s(1-\hat{y}_s)}$$

$$\frac{\partial \hat{y}_s}{\partial \hat{y}} = \frac{1}{2} \qquad \frac{\partial \hat{y}}{\partial z} = 1-\hat{y}^2 \qquad \frac{\partial z}{\partial \theta_i} = x_i$$

$$\frac{\partial L}{\partial \theta_i} = x_i \frac{(\hat{y}_s - y)(1-\hat{y}^2)}{2\hat{y}_s(1-\hat{y}_s)}$$

$$\frac{\partial L}{\partial \theta_i} = x_i \frac{(\frac{\hat{y}+1}{2} - y)(1-\hat{y}^2)}{2\frac{\hat{y}+1}{2}(1-\frac{\hat{y}+1}{2})}$$

$$\frac{\partial L}{\partial \theta_i} = x_i \frac{(\hat{y}+1-2y)(1-\hat{y}^2)}{(\hat{y}+1)(1-\hat{y})}$$

$$\frac{\partial L}{\partial \theta_i} = x_i(\hat{y}+1-2y)$$

⬅

$$\frac{\partial^2 L}{\partial \theta_i^2} = \frac{\partial}{\partial \theta_i}[x_i(\hat{y}+1-2y)] = x_i^2(1-\hat{y}^2)$$

